

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 321 — Past Questions & Materials
by Renance

Course: PEE 321
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 300

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Paper 1 of 4 — PEE 321 Petroleum Drilling Technology

Level 300 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
PROGRAMME: B.ENG IN PETROLEUM ENGINEERING
SESSION: 2023/2024, SEMESTER: SECOND SEMESTER EXAMINATION.
COURSE TITLE: | PETROLEUM DRILLING TECHNOLOGY
COURSE CODE: | PEE 321.

COURSE UNIT: | 3 1

| DATE OF EXAM: | 23rd SEPTEMBER, 2024 _ cen

INSTRUCTIONS: Attempt Question i and any other 3 TIME: 3 Hours

SECTION A: Attempt All

Question 1

- (a) What are the two main advantages of PDC bit over roller cone bit?
- (b) One of the functions of drilling fluid is its use in formation evaluation. How?
- (c) How are the pump or circulating pressure, the pressure drop across the bit nozzles and the pressure loss in the system related?
- (d) What are the shortcomings of drilling bit cost equation?
- (e) Why is circulation often lost while drilling large diameter top hole?
- (f) What can be done to prevent the situation in (e) above?
- (g) Calculate the pump output required, given the following data: Annular Velocity = 100 ft/min. Hole Diameter = 12.25 inches, Drill Pipe OD = 5 inches

Pod SECTION B: Attempt any 3

(yesdo is

- (a) What are the main components of a rotary drilling rig?
- (b) What do you understand by the following? (i) making a connection (ii) making a trip (iii) kick (iv) drilling-in (v) a stand.
- (c) What is the function of each of the following? (i) swivel (ii) Kelly saver sub (iii) drill collars (iv) stabilizer subs.

Question 5

- (a) Succinctly explain any three well control methods you know, highlighting any three possible advantages.
- (b) A pump is being operated at a rate 800 gal/min and a pressure 3000 psig. The density of the drilling fluid is 15 ppg, and the nozzle area of the bit is 0.589 square inch. (i) compute the power developed by the pump. (ii) compute the power loss to viscous effects. What happened to this energy? (iii) compute the impact force of the jets of fluid against the bottom of the

hole.

Question 4

(a) You are given the following information about a well. Complete the table. Depth = 8000 ft. Bit Cost = N200, Rig Cost = N100 per hour, Round Trip Time = 2 hr per 1000 ft.

T (hrs) R (ft/hr) F (ft) Total Cost/ft

5 25 137.50
10 20 236.00
12 18 288.00
14 16 322.00
16 14 352.00
18 12 378.00
20 10 400.00

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b) A drillstring i .

j (0) avin Fe Oth onacg of 9000 ft of 5-in. 19.5 lbm/ft drillpipe and 1000 ft of drill collars collars in bbls (ii) -in ID. Calculate: (i) capacity of drillpipe in bbls (ii) capacity of drill is a duplex d i number of pump cyc:es. required to pump surface mud to the bit (the pump golumetiia eaienarat ata be 6-in liners, 2.5-in rods, 16-in strokes and operates at a : ' yo 0) (iv) displacement of the drillpipe i F i of drill collars in bbl/R. ID of drillpipe = 4.276-in e drillpipe in bbl/ft. (v) displacement

Question 5

(a) er bis the pies bits had the best (least) cost/ft pean assuming the Rig Cost is \$4, e 1 Vike 9 88, "D tite trove per day ae "igen = \$290 pe how.

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| S/No | Bit No | Bit Bit Depth | Footage | Time Trip | Bit |
Size Type | Out ({t) Rotating | Time | Cost
| (\$)

QQ 1 n-9 Big Bad 1500 12] 12 10 \$925

ae" 2 a8 |Big | Good | 15,948 [148 | 18 HI \$825 _|

ft 3 n-7 Big Poor 16,080 | 132 14 5! \$ 1050

¢ 4 n-6 Big Fair 16,250 | 170 22 12 \$ 2100

yo

ed, (ii) rotating time, (iii) bearing wear,

(b) Calculate the expected value of (i) footage expect connection time, Tt = 7 hrs, Bit cost, Cb

et (v) cost per foot for the data given below: Trip and

= \$2000, Rig cost = \$2500/hr, weight on bit (W) = 60,000 lb, Rotating speed, N = 80 rpm, Flowrate, Q = 400 gal/min, equivalent nozzle diameter, dn = 0.8679 inches, Tooth wear, T-

$7 = 0.875$, Drillability constant, $K = 8.51$ {t/hr, Formation abrasiveness constant, $A_f = 24.9$ hrs, Bearing constant, $S = 32.77$ hrs. Bit formation and mud type (i.e. dullness computation) $a_1 = 1.2$, $a_2 = 0.6$, $a_3 = 0.4$. Formation bit factors: $D_b = 9.875$, $H_1 = 1.84$, $H_2 = 6$, $H_3 = 0.8$
 $YJ / D_b y_{max} = 8$

Question &

(a) (i) Give five reasons why it may be necessary to drill directional wells (ii) List the areas of application of directional wells in the industry. (iii) What is a multi-lateral well and list the types that are currently used in the industry with diagrams to describe them.

(b) (i) State four major components to be considered when planning a well trajectory. (ii) State briefly the difference between a vertical well, deviated well, horizontal well and multi-lateral well.

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Paper 2 of 4 — PEE 321 PETROLEUM DRILLING TECHNOLOGY

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
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COURSE UNIT:	3
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TIME: 3 Hours

SECTION A: Attempt All

Question 1

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- One of the functions of drilling fluid is its use in formation evaluation. How?
- How are the pump or circulating pressure, the pressure drop across the bit nozzles and the pressure loss in the system related?
- What are the shortcomings of drilling bit cost equation?
- Why is circulation often lost while drilling large diameter top hole?
- What can be done to prevent the situation in (e) above?
- Calculate the pump output required, given the following data: Annular Velocity = 100 ft/min. Hole Diameter = 12.25 inches, Drill Pipe OD = 5 inches

SECTION B: Attempt any 3

Question 2

- What are the main components of a rotary drilling rig?
- What do you understand by the following? (i) making a connection (ii) making a trip (iii) kick (iv) drilling-in (v) a stand.
- What is the function of each of the following? (i) swivel (ii) Kelly saver sub (iii) drill collars (iv) stabilizer subs.

Question 3

- Succintly explain any three well control methods you know, highlighting any three possible advantages.
- A pump is being operated at a rate 300 gal/min and a pressure 3000 psig. The density of the drilling fluid is 15 ppg, and the nozzle area of the bit is 0.589 square inch. (i) compute the power developed by the pump. (ii) compute the power loss to viscous effects. What happened to this energy? (iii) compute the impact force of the jets of fluid against the bottom of the hole.

Question 4

- You are given the following information about a well. Complete the table. Depth = 8000 ft, Bit Cost = N200, Rig Cost = N100 per hour, Round Trip Time = 2 hr per 1000 ft.

T (hrs)	R (ft/hr)	F (ft)	Total Cost/ft
5	25	137.50	
10	20	236.00	
12	18	288.00	
14	16	322.00	
16	14	352.00	
18	12	378.00	
20	10	400.00	

$D = 3.416 r^2 \times S \times C$ C = number of cylinder
S = length of piston stroke r = radius

- (b) A drillstring is made up of 9000 ft of 5-in. 19.5 lbm/ft drillpipe and 1000 ft of drill collars having 8-in OD and 3-in ID. Calculate: (i) capacity of drillpipe in bbls (ii) capacity of drill collars in bbls (iii) number of pump cycles required to pump surface mud to the bit (the pump is a duplex double acting pump with 6-in liners, 2.5-in rods, 16-in strokes and operates at a volumetric efficiency of 85%) (iv) displacement of the drillpipe in bbl/ft. (v) displacement of drill collars in bbl/ft. ID of drillpipe = 4.276-in.

Question 5

- (a) Which of the following bits had the best (least) cost/ft performance, assuming the Rig Cost is \$4,800 per day

$\frac{\$4800 \times \text{day}}{24 \text{ hrs}} = \200 per hour.

S/No	Bit No	Bit Size	Bit Type	Depth Out	Footage (ft)	Time Rotating	Trip Time	Bit Cost (\$)
1	n-9	Big	Bad	1500	121	12	10	\$ 925
2	n-8	Big	Good	15,948	148	18	11	\$ 825
3	n-7	Big	Poor	16,080	132	14	11	\$ 1050
4	n-6	Big	Fair	16,250	170	22	12	\$ 2100

- (b) Calculate the expected value of (i) footage expected, (ii) rotating time, (iii) bearing wear, (v) cost per foot for the data given below: Trip and connection time, $T_t = 7$ hrs, Bit cost, $C_b = \$2000$, Rig cost = \$2500/hr, weight on bit (W) = 60,000 lb, Rotating speed, $N = 80$ rpm, Flowrate, $Q = 400$ gal/min, equivalent nozzle diameter, $d_n = 0.8679$ inches, Tooth wear, $T_7 = 0.875$, Drillability constant, $K = 8.51$ ft/hr, Formation abbrassiveness constant, $A_f = 24.9$ hrs, Bearing constant, $S = 32.77$ hrs. Bit formation and mud type (i.e. dullness computation) $a_1 = 1.2$, $a_2 = 0.6$, $a_3 = 0.4$. Formation bit factors: $D_b = 9.875$, $H_1 = 1.84$, $H_2 = 6$, $H_3 = 0.8$ ($\sqrt{W/D_b}$)_{max} = 8

Question 6

- (a) (i) Give five reasons why it may be necessary to drill directional wells (ii) List the areas of application of directional wells in the industry. (iii) What is a multi-lateral well and list the types that are currently used in the industry with diagrams to describe them.
- (b) (i) State four major components to be consider when planning a well trajectory. (ii) State briefly the difference between a vertical well, deviated well, horizontal well and multi-lateral well.

$$C_T = \frac{B}{F} + \frac{R(T)}{F}$$

$$C_T = \frac{B}{F} + \frac{C_b T_t}{F} + \frac{C_r T_r}{F}$$

$$C_T = \frac{B + C_b (T_t + T_r)}{F}$$

$$T_t = 7 \text{ hrs}$$

$$C_b = \$2000$$

$$C_r = \$2500/\text{hr}$$

$$W = 60000 \text{ lb}$$

$$N = 80 \text{ rpm}$$

$$Q = 400 \text{ gal/min}$$

$$d_n = 0.8679 \text{ inches}$$

$$T_7 = 0.875$$

$$K = 8.51 \text{ ft/hr}$$

$$A_f = 24.9 \text{ hrs}$$

$$S = 32.77 \text{ hrs}$$

$$a_1 = 1.2 \quad a_2 = 0.6 \quad a_3 = 0.4$$

$$D_b = 9.875 \quad H_1 = 1.84 \quad H_2 = 6$$

$$H_3 = 0.8 \quad \left(\frac{W}{D_b}\right)_{\text{max}} = 8$$

$$\begin{array}{r} F=7 \\ B=7 \\ C_b=7 \\ C_r=7 \end{array}$$

$$\begin{array}{r} 34 \\ \times 2 \\ \hline 68 \end{array}$$

$$\begin{array}{r} 129 \\ \times 5 \\ \hline 645 \end{array}$$

Paper 3 of 4 — PEE321 past question

Level 300 • Second Semester • PastQuestion

SECTION B

QUESTION 2 (15 Marks)

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(H) ccececereces i @ Surface connecting positions of points on the wave that are doing
the same thing a:

is also known as incompressibility in elastic constant definition

Pe The three (3) types of emissions manifest in radioactive elements are, 2-----

(v) eee os BIS Bisests bo aeu ig viet on emitted from an excited nucleus.

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are electromagnetic (EM) radiati

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xi) The loss of energy at grain boundaries results in a dec

tease in amplitude is known

_(xii) The total energy of the reflected and transmitted waves equals the..... Ew ie
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(xiv) The proportion of energy reflected and or transmitted are determined/depends on
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(xvi) At critical angle, the refracted wave will travel.....0..cc.cccccccceecsescesdennee With
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ss Zeb Tai esh Rees csligysccs ses is the distance at which the travel time of direct wave equals
the travel time of refracted wave

CTION C

QUESTION 3 (20 marks)

(a) Briefly describe the gravity exploration method under the following Sub-headings;

(i) Measured physical property

(ii) — Principle

(iii) | Application (10 marks)

(b) Copy and complete the table below; (6 marks)

S/N	Method	Measured Parameter	Operative physical property
i.	Spatial variations in the		strength of

strength of

the gravitational field of the Earth

ii. _ | Magnetic

iii, Electrical Resistivity

(c) What is Data reduction? (4 marks)

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—(b) A drillstring is made up of 9000 ft of 5-in. 19.5 lbm/ft drillpipe and 1000 ft of drill collars having 8-in OD and 3-in ID, Calculate: (i) capacity of drillpipe in bbls (ii) capacity of drill —g" collars in bbls (iii) number of pump cycles required to pump surface mud to the bit (the pump ____ isa duplex double acting pump with 6-in liners, 2.5-in rods, 16-in strokes and operates at a ____ volumetric efficiency of 85%) (iv) displacement of the drillpipe in bbl/ft. (v) displacement ' of drill collars in bbl/ft. ID of drillpipe = 4.276-in. a Cae

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Bit No | Bit Bit Depth | Footage | Time Trip | Bit

Size Type | Out (ff) Rotating | Time | Cost

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of Big [Bad {1500 [121 | 12 101 \$925

| Good | 15,948 | 148 | 18 11 [\$825

Poor | 16,080 | 132 | 14 11 | \$1050

ome 16,250 |170 [22 12 [\$2100

al = 1.2, a2 = 0.6, a3 = 0.4. Formation ; P 'H3=0.8 ei

(W/Db)max = 8 . .

Question 6

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Paper 4 of 4 — PEE 321

Level 300 • Second Semester • Handout

DIRECTIONAL DRILLING

. INTRODUCTION

In the early days of land drilling most wells were drilled vertically, straight down into the reservoir. Although these wells were considered to be vertical, they rarely were. Some deviation in a wellbore will always occur, due to formation effects and bending of the drillstring. The first recorded instance of a well being deliberately drilled along a deviated course was in California in 1930. This well was drilled to exploit a reservoir which was beyond the shoreline underneath the Pacific Ocean. It had been the practise to build jetties out into the ocean and build the drilling rig on the jetty. However, this became prohibitively expensive and the technique of drilling deviated wells was developed. Since then many new techniques and special tools have been introduced to control the path of the wellbore.

An operating company usually hires a directional drilling service company to: provide expertise in planning the well; supply special tools; and to provide onsite assistance when operating the tools. The operator may also hire a surveying company to measure the inclination and direction of the well as drilling proceeds.

In this chapter we will discuss: the applications of directional well drilling; the design of these wells; and the techniques used to drill a well with controlled deviation from the vertical. The next chapter will discuss the tools and techniques used to survey the position of the well (determine the three dimensional position of all points in the wellbore relative to the wellhead).

2. APPLICATIONS

There are many reasons for drilling a non-vertical (deviated) well. Some typical applications of directionally controlled drilling are shown in Figure 1.

(a) Multi-well Platform Drilling

Multi-well Platform drilling is widely employed in the North Sea. The development of these fields is only economically feasible if it is possible to drill a large number of wells (up to 40 or 60) from one location (platform). The deviated wells are designed to intercept a reservoir over a wide aerial extent. Many oilfields (both onshore and offshore) would not be economically feasible if not for this technique.

(b) Fault Drilling

If a well is drilled across a fault the casing can be damaged by fault slippage. The potential for damaging the casing can be minimised by drilling parallel to a fault and then changing the direction of the well to cross the fault into the target.

(c) Inaccessible Locations

Vertical access to a producing zone is often obstructed by some obstacle at surface (e.g. river estuary, mountain range, city). In this case the well may be directionally drilled into the target from a rig site some distance away from the point vertically above the required point of entry into the reservoir.

(d) Sidetracking and Straightening

It is in fact quite difficult to control the angle of inclination of any well (vertical or deviated) and it may be necessary to 'correct' the course of the well for many reasons. For example, it may be necessary in the event of the drillpipe becoming stuck in the hole to simply drill around the stuckpipe (or fish), or plug back the well to drill to an alternative target.

(e) Salt Dome Drilling

Salt domes (called Diapirs) often form hydrocarbon traps in what were overlying reservoir rocks. In this form of trap the reservoir is located directly beneath the flank of the salt dome. To avoid potential drilling problems in the salt (e.g. severe washouts, moving salt, high pressure blocks of dolomite) a directional well can be used to drill alongside the Diapir (not vertically down through it) and then at an angle below the salt to reach the reservoir.

(f) Relief Wells

If a blow-out occurs and the rig is damaged, or destroyed, it may be possible to kill the "wild" well by drilling another directionally drilled well (relief well) to intercept or pass to within a few feet of the bottom of the "wild" well. The "wild" well is killed by circulating high density fluid down the relief well, into and up the wild well.

3. DEPTH REFERENCE AND GEOGRAPHICAL REFERENCE SYSTEMS

The trajectory of a deviated well must be carefully planned so that the most efficient trajectory is used to drill between the rig and the target location and ensure that the well is drilled for the least amount of money possible. When planning, and subsequently drilling the well, the position of all points along the wellpath and therefore the trajectory of the well must be considered in three dimensions (Figure 2). This means that the position of all points on the trajectory must be expressed with respect to a three dimensional reference system. The three dimensional system that

is generally used to define the position of a particular point along the wellpath is:

- The vertical depth of the point below a particular reference point
- The horizontal distance traversed from the wellhead in a Northerly direction
- The distance traversed from the wellhead in an Easterly direction

The depth of a particular point in the wellpath is expressed in feet (or meters) vertically below a reference (datum) point and the Northerly and Easterly displacement of the point is expressed in feet (or meters) horizontally from the wellhead.

3.1 Depth Reference Systems

There are a number of datum systems used in the depth reference systems. The datum systems which are most widely used are :

- Mean Sea Level, MSL
- Rotary Table Elevation, RTE
- 20" Wellhead Housing

The Mean Sea Level, MSL is a permanent, national and well documented datum where as datum such as the Rotary Table Elevation, RTE only exists when the drilling rig is on site. The top of the 20" Wellhead Housing is only available when the wellhead housing has been installed and will be removed when the well is abandoned. Hence, since the only permanent datum is the MSL (the rig

will be removed and the wellhead may be removed on abandonment) the distance between the MSL and the rotary table on the drillfloor and the MSL and the wellhead housing must be measured and recorded carefully on the well survey documents. The elevation of the rotary table above the MSL will be measured when the drilling rig is placed over the drilling location.

The depths of the formations to be penetrated are generally referenced, by the geologists and reservoir engineers, to MSL since the Rotary Table Elevation will not be known until the drilling rig is in place. In most drilling operations the Rotary Table elevation (RTE) is used as the working depth reference since it is relatively simple, for the driller for instance, to measure depths relative to this point. The elevation of the RTE is also referred to as Derrick Floor Elevation (DFE). Depths measured from these references are often called depths below rotary table (BRT) or below derrick floor (BDF). The top of the kelly bushing is also used as a datum for depth measurement. In this case the depths are referred to as depths below rotary kelly bushing (RKB). The depth of any point in the wellpath can be expressed in terms of the Along Hole Depth (AHD) and the True Vertical Depth (TVD) of the point below the reference datum. The AHD is the depth of a point from the surface reference point, measured along the trajectory of the borehole. Whereas the TVD is the vertical depth of the point below the reference point. The AHD will therefore always be greater than the TVD in a deviated well. Since there is no direct way of measuring the TVD, it must be calculated from the information gathered when surveying the well. The techniques used to survey the well will be discussed in the chapter on wellbore surveying.

3.2 Geographical Reference Systems

The position of a point in the well can only be defined in three dimensions when, in addition to the TVD of the point, its lateral displacement and the direction of that displacement is known. The lateral displacement is expressed in terms of feet (or meters) from the wellhead in a Northerly and Easterly direction or in degrees of latitude and longitude. All displacements are referenced to the wellhead position. The position of the wellhead is determined by land or satellite surveying techniques and quoted in latitude and longitude or an international grid co-ordinate system (e.g. Universal Transverse Mercator UTM system). Due to the large number of digits in some grid co-ordinate systems, a local origin is generally chosen and given the co-ordinates zero, zero (0,0). This can be the location of the well being drilled, or the centre of an offshore platform. When comparing the position of points in a well, and in particular for anti-collision monitoring, it is important that all coordinate data are ultimately referenced to a single system.

4. PLANNING THE PROFILE OF THE WELL

There are basically three types of deviated well profile (Figure 3):

- Build and Hold
- S-shaped
- Deep kick-off

The build and hold profile is the most common deviated well trajectory and is the most simple trajectory to achieve when drilling. The S-shaped well is more complex but is often required to ensure that the well penetrates the target formation vertically.

This type of trajectory is often required by reservoir engineers and production technologists in exploration and appraisal wells since it is easier to assess the potential productivity of exploration

wells, or the efficiency of stimulation treatments when the productive interval is entered vertically, at right angles to the bedding planes of the formation. The deep kick-off profile may be required when drilling horizontal wells or if it is necessary to drill beneath an obstacle such as the flank of a Salt Diapir. This well profile is the most difficult trajectory to drill since it is necessary to initiate the deviated trajectory in deeper, well compacted formations.

4.1 PARAMETERS DEFINING THE WELLPATH

There are three specific parameters which must be considered when planning one of the trajectories

shown in Figure 3. These parameters combine to define the trajectory of the well and are the:

- Kick-off Point
- Buildup and Drop off Rate and
- Tangent Angle of the well

(a) The Kickoff Point (KOP)

The kick off point is the along hole measured depth at which a change in inclination of the well is initiated and the well is orientation in a particular direction (in terms of North, South, East and West). In general the most distant targets have the shallowest KOPs in order to reduce the inclination of the tangent section of the well (see below). It is generally easier to kick off a well the shallow formations than in deep formations. The kick-off should also be initiated in formations which are stable and not likely to cause drilling problems, such as unconsolidated clays.

(b) Buildup Rate (BUR) and Drop Off Rate (DOR)

The build up rate and drop off rate (in degrees of inclination) are the rates at which the well deviates from the vertical (usually measured in degrees per 100 ft drilled). The build-up rate is chosen on the basis of drilling experience in the location and the tools available, but rates between 1 degree and 3 degree per 100ft Directional Drilling of hole drilled are most common in conventional wells. Since the build up and drop off rates are constant, these sections of the well, by definition, form the arc of a circle. Build up rates in excess of 3 degrees per 100 ft are termed doglegs when drilling conventional deviated wells with conventional drilling equipment.

The build up rate is often termed the dogleg severity.

(c) Tangent (or Drift) Angle

The tangent angle (or drift angle) is the inclination (in degrees from the vertical) of the long straight section of the well after the build up section of the well. This section of the well is termed the tangent section because it forms a tangent to the arc formed by the build up section of the well. The tangent angle will generally be between 10 and 60 degrees since it is difficult to control the trajectory of the well at angles below 10 degrees and it is difficult to run wireline tools into wells at angles of greater than 60 degrees.

4.2 DEFINING THE POINTS ON THE WELLPATH

Having fixed the target and the rig position, the next stage is to plan the geometrical profile of the well to reach the target. The most common well trajectory is the build and hold profile, which consists of 3 sections - vertical, build-up and tangent.

The trajectory of the wellbore can be plotted when the following points have been defined :

- KOP (selected by designer)

- TVD and horizontal displacement of the end of the build up section.
- TVD and horizontal displacement of the target (defined by position of rig and target)

Since the driller will only be able to determine the along hole depth of the well the following information will also be required:

- AHD of the KOP (same as TVD of KOP)
- Build up rate for the build up section (selected by designer)
- Direction in which the well is to be drilled after the KOP in degrees from North (defined by position of rig and target)
- AHD at which the build up stops and the tangent section commences and
- AHD of the target

5. CONSIDERATIONS WHEN PLANNING THE DIRECTIONAL WELL PATH

When planning a directional well a number of technical constraints and issues will have to be considered. These will include the:

- Target location
- Target size and shape
- Surface location (rig location)
- Subsurface obstacles (adjacent wells, faults etc.)

In conjunction with the above constraints the following factors must be considered in the geometrical design of the well:

- Casing and mud programmes
- Geological section

(a) Target Location

The location of the target is chosen by the geologists and/or the reservoir engineers. The target location will be specified in terms of a geographical co-ordinate system such as longitude and latitude or a grid co-ordinate system such as the UTM system. The grid reference system, in which the co-ordinates are expressed in terms of feet (or meters) north and east of a local or national reference point, is particularly useful when planning the directional well path, since the displacement of all points on the wellpath can be easily calculated.

The depth of the target is generally expressed by the geologist in terms of true vertical depth, TVD below a national reference datum such as Mean Sea Level. The difference between this national reference point and the drilling reference datum (such as the Rotary table) must be computed so that the driller can translate the computed TVD of the borehole below the rotary table elevation, into depth below mean sea level, and therefore proximity to the target.

(b) Specification of Target, Size and Shape

The size and shape of the target is also chosen by geologists and/or reservoir engineers. The target area will be dictated by the shape of the geological structure and the presence of geological features, such as faults. In general the smaller the target area, the more directional control that is required, and so the more expensive the well will be.

(c) Rig Location

The position of the rig must be considered in relation to the target and the geological formations to be drilled (e.g. salt domes, faults etc.). If possible the rig will be placed directly above the target

location. When developing a field from a fixed platform the location of the platform will be optimised so that the directionally drilled wells can reach the full extent of the reservoir.

(d) Location of Adjacent Wells

Drilling close to an existing well can be very dangerous, particularly if the existing well is on production. This is especially true just below the seabed on offshore platforms, where the wells are very closely spaced. The proposed wellpath must be designed so that it avoids all other wells in the vicinity. It is essential that the possible errors in determination of the existing and proposed wells are considered when the trajectory of the new well is designed.

(e) Geological Section

The equipment and techniques involved in controlling the deviated wellpath are not suited to certain types of formation. It is for example difficult to initiate the deviated portion of the well (kickoff the well) in unconsolidated mudstone. The engineer may therefore decide to drill vertically through the problematic formation and commence the deviated part of the well once the well has entered the next most suitable formation type. The vertical depth of the formation tops will be provided by the geologists.

(f) Casing and Mud Programmes

The trajectory of the well will be designed so that the most difficult parts of the well are drilled through competent formations, minimising problems whilst drilling the well. It is very common to initiate the kick-off just below the surface casing and possibly to change out to oil-based mud when drilling the build-up section. In highly deviated wells the build-up section of the well may also be cased off before drilling the long, tangent section of the well. Oil-based mud may also be used in the long tangent sections of the well. The trajectory of the well will therefore be designed so that these operations correspond to the casing setting depths which have been selected for many other reasons. This is an iterative process taking into account all of the considerations when designing the well.

6. DEFLECTION TOOLS

There are a number of tools and techniques which can be used to change the direction in which a bit will drill. These tools and techniques can be used to change the inclination or the azimuthal direction of the wellbore or both. All of these tools and techniques work on one of two basic principles. The first principle is to introduce a bit tilt angle into the axis of the BHA just above the bit and the second is to introduce a sideforce to the bit (See Figure 5). The introduction of a tilt angle or sideforce to the bit will result in the bit drilling off at an angle to the current trajectory. The major tools currently used for this purpose are:

- Bent Sub and Positive Displacement Motor
- Non-Rotating Steerable Drilling Systems
- Rotary Steering System
- Directional Bottom Hole Assemblies (BHA)
- Whipstocks

6.1 Bent Sub and Mud Motor

The most commonly used technique for changing the trajectory of the wellbore uses a piece of equipment known as a “bent sub” (Figure 6) and a Positive Displacement (mud) motor. A bent sub

is a short length of pipe with a diameter which is approximately the same as the drillcollars and with threaded connections on either end. It is manufactured in such a way that the axis of the lower connection is slightly offset (less than 3 degrees) from the axis of the upper connection. When made up into the BHA it introduces a “tilt angle” to the elements of the BHA below it and therefore to the axis of the drillbit. However, the introduction of a bent sub into the BHA means that the centre of the bit is also offset from the centre line of the drillstring above the bent sub and it is not possible therefore to rotate the drillbit by rotating the drillstring from surface. Even if this were possible, the effect of the tilt angle would of course be eliminated since there would be no preferential direction for the bit to drill in.

The bent sub must therefore be used in conjunction with a Positive Displacement Motor, PDM or a Drilling Turbine. The PDM is often called a mud motor and is used in far more wells than the turbine. A detailed description, and some technical specifications, of mud motors and turbines is provided in the Appendix at the end of this chapter.

The mud motor is made up into the BHA of the drillstring below the bent sub, between the bent sub and drillbit (See Figure 7). When drilling fluid is circulated through the drillstring the inner shaft of the mudmotor, which is connected to the bit, rotates and therefore the bit rotates. It is therefore not necessary to rotate the entire drillstring from surface if a mud motor is included in the BHA. Mud motors and turbines are rarely used when not drilling directionally because they are expensive pieces of equipment and do wear out. A scribe line is marked on the inside of the bend of the bent sub, and this indicates the direction in which the bit will drill (this direction is known as the “tool face”).

A directional surveying tool (quite often an MWD tool) is generally run as part of the BHA, just above the bent sub so that the trajectory of the well can be checked periodically as the well is deviating. The bent sub and PDM can of course only be used in the build up or dropoff portion of the well since the bit will continue to drill in the direction of the tilt angle as long as the bent sub is in the assembly and the mud motor is being used to rotate the bit. This leads to the major disadvantage of using a bent sub and PDM to change the trajectory of the well.

When drilling a well, the “conventional” assembly (without bent sub and mud motor) used to drill the straight portion of the well must be pulled from the hole and the bent sub and PDM assembly run in hole before the well trajectory can be changed.

The bent sub and motor will then be used to drill off in a particular direction. When the well is drilling in the required direction (inclination and azimuth), the bent sub and PDM must then be pulled and the conventional assembly re-run. Otherwise the drillbit would continue to change direction. This is a very time consuming operation (taking approximately 8 hrs at 10,00 ft depth for each trip out of, and into, the hole). Remember however that the build up section of a well can be 1-2000 ft long depending on the build up rate (typically 1-3 degrees/100ft) and the required inclination and therefore the bent sub and mud motor will be, depending on the rate of penetration, in the well for quite a long time.

6.2 Steerable Drilling Systems

A steerable drilling system allows directional changes (azimuth and/or inclination) of the well to be performed without tripping to change the BHA, hence its name.

It consists of: a drill bit; a stabilized positive displacement steerable mud motor; a stabilizer; and a directional surveying system which monitors and transmits to surface the hole azimuth, inclination and toolface on a real time basis (See Figure 8).

The capability to change direction at will is made possible by placing the tilt angle very close to the bit, using a navigation sub on a standard PDM. This tilt angle can be used to drill in a specific direction, in the same way as the tilt angle generated Directional Drilling by a bent sub with the the drillbit being rotated by the mud motor when circulating.

However, since the tilt angle is much closer to the bit than a conventional bent sub assembly, it produces a much lower bit offset and this means that the drill bit can also be rotated by rotating the entire string at surface (in the same way as when using a conventional assembly). Hence the steerable assembly can be used to drill in a specific direction by orienting the bent sub in the required direction and simply circulating the fluid to rotate the bit (as in the bent sub assembly) or to drill in a straight line by both rotating and circulating fluid through the drillstring. When rotating from surface we will of course be circulating fluid also and therefore the rotation of the bit generated by the mud motor will be super-imposed on the rotation from surface. This does not alter the fact that the effect of the bit tilt angle will be eliminated by the rotation of the entire assembly.

When using the navigation sub and mud motor to drill a deviated section of hole (such as build up or drop off section of hole) the term “oriented or sliding” drilling is used to describe the drilling operation. When drilling in a straight line, by rotation of the assembly, the term “rotary” drilling is used to describe the drilling operation. The directional tendencies of the system are principally affected by the navigation sub tilt angle and the size and distance between the PDM stabilizer and the first stabilizer above the motor.

The steerable drilling systems are particularly valuable where: changes in the direction of the borehole are difficult to achieve; where directional control is difficult to maintain in the tangent sections of the well (such as in formations with dipping beds) or where frequent changes may be required.

The steerable systems are used in conjunction with MWD tools which contain petrophysical and directional sensors. These types of MWD tools are often called Logging Whilst Drilling, LWD tools. The petrophysical sensors are used to detect changes in the properties of the formations (lithology, resistivity or porosity) whilst drilling and therefore determine if a change in direction is required. Effectively the assembly is being used to track desirable formation properties and place the wellbore in the most desirable location from a reservoir engineering perspective.

The term “Geosteering” is often used when the steerable system is used to drill a directional well in this way.

6.2.1 Components

There are five major components in a Steerable Drilling System (Figure 11). These components are:

- (a) Drill Bit
- (b) Mud Motor
- (c) Navigation Sub

- (d) Navigation Stabilizers
- (e) Survey System
- (a) Drill Bit

Steerable systems are compatible with either tricone or PDC type bits. In most cases, a PDC bit will be used since this eliminates frequent trips to change the bit.

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SECTION 3

CASING DESIGN

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1.0 Introduction

Casing design is about achieving the total depth of the well safely, with the most cost effective number of casing or liner strings.

1.1 Purpose of Installing Casing

In order to allow the drilling and completing of a well, it is necessary to line the drilled open hole with steel pipe / casing. Once in place this pipe is cemented, supporting the casing and sealing the annulus in order to:-

- Strengthen the hole.
- Isolate unstable / flowing / underbalanced / overbalanced formations.
- Prevent the contamination of freshwater reservoirs.
- Provide a pressure control system.
- Confine and contain drilling / completion / produced fluids and solids.
- Act as a conduit for associated operations (drilling, wireline, completion and further casing / tubing strings) with known dimensions (ID's etc).
- Support wellhead and additional casing strings.
- Support the BOP and Xmas tree.

There are primarily 6 types of casing installed in an onshore / offshore well:

- Stove Pipe, Marine Conductor, Foundation Pile.
- Conductor String.
- Surface Casing.
- Intermediate Casing.
- Production Casing.
- Liners.

1.1.1 Stove Pipe, Marine Conductor, Foundation Pile

Stove Pipe: is used for onshore locations and is either driven or cemented into a pre-drilled hole. The pipe protects the immediate soil at the base of the rig from erosion caused by the drilling fluid.

Marine Conductor: is a feature of offshore drilling operations where the BOP stack is above the water. It provides structural strength and guides drilling and casing strings into the hole. It is usually driven or cemented in a pre-drilled hole. The string helps isolate shallow unconsolidated formations and protects the base of the structure from erosion by the drilling fluid.

Foundation Pile: is usually jetted in or cemented into a pre-drilled hole from a floating drilling unit - where the BOP stack is on the sea floor. Again the string isolates unconsolidated formations and supports the guide base for the BOP stack / Xmas tree / flowbase and guides drilling and casing strings into the hole.

1.1.2 Conductor String

This string is used to support unconsolidated formations, protect freshwater sands from contamination and case off any shallow gas deposits. The string is usually cemented to the surface onshore and to the seabed offshore. This is the first string onto which the BOP is installed. If surface BOP's are used (i.e. Jack ups) the conductor string also supports the wellhead, the Christmas tree and subsequent casing strings.

1.1.3 Surface Casing.

Provides blow-out protection for deeper drilling, structural support for the wellhead / subsequent casing strings, and is often used to isolate troublesome formations. The string is either cemented to the surface or to inside the conductor string.

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1.1.4 Intermediate Casing

Again this string provides blowout protection for deeper drilling and isolates troublesome formations that could impair well safety and / or hamper drilling operations. An intermediate casing string is commonly set when a well is likely to encounter an influx and / or loss of circulation in the open hole thus providing blowout protection by upgrading the strength of the well. The cement height is determined by the design requirement to seal off any hydrocarbon / flowing salt zones. The top of cement does not need to be inside the surface string.

1.1.5 Production Casing

This is the name applied to the casing that has the production tubing run within it and could potentially be exposed to reservoir fluids. It can either extend to the surface as an integral string or be a combination of a production liner (7") and the previously set production casing (9-5/8"). The purpose of production casing is to isolate the producing zones, allow for reservoir control, act as a conduit for safe transmittal of fluids / gas / condensate to the surface and prevent the influx of unwanted fluids.

1.1.6 Liners

A liner will be suspended a short distance above the previous casing shoe and will be cemented along its whole length to insure a good seal isolating the annulus. Often a liner top packer can be set as a precautionary second barrier. HP / HT wells that incorporate a long liner may only cement the shoe and squeeze the liner lap. Liners permit deeper drilling, separate productive zones from reservoir formations and can thus be installed for testing purposes.

Drilling liners are set:

- to provide a deeper shoe
- isolate unstable formations
- to achieve a drilling casing at a reduced cost
- due to rig limitations

Production liners are set:

- to complete the well at a reduced cost.
- allow for a larger production conduit providing a range of choice for the tubing.

- due to rig limitations.

2.0 Casing Properties

Casing is usually specified by the following properties

- Outside diameter and wall thickness
- Weight per unit length
- Grade of steel
- Type of connection
- Length of joint

2.1 Outside Diameter and Wall Thickness

The outside diameter refers to the pipe body and not to the coupling. Coupling diameter is important as it determines the minimum hole size that the casing can be run into.

Wall thickness determines the inside diameter of the pipe and hence the maximum bit size that can be run through the pipe.

The permitted tolerance on outside diameter and wall thickness is given in API Spec 5CT. As a general rule:

Casing outside diameter $\geq 4\frac{1}{2}$ " Tolerance + 1.00%, - 0.50%

Casing outside diameter $< 4\frac{1}{2}$ " Tolerance $\pm 0.031\%$

Wall Thickness Tolerance - 12.5%

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2.2 Weight per Unit Length

The nominal weight of casing is used primarily to identify casing during ordering. Nominal weights are not exact and are based on the calculated theoretical weight of a 20 foot length of threaded and coupled pipe.

2.3 Grade of Steel

The mechanical and physical properties of casing are dependent upon the chemical composition of the steel and the heat treatment it receives during manufacture.

API defines nine grades of steel for casing.

H40 J55 K55 C75 L80 N80 C95 P110 Q125

The number in the designation gives the API minimum yield strength in thousands of psi. Hence L80 casing has a yield strength of 80,000 psi.

The letter in the designation gives an indication of the type of steel and the treatment it received during manufacture.

A more detailed section on Material Selection can be found later.

2.4 Type of Connection

There are a host of available connection types available on the market today. Selection of a suitable connection should be based upon the intended application, the required performance and cost.

The table below can act as a rough guide as whether API or Premium threads should be used.

Production Tubing

Liquids API <5000 Premium

Threads psi> Threads

Gas API <3500 Premium

Threads psi> Threads

Production Casing

Liquids API <7500 Premium

Threads psi> Threads

Gas API <5000 Premium

Threads psi> Threads

Surface and Intermediate Casing

If the pressure differential across the connection is $\geq 7,500$ psi a premium thread is the preferred option. An API thread with an enhanced coupling design can be used although its sealing qualities are not as reliable. Leak resistance values for API connections can be found in API bulletin 5C2.

Connection Properties

The connection collapse, burst and tensile properties should be compared with the pipe body properties. Whichever are the lowest should be used in all casing design connections.

In addition some connections have a very low capacity in compression when compared to their tensile strengths. If compression or compression / bending is a critical load, query the manufacturer on their coupling capacity under these conditions (i.e. Vam SC has only 25% cap in compression versus tension)

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Six generic connection types are available. These are shown below with some general characteristics.

2.4.1 API 8-Round, STC or LTC

- Good availability and price
- Liquid sealability up to about 210°F
- Sealing is a combination of connection geometry and thread dope
- Poor gas tightness
- Gauges and expertise are widely available for re-work and refurbishment
- Prone to galling and cross-threading due to out of roundness, especially in larger OD's
- High assembly circumferential (hoop) stress in coupling
- Tensile efficiency 70-75% depending on thread type
- Leak resistance must be verified per API Bulletin 5C3

2.4.2 API BTC

- Good availability and price.
- Poor gas tightness

- Liquid sealability up to about 210°F
- Sealing is a combination of connection geometry and thread dope
- Tin plating improves leak resistance
- Gauges and expertise are widely available for re-work and refurbishment
- Prone to galling and to cross-threading due to out of roundness, especially in larger OD's
- High assembly circumferential (hoop) stress in coupling
- Tensile efficiency is generally 85 - 95% of pipe body
- Leak resistance of BTC must be verified per API Bulletin 5C3

2.4.3 Metal-to-Metal Seal, Threaded & Coupled

- Availability to depend on propriety type, e.g., Vam, Fox, NS-CC etc.
- Good gas tightness, generally.
- Special clearance couplings manufactured from some or higher grade material are available to improve hole clearance.
- Susceptible to handling damage if not treated with care. Pins must be bored concentric to seals for effective gas sealing.
- Particularly suited to use on cold worked high alloys that cannot be upset.
- Generally good make-up characteristics due to reduced thread interference compared to API connections.
- Gauges and expertise are available, depending on type, for re-work and refurbishment and can readily be re-cut.
- Assembly circumferential (hoop) stress in coupling can be controlled by reduced thread interference since sealing in the thread is not a requirement.
- Tensile efficiency is generally at least equal to BTC and in many instances equal to or exceeds pipe body.

2.4.4 Metal-to-Metal Seal, Upset & Integral (or Coupled)

- Poor availability of couplings and limited upset re-cuts for pipe refurbishment.
- Costly, especially upsetting.
- Good gas tightness.
- Usually exhibiting very good repeated make/break capabilities.
- Susceptible to handling damage if not treated with care.
- Pins must be bored concentric to seals for effective gas sealing.
- Tensile efficiency at least equal to or greater than pipe body.

2.4.5 Metal-to-Metal Seal, Formed and Integral (Flush)

- Hole clearance characteristics excellent, flush pipe OD

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- Reasonable availability, easy to refurbish/recut, no requirement for couplings.
- Good gas tightness.
- Pins must be bored concentric to seals for effective gas sealing.
- Tensile efficiency = 50 - 75% of pipe body depending on type of connections.

- Connections may be weaker than the pipe body for internal pressure rating.

2.4.6 Weld on, Upset and Integral

- Very costly (connector, weld and NDT).
- Elimination of mill end with weld on box.
- Coarse threads to resist cross threading or galling.
- Continuous threaded product resists disengagement under severe bending.
- Grades limited to weldable (linepipe) or H-40, K/J-55.
- Tensile efficiency generally greater than pipe body.

2.5 Length of Joint

Casing joints are not manufactured in exact lengths. API has specified three ranges in which pipe lengths must lie.

Range	Length (ft)	Average Length (ft)
1	16 – 25	22
2	25 – 34	31
3	> 34	42

3.0 The Casing Design Operation

There are two phases of casing design.

1. The first takes place during the Preliminary Well Design and involves the casing scheme selection and casing setting depth determination.
2. The second takes place during the Detailed Well Design and involves a determination of the loads that the casing will be exposed to during the life of the well and the selection of tubulars with suitable mechanical and physical properties that can withstand the predicted loads.

4.0 Preliminary Design

4.1 Casing Setting Depth Determination

The initial selection of casing setting depths is based on the anticipated pore pressure and fracture gradients. The drilling engineer is responsible for ensuring that, as far as possible, all the relevant offset data has been considered in the estimation of pore pressure and fracture gradients, and that, for directional wells, the effect of hole angle on offset fracture gradient data has been considered.

The total depth of the well, and hence the setting depth of the production casing or liner, is driven by logging, testing, and completion requirements. The shoe must be set deep enough to give an adequate sump for logging, perforating, and test on production activities.

The initial estimate of determining casing setting depths is best determined graphically, as follows, plotting pore pressure and fracture gradient, expressed in equivalent density, against depth.

1. Draw the mean pore pressure gradient curve along with lithology, if available. Note any intervals which are potential problem areas such as differential sticking, loss circulation or high pressure gas zones.
2. Draw the mud weight curve. The mud weight curve should include a trip margin of around

200 to 400 psi.

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3. Draw the predicted fracture gradient curve. Draw a fracture gradient design curve, which parallels the predicted fracture gradient curve with a reduction of 0.3 to 0.5 ppg for kicks and ECD during cementing.

4. Plot offset mud weights and LOT's to provide a check of the pore pressure predictions or highlight the need for further investigation.

A typical plot is attached. The initial casing setting depths can be determined as follows.

Casing

Conductor

Normal Pressure

Mud Weight Surface

Curve

Intermediate

Base Fracture Gradient

E D Design Curve

(inc. 3ppg Kick &

Cementing Margin)

Depth

Pore Pressure Production

Gradient

C

B

Production

Liner

A

Equivalent Mud Weights

1. Working from the bottom up enter the mud weight curve at Point A.
2. Move up to Point B which determines the initial estimated setting depth for the production casing.
3. Move across to Point C, which identifies the mud weight requirement for that depth.
4. Move up to Point D which determines the initial estimated setting depth for the intermediate casing.
5. Move across to Point E to identify the mud weight required at that depth. For the example shown, Point E is the normal pressure range and no further casing is required to withstand the associated mud weight. However, a conductor and surface casing are required and the setting depth for these casings is discussed later.

Other factors that may impact casing depth selection in addition to pore pressure and fracture

pressures are:

- Shallow gas zones.
- Lost circulation zones.
- Formation stability which is sensitive to exposure time or mud weight.

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- Directional well profile. It is important to line out the well trajectory before setting casing and attempt to achieve a consistent survey ahead of a tangent section. Also, long, open hole sections may require casing to reduce the occurrence of stuck pipe and the level of torque.
- Sidetracking requirements as specified in the Basis of Design e.g. 13-3/8" casing might be set high to allow 9-5/8" casing to be cut and pulled for a sidetrack in 12-1/4" hole.
- Fresh water sands (drinking water).
- Hole cleaning, particularly if a long section of 17½" hole is required.
- Salt sections.
- High pressure zones.
- Lithology - casing shoes should, where practicable, be set in competent impermeable formations.
- Uncertainty in depth estimating due to seismic uncertainty.

All of the above need to be considered and the initial casing setting depths adjusted accordingly.

4.2 Kick Tolerance

Once the initial casing setting depths have been selected, the kick tolerance associated with those depths should be calculated. Starting from TD up to surface the kick tolerance and preferred setting depth for each casing string should be calculated.

Kick tolerance is the maximum kick size that can be taken into the wellbore and circulated to the shoe without breaking down the formation. It is dependent on the mud weight in use, the open hole weak point (normally assumed to be the previous casing shoe), the formation pressure, the size and density of the influx and the hole geometry.

There are two methods for calculating kick tolerance. The first calculates a kick intensity and the second a kick volume.

Note that both methods neglect any temperature effects and assume an ideal gas.

4.2.1 Kick Intensity

Kick intensity (as shown in the Well Control Manual) is a measure of how much the mud weight can be raised for a given kick volume. In other words if you drill into an overpressured zone by how much can you raise the mud and still circulate out the kick.

For casing design purposes the kick volume is assumed to be 25 bbls and the minimum acceptable kick intensity is 0.5 ppg. If the kick intensity is below this value, then further approval should be sought.

Kick intensity is calculated using the following equation.

$$KI = MAASP - (MW \times 0.052 \times Hi) \text{ (ppg)}$$

$$0.052 \times TVD$$

Where:

KI = kick intensity (ppg)

MAASP = maximum allowable annular surface pressure (psi)

MW = mud weight in the hole (ppg)

Hi = height of influx (ft)

TVD = true vertical depth of well (ft)

Example

12¼" hole TD 13,123 ft

BHA 697 ft x 8" DC

DP 5"

Mud weight 13.2 ppg

Previous casing shoe 8,842 ft

LOT at shoe 14.3 ppg EMW

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$$MAASP = 8,842 \text{ ft} \times (14.3 \text{ ppg} - 13.2 \text{ ppg}) \times 0.052$$

$$= 506 \text{ psi}$$

$$2 \text{ 2}$$

$$Hi = 25 \text{ bbls} / [(12.25 - 8) \times 0.0009714]$$

$$= 300 \text{ ft}$$

(Note that this is the height of the influx around the drill collars, as this volume passes up around the drill pipe the height will reduce due to a larger annular capacity between DP and open hole)

$$KI = [506 \text{ psi} - (13.2 \text{ ppg} \times 0.052 \times 300 \text{ ft})] / (0.052 \times 13,123 \text{ ft})$$

$$= 0.44 \text{ ppg}$$

4.2.2 Kick Volume

The kick volume determines the maximum kick size that can be taken into the wellbore and circulated to the shoe.

The following process can be used to calculate kick volume.

1. Estimate the safety margin to be applied to the leakoff pressure at the open hole weak point

When the influx is displaced from the hole, there will be additional pressures acting in the wellbore. The following are possible causes of such additional pressure during circulation:

- Annulus friction
- Choke operator error
- Choke line losses (if not compensated for)

The total safety margin to be applied to the leakoff pressure will be the sum of these additional

pressures. The maximum allowable static weak point pressure can therefore be determined. (This is the maximum allowable weak point pressure before circulation is initiated.)

The Drilling Engineer must use his/her judgement to determine the most appropriate safety factor to be applied to the leakoff pressure at the open hole weak point. This safety factor should be based on operating area experience.

2. Calculate the maximum allowable static weak point pressure (P_{max}).

The maximum allowable pressure is given by:

$$P_{max} = P_{lo} - (\text{safety margin}) \text{ (psi)}$$

Where:

P_{max} = maximum allowable weak point pressure (psi)

P_{lo} = leakoff pressure at the open hole weak point (psi)

3. Calculate the maximum allowable height of influx in the open hole section

The maximum height of influx that can be taken in the open hole without exceeding P_{max} at Dwp is given by:

$$H = \frac{P_{max} - P_f - (TD - D_{wp}) MW}{0.052 \text{ (ft)}}$$

Where:

H = height of influx (ft)

P_{max} = maximum allowable pressure at the open hole weak point (psi)

MW = mud weight in the hole (ppg)

gg = gas gradient psi/ft

TD = bit depth (ft)

P_f = formation pressure at TD psi

Dwp = depth of shoe or weak point

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4. Calculate the volume that this height corresponds to at initial shut-in conditions

At initial shut-in conditions, this can be converted to an influx volume as follows:

$$V_1 = H \times C_1 \text{ (bbl)}$$

Where:

V_1 = kick tolerance for initial influx (bbl)

C_1 = annular capacity at the BHA (bbl/ft)

C_1 must be determined bearing in mind the hole dimensions in relation to the height of the influx, H. For example, if H is greater than the height of the BHA, both the capacity of the drillpipe /open hole annulus and the drill collar/open hole annulus must be used to calculate the kick tolerance, V.

5. Calculate the volume that this height corresponds to when the top of the influx has been circulated to the open hole weak point

This height corresponds to a volume at the open hole weak point given by:

$$V_{wp} = H \times C2 \text{ (bbl)}$$

Where:

$C2$ = annular capacity below the open hole weak point (bbl/ft)

$C2$ must be determined bearing in mind the hole dimensions immediately below the open hole weak point in relation to the height of the influx, H .

6. Calculate what V_{wp} (as calculated in (5)) would be at initial shut-in conditions

Using Boyle's Law to convert this volume to its original volume at initial shut-in conditions ($V2$):

$$P1 \times V1 = P2 \times V2 \text{ or in this case:}$$

$$P_f \times V2 = P_{max} \times V_{wp}$$

$$V2 = P_{max} \times V_{wp}$$

P_f

7. The kick tolerance is the lower of $V1$ and $V2$

As a general rule, a kick tolerance of less than 100bbl should be justified by a review of type of well, rig equipment for kick detection and operator/driller's experience, area experience and geology.

The proposed kick tolerance for casing seat design should be noted for each casing, approved by the Drilling Superintendent, and recorded on the drilling program.

Sensitivity studies should be considered to identify the contingency provisions required to accommodate pore or fracture pressures during drilling which differ from those assumed during design.

Software is available to calculate kick tolerance and is available as a spreadsheet from the Well Construction website on the TSF intranet.

Example

12¼" hole TD 13,123 ft

BHA 697 ft x 8" DC

DP 5"

Mud weight 13.2 ppg

Previous casing shoe 8,842 ft

LOT at shoe 14.3 ppg EMW

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1. Assume a 150 psi safety margin

$$2. P_{max} = (14.3 \text{ ppg} \times 8,842 \text{ ft} \times 0.052) - 150 \text{ psi} \\ = 6,424 \text{ psi}$$

$$3. P_f = 13,123 \text{ ft} \times 13.2 \text{ ppg} \times 0.052 \\ = 9,008 \text{ psi}$$

Note that this assumes the well is on balance

$$H = 6,424 \text{ psi} - 9,008 \text{ psi} + [(13,123 \text{ ft} - 8,842 \text{ ft}) \times 13.2 \text{ ppg} \times 0.052]$$

$$(0.052 \times 13.2) - 0.1$$

$$= 604 \text{ ft}$$

$$2^2$$

$$4. V1 = 604 \text{ ft} \times [(12.25 - 8) \times 0.0009714]$$

$$= 50.5 \text{ bbls}$$

$$2^2$$

$$5. V_{wp} = 604 \text{ ft} \times [(12.25 - 5) \times 0.0009714]$$

$$= 73.4 \text{ bbls}$$

$$6. V2 = 6,424 \text{ psi} \times 73.4 \text{ bbls} / 9,008 \text{ psi}$$

$$= 52.3 \text{ bbls}$$

$$7. \text{ Kick volume, in this case, is } V1 = 50.5 \text{ bbls}$$

4.3 Surface and Conductor Setting Depth Design

The minimum setting depth for surface and conductor casings is the depth at which the bottom hole pressure created by the circulating drilling fluid (ECD) is exceeded by the fracture value of the formation.

The ECD can be significantly affected in large diameter holes by high ROP and poor hole cleaning.

Water depth can play a significant part and results in the depth that the ECD is exceeded by the fracture value being pushed that much deeper and can result in additional casing strings being run.

In deepwater areas bending and axial loading are primary considerations in the design of the conductor casing. Due to the complexity of the interaction of the various parameters that affect bending and axial loading a computer program is needed to accurately model the loads and the behaviour of the conductor.

In addition the load bearing capacity (which directly relates to the soil strength below the mudline) must be determined.

5.0 Detailed Design

The detailed design stage is about determining the loads each casing string will be exposed to during the life of the well and the selection of tubulars with suitable mechanical and physical properties that can withstand the predicted loads.

The major steps in detailed design for each casing string are shown below.

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Determine Burst

Define Load Define Initial Determine Tensile
and Collapse

Cases Casing String Loads

Loads

Adjust Initial Determine Triaxial Finalise Casing

Casing String Loads (if required) String

5.1 Design Load Cases

The anticipated design load cases should be calculated in the order in which they occur. This helps identify all the loads to which a casing string may be exposed.

Note that not all of these load cases will be applicable to every casing string. For example production loads do not have to be considered for an intermediate casing string on an exploration well.

5.1.1 Installation Loads

Typical installation loads include

- Casing running
- Cementing
- Conventional cementing, stab-in, etc.
- Plug bump

5.1.2 Drilling Loads

Typical drilling loads include

- Pressure testing after waiting on cement
- Maximum mud weight
- Well control
- Lost circulation

5.1.3 Production Loads

Typical production loads include

- Pressure testing with completion or kill weight fluid
- Functioning DST tools
- Near surface tubing leak
- Collapse due to plugged perforations
- Special production operations (stimulation, gas lift, injection)

5.2 Design Factors

To account for factors that are improperly addressed or not directly accounted for, the properties of the casing are downrated by a design factor before being compared with the calculated design loads.

Typical design factors used are as follows

Collapse 1.0

Burst 1.1

Tension 1.3

Triaxial 1.25

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Local legislation should be checked to ensure that other more stringent design factors are not stipulated.

5.3 Collapse Design

For all casing strings a collapse load occurs when the external pressure is greater than the internal pressure.

Collapse design focuses on the internal and external pressure profiles.

Generally speaking the collapse load will be highest at the casing shoe.

5.3.1 Collapse Installation Loads

The worst case collapse load during installation occurs during cementing.

For conventional cement jobs the worst case occurs with the cement column on the outside of the casing.

Example

9-5/8" Production Casing set at 11,450 ft

Mud Weight 11 ppg

Top of Cement at 7,000 ft

Cement Weight 16 ppg

Top of Spacer at 5,500 ft

Spacer Weight 13 ppg

Internal Pressure Profile

At surface = 0 psi

At casing shoe = $11,450 \text{ ft} \times 11 \text{ ppg} \times 0.052$

= 6,549 psi

External Pressure Profile

At surface = 0 psi

At top of spacer = $5,500 \text{ ft} \times 11 \text{ ppg} \times 0.052$

= 3,146 psi

At top of cement = $3,146 \text{ psi} + [(7,000 \text{ ft} - 5,000 \text{ ft}) \times 13 \text{ ppg} \times 0.052]$

= 4,160 psi

At casing shoe = $4,160 \text{ psi} + [(11,450 \text{ ft} - 7,000 \text{ ft}) \times 16 \text{ ppg} \times 0.052]$

= 7,862 psi

Net collapse load at casing shoe = $7,862 \text{ psi} - 6,549 \text{ psi}$

= 1,313 psi

The net collapse load is best represented graphically, as shown below, by plotting the internal and external pressure profiles and the net collapse load.

For stab-in cement jobs the possibility of bridging needs to be taken into account. If bridging occurs then the external hydrostatic pressure will be increased by the circulating pressure, with a subsequent increase in the net collapse load.

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Internal Hydrostatic

2,000

Mud

Mud 4,000

Depth (TVD) ft

6,000

Spacer

TOC

8,000

External Hydrostatic

Cement 10,000

Net Collapse Load

12,000

Casing

1,000 2,000 3,000 4,000 5,000 6,000 7,000 8,000

Shoe

Pressure (psi)

Collapse - Installation

Example

20" Surface Casing set at 1,500 ft

Mud Weight 9.5 ppg

Cement to surface

Cement Weight 16 ppg

Bridge occurs with 1000 psi surface pressure

Internal Pressure Profile

At surface = 0 psi

At casing shoe = 1,500 ft x 9.5 ppg x 0.052

= 741 psi

External Pressure Profile

At surface = 1,000 psi

At casing shoe = 1,000 psi + (1,500 ft x 16 ppg x 0.052)

= 2,248 psi

Net collapse load at casing shoe = 2,248 psi - 741 psi

= 1,507 psi

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Internal Pressure

Mud

500 External Pressure

Depth (TVD) ft

Inner

String

1000

Net Collapse Load

Cement

1500

Stab In

1,000 2,000 3,000 4,000 5,000 6,000

Shoe

Pressure (psi)

Collapse - Installation (stab-in)

5.3.2 Collapse Drilling Loads

The worst case collapse load during drilling occurs if lost circulation is encountered and the internal hydrostatic pressure decreases.

By convention, the external fluid is deemed to be the mud that was in place when the casing was run. This is due to the uncertainty of complete cement isolation around the casing caused by channelling or washouts.

The level that the internal fluid drops to can be anything from hundreds of feet to total evacuation of the casing and depends upon the internal mud weight in use and the pore pressure of the loss zone.

Example

13-3/8" Intermediate Casing set at 9,750 ft

External Mud Weight 11 ppg

Internal Mud Weight 11.2 ppg

Drilling ahead 12-1/4" hole at 13,360 ft.

Experienced total losses and fluid level dropped to 2,528 ft

Internal Pressure Profile

At surface = 0 psi

At 2,528 ft = 0 psi

At casing shoe = $(9,750 \text{ ft} - 2,528 \text{ ft}) \times 11.2 \text{ ppg} \times 0.052$
 $= 4,206 \text{ psi}$

External Pressure Profile

At surface = 0 psi

At casing shoe = $9,750 \text{ ft} \times 11 \text{ ppg} \times 0.052$
 $= 5,577 \text{ psi}$

Net collapse load at casing shoe = $5,577 \text{ psi} - 4,206 \text{ psi}$
 $= 1,371 \text{ psi}$

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Fluid 2,000

Level

4,000

Depth (TVD) ft

6,000

Drill External Pressure

String

8,000

10,000

Internal Pressure

Net Collapse Load

12,000

Casing

1,000 2,000 3,000 4,000 5,000 6,000

Shoe

Pressure (psi)

Collapse - Drilling

5.3.3 Collapse Production Loads

Collapse loads that production casing and liners will be exposed to need to be considered for the entire life of the well. This depends upon what the well will be used for but consideration should be given to the following as applicable:

- DST operations
- Stimulation techniques
- Gas Lift
- Drawdown

By convention, the external fluid is deemed to be the mud that was in place when the casing was run. After a period of time (typically one year) this can be relaxed as follows:

Position In Well External Fluid

Uncemented Casing / Casing Annulus Mud that casing was run in

Cemented Casing / Casing Annulus Mud that casing was run in

Cemented Casing / Open Hole Annulus Pore Pressure

The internal hydrostatic pressure will vary, depending upon the position of the production packer and the collapse considerations should be separated into above the packer or below the packer.

Example

9-5/8" Production Casing set at 15,700 ft

Production Packer set at 12,000 ft

Completion Fluid Weight 9.2 ppg

Mud Weight behind casing 11 ppg

Gas Gradient 0.1 psi / ft

Perforations have plugged and the well has been drawn down to 0 psi at surface

Above Packer

Internal Pressure Profile

At surface = 0 psi

At packer = 12,000 ft x 9.2 ppg x 0.052

= 5,741 psi

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External Pressure Profile

At surface = 0 psi

At packer = 12,000 ft x 11 ppg x 0.052

= 6,864 psi

Net collapse load at packer = 6,864 psi - 5,741 psi

= 1,123 psi

Below Packer

Internal Pressure Profile

At packer = 12,000 ft x 0.1 psi/ft

= 1,200 psi

At casing shoe = 15,700 ft x 0.1 psi/ft

= 1,570 psi

External Pressure Profile

At packer = 12,000 ft x 11 ppg x 0.052

= 6,864 psi

At casing shoe = 15,700 ft x 11 ppg x 0.052

= 8,980 psi

Net collapse load at packer = 6,864 psi - 1,200 psi

= 5,664 psi

Net collapse load at casing shoe = 8,890 psi - 1,570 psi

= 7,320 psi

2,000

Tubing 4,000 Internal Pressure

(above Packer)

6,000

Depth (TVD) ft

8,000

Completion External Pressure

Fluid 10,000

Net Collapse Load

Packer

12,000

14,000

Gas Internal Pressure

(below Packer)

16,000

1,000 2,000 3,000 4,000 5,000 6,000 7,000 8,000 9,000

Pressure (psi)

Collapse - Production

5.3.4 Selecting a Casing that Meets the Collapse Loads

From the net collapse loads generated by looking at the various installation, drilling and production load cases, the worst case collapse load that the casing will have to tolerate can be determined.

From casing tables it is then possible to choose a casing or series of casings that match the worst case collapse load. This stage is often ignored until the burst loads have been calculated. The casing collapse figures that are quoted in most casing tables are generated from a series of equations detailed in API Bulletin 5C3 and are a function of the casing OD, the wall thickness and the casing yield strength.

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5.3.5 Biaxial Loading

The collapse figures determined using API Bulletin 5C3 are for casings that are under zero axial load. In practice, due to the weight of the casing or to the combined action of internal and external pressures, it is rare for casing to be under zero axial load.

The effect of axial load is to decrease the collapse strength of casing.

For casing design purposes, unless the collapse strength is critical, the reduction in collapse strength due to axial load is normally ignored.

5.3.6 Other Considerations for Collapse Design

Casing Wear

If casing wear is anticipated to be an issue (i.e. high sidewall forces, extended drillstring to casing contact, localised doglegs, abrasive or damaged hardbanding in use, etc.) then this needs to be accounted for.

As the collapse strength of casing is related to the wall thickness, then if casing wear estimates that 20% of the wall thickness will be worn away during drilling, then the residual collapse strength of the casing will be 80% of that of new casing.

More detail on casing wear can be found later.

5.4 Burst Design

For all casing strings a burst load occurs when the internal pressure is greater than the external pressure.

As with collapse, burst design focuses on the internal and external fluids and the hydrostatic pressures that they exert.

5.4.1 Burst Installation Loads

The worst case burst load during installation occurs during cementing.

Two cases need to be considered here.

- During the displacement immediately prior to the spacer exiting the shoe
- Bumping the plug

Example – Cement Displacement

13-3/8" Intermediate Casing set at 9,750 ft

Mud Weight 11 ppg

5,000 ft of 16 ppg cement

2,000 ft of 13 ppg spacer

Internal Pressure Profile

At surface = 0 psi

At top of cement = $(9,750 \text{ ft} - 7,000 \text{ ft}) \times 11 \text{ ppg} \times 0.052$
= 1,573 psi

At top of spacer = $1,573 + (5,000 \text{ ft} \times 16 \times 0.052)$
= 5,733 psi

At casing shoe = $5,733 \text{ psi} + (2,000 \text{ ft} \times 13 \text{ ppg} \times 0.052)$
= 7,085 psi

External Pressure Profile

At surface = 0 psi

At casing shoe = $9,750 \text{ ft} \times 11 \text{ ppg} \times 0.052$
= 5,577 psi

Net burst load at surface = 0 psi

Net burst load at casing shoe = $7,085 \text{ psi} - 5,577 \text{ psi}$
= 1,508 psi

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The net burst load is best represented graphically, as shown below, by plotting the internal and external pressure profiles and the net collapse load.

Mud

2,000

Depth (TVD) ft

4,000

Internal Pressure

Cement

6,000

Net Burst Load

8,000

External Pressure

Spacer

10,000

1,000 2,000 3,000 4,000 5,000 6,000 7,000 8,000 9,000

Casing

Shoe

Pressure (psi)

Burst - Installation - Cement Displacement

Example – Plug Bump

13-3/8" Intermediate Casing set at 9,750 ft

Mud Weight 11 ppg

Top of Spacer at 1,750 ft

Spacer Weight 13 ppg

Top of Cement at 3,000 ft

Cement Weight 16 ppg

Plug Bump Pressure 2,500 psi

Internal Pressure Profile

At surface = 2,500 psi

At casing shoe = 2,500 psi + (9,750 ft x 11 ppg x 0.052)

= 8,077 psi

External Pressure Profile

At surface = 0 psi

At top of spacer = 1,750 ft x 11 ppg x 0.052

= 1,001 psi

At top of cement = 1,001 psi + [(3,000 ft – 1,750 ft) x 13 ppg x 0.052]

= 1,846 psi

At casing shoe = 1,846 psi + [(9,750 ft – 3,000 ft) x 16 ppg x 0.052]

= 7,462 psi

Net burst load at surface = 2,500 psi

Net burst load at casing shoe = 8,077 psi – 7,462 psi

= 615 psi

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2,500 psi

Mud

Internal Pressure

Spacer Mud 2,000

Depth (TVD) ft

4,000

6,000

Cement 8,000

Net Burst Load External Pressure

10,000

Casing

1,000 2,000 3,000 4,000 5,000 6,000 7,000 8,000

Shoe

Pressure (psi)

Burst - Installation - Plug Bump

5.4.2 Burst Drilling Loads

The worst case burst drilling loads occur either during pressure testing or during a well control event.

Example - Pressure Test

13-3/8" Intermediate Casing set at 9,750 ft

Mud Weight 11.5 ppg

Top of Cement at 3,000 ft

Previous Casing Shoe at 1,500 ft

Normally Pressured

Pressure Test 3,000 psi

Internal Pressure Profile

At surface = 3,000 psi

At casing shoe = $3,000 \text{ psi} + (9,750 \text{ ft} \times 11.5 \text{ ppg} \times 0.052)$

= 8,831 psi

External Pressure Profile

At surface = 0 psi

At top of cement = $3,000 \text{ ft} \times 8.33 \text{ ppg} \times 0.052$

= 1,299 psi

At casing shoe = $1,299 \text{ psi} + [(9,750 \text{ ft} - 3,000 \text{ ft}) \times 8.6 \text{ ppg} \times 0.052]$

= 4,318 psi

Net burst load at surface = 3,000 psi

Net burst load at casing shoe = $8,831 \text{ psi} - 4,318 \text{ psi}$

= 4,513 psi

NOTE: The external pressure profile after cement has set is a matter of great debate, and requires careful consideration.

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In the example above it has been assumed that for casing in contact with formation via cement the external pressure is equal to the expected pore pressure. For casing exposed to formation

via an uncemented section (top of cement below previous casing shoe) the external pressure is equal to a column of mud mix fluid with zero surface pressure (i.e. the mud has wholly settled out - in the above example a water based mud was assumed).

3,000 psi

Mud

Top of

Cement

Mud 2,000 Internal Pressure

Depth (TVD) ft

4,000

6,000

External Pressure

8,000

Net Burst Load

10,000

Casing

Shoe 1,000 2,000 3,000 4,000 5,000 6,000 7,000 8,000 9,000

Pressure (psi)

Burst - Drilling

5.4.3 Burst Production Loads

Burst loads that need to be considered during production include pressure testing with a completion or kill weight fluid, a near surface tubing leak.

The internal load is the hydrostatic pressure of the fluid plus any applied pressure.

The external load will be as for the Burst Drilling Load.

Example - Near Surface Tubing Leak

9-5/8" Production Casing set at 15,700 ft

Production Packer set at 12,000 ft

Completion Fluid Weight 9.2 ppg

Mud Weight behind casing 11 ppg

Gas Gradient 0.1 psi / ft

Top of cement 8,000 ft

Pore pressure 9.0 ppg EMW

SITHP 5,778 psi

Above Packer

Internal Pressure Profile

At surface = 5,778 psi

At packer = 5,778 psi + (12,000 ft x 9.2 ppg x 0.052)

= 11,519 psi

External Pressure Profile

At surface = 0 psi

At TOC = 8,000 ft x 8.33 ppg x 0.052

= 3,465 psi

At packer = 3,465 psi + [(12,000 - 8,000 ft) x 9.0 ppg x 0.052]

= 5,337 psi

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Net burst load at packer = 11,519 psi - 5,337 psi

= 6,182 psi

5.4.4 Selecting a Casing that Meets the Burst Loads

From the net burst loads generated by looking at the various installation, drilling and production load cases the worst case burst load that the casing will have to tolerate can be determined.

From casing tables it is then possible to choose a casing or series of casings that match the worst case burst and worst case collapse loads.

Remember that the burst or internal yield of the casing needs to be down rated by the Design Factor.

Example - 13-3/8" Intermediate Casing

From the previous examples, the worst case collapse (during drilling) and the worst case burst (during drilling) are plotted on the same graph.

Assuming that the following 13-3/8" casing is available in stock

13-3/8" 68 lb/ft K55

13-3/8" 72 lb/ft N80

The collapse ratings for both of these casings are determined and plotted on the graph. From this it can be seen that either of these casings is acceptable from a collapse standpoint.

The burst ratings for both of these casings are determined, downrated by the design factor of 1.1 and then plotted on the graph. From this it can be seen that the 68 lb/ft K55 casing can only be used for the top 1,500 ft.

In practice it is more likely that a full string of 72 lb/ft N80 casing would be run.

See attached graph.

↩ - ® -

68# 68#

K55 K55

2,000

Depth (TVD) ft

Worst Case 72#

Collapse N80

4,000

6,000

8,000

Worst Case

Burst 72#

N80

10,000

1,000 2,000 3,000 4,000 5,000 6,000

Pressure (psi)

1 13.3/8" 68 lbft K55 Collapse = 1,950 psi

2 13.3/8" 72lbft N80 Collapse = 2,670 psi

3 13.3/8" 68lbft K55 Burst = 3,450 psi

= 3,136 psi (downrated by design factor)

4 13.3/8" 72lbft N80 Burst = 5,380 psi

= 4,891 psi (downrated by design factor)

Selecting casing for Collapse & Burst

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5.4.5 Other Considerations for Burst Design

Casing Wear

As for collapse, if casing wear is anticipated to be an issue then this needs to be accounted for.

To allow for casing wear, the API burst rating should be multiplied by a casing wear factor (CWF) as shown in the following table. These values of CWF have been derived from finite element analysis of different sizes, weights and grades of casing subjected to different wear grooves.

Percent Casing Wear CWF

0 0

10 0.9

20 0.8

30 0.7

40 0.6

More detail on casing wear can be found later.

5.5 Tensile Design

Using the casing selected that meets the collapse and burst loads, it is then necessary to confirm that this casing will also meet the requirements of the tensile design.

5.5.1 Tensile Installation Loads

This stage involves assessing the suitability of the selected casing for withstanding running loads, cementing loads and any pressure testing.

It is assumed that the casing is fixed at surface but free to move at the shoe.

The loads that need to be considered are as follows.

Weight in Air

The weight of casing in air is simply the nominal casing weight multiplied by the true vertical depth of the casing.

$$F_{air} = W \times TVD$$

where W = Nominal casing weight (lb / ft)

TVD = TVD below point of interest to casing shoe (ft)

Buoyancy

Buoyancy can be calculated using the pressure area method and is normally the hydrostatic pressure multiplied by the casing cross sectional area.

Care needs to be taken if tapered strings of casing are used as the buoyancy force will vary depending upon the depth and outside and inside diameters.

Buoyancy is always subtracted.

When the same fluid is on the inside and outside of the casing (i.e. when casing is being run) the following equation can be used.

$$F_{buoy} = P_e \times (A_o - A_i)$$

where P_e = Hydrostatic pressure at the bottom of the casing (psi)

2

A_o = Area of the outside diameter (in²)

2

A_i = Area of the inside diameter (in²)

When different fluids are on the inside and outside of the casing (i.e. during cementing) the following equation can be used.

$$F_{buoy} = (P_e \times A_o) - (P_i \times A_i)$$

Where P_e = External hydrostatic pressure at the bottom of the casing

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2

A_o = Area of the outside diameter (in²)

P_i = Internal hydrostatic pressure at the bottom of the casing

2

A_i = Area of the inside diameter (in²)

Bending

Whenever pipe is forced around a bend or curve in the well, a bending force will occur. The bending force is a tensile load occurring on the outer wall and compressive loads on the inner wall of the casing.

Bending loads are calculated using the following formula

$$F_{bend} = 64 \times DLS \times OD \times W$$

where DLS = dog leg severity (° / 100 ft)

OD = casing outside diameter (in)

In nominally vertical wells the DLS can assumed to be 1° / 100 ft. For deviated wells the

bending load only applies to the point where curvature exists i.e. in build sections.

Drag

Drag is the result of sliding resistance between the well bore and the pipe. It occurs in deviated wells and in sticky tight holes. It is not easy to compute manually and is best left to computer simulations.

Shock

Shock loading is the load resulting from movement of the casing as it is run in the hole or when the slips are set or when the casing encounters a ledge downhole.

Shock loads are calculated using the following formula

$$F_{\text{shock}} = 1780 \times V \times A_s$$

where V = instantaneous velocity (ft / sec)

2

$$A_s = A_o - A_i \text{ (in }^2\text{)}$$

Care should be taken that the instantaneous velocity used in this calculation is not exceeded during rig operations.

Pressure Testing

The purpose of a casing test is to verify that the casing string can withstand the maximum anticipated burst loads. Therefore it should exceed the greatest predicted loads during both drilling and production operations.

$$F_{\text{ptest}} = P_{\text{ptest}} \times A_i$$

Where P_{ptest} = Plug bump pressure or applied pressure test (psi)

Example

Running 13-3/8" 72 lb/ft Intermediate Casing to 9,750 ft

Inside Diameter 12.347"

Mud Weight 11 ppg

Instantaneous Velocity 5 ft/sec

$$F_{\text{air}} = 9,750 \text{ ft} \times 72 \text{ lb/ft}$$

$$= 702,000 \text{ lbs}$$

$$= 9,750 \text{ ft} \times 11 \text{ ppg} \times 0.052 \times [(13.375 - 12.347) \times \pi / 4]$$

2 2

$$F_{\text{buoy}}$$

$$= 115,821 \text{ lbs}$$

$$F_{\text{bend}} = 64 \times 1^3 / 100 \text{ ft} \times 13.375" \times 72 \text{ lb/ft}$$

$$= 61,632 \text{ lbs}$$

$$= 1780 \times 5 \text{ ft/sec} \times [(13.375 - 12.347) \times \pi / 4]$$

2 2

$$F_{\text{shock}}$$

$$= 184,832 \text{ lbs}$$

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$$\begin{aligned}
 F_{\text{total}} &= F_{\text{air}} - F_{\text{buoy}} + F_{\text{bend}} + F_{\text{shock}} \\
 &= 702,000 - 115,821 + 61,632 + 184,832 \text{ lbs} \\
 &= 832,643 \text{ lbs}
 \end{aligned}$$

Example

Cementing 13-3/8" 72 lb/ft Intermediate Casing at 9,750 ft

Inside Diameter 12.347"

Mud Weight 11 ppg

Top of lead slurry at 3,000 ft

Weight of lead slurry 11.6 ppg

Top of tail slurry at 9,000 ft

Weight of tail slurry 15.8 ppg

Plug bump pressure 3,000 psi

$$F_{\text{air}} = 9,750 \text{ ft} \times 72 \text{ lb/ft}$$

$$= 702,000 \text{ lbs}$$

$$\begin{aligned}
 \text{External Hydrostatic} &= (3,000 \text{ ft} \times 11 \text{ ppg} \times 0.052) + (6,000 \text{ ft} \times 11.6 \text{ ppg} \times 0.052) + \\
 &(750 \text{ ft} \times 15.8 \times 0.052)
 \end{aligned}$$

$$= 5,951 \text{ psi}$$

$$\text{Internal Hydrostatic} = 9,750 \text{ ft} \times 11 \text{ ppg} \times 0.052$$

$$= 5,577 \text{ psi}$$

$$= (5,951 \text{ psi} \times 13.375 \times \pi / 4) - (5,577 \text{ psi} \times 12.347 \times \pi / 4)$$

$$22$$

F_{buoy}

$$= 168,425 \text{ lbs}$$

$$F_{\text{bend}} = 64 \times 1^{\circ} / 100 \text{ ft} \times 13.375" \times 72 \text{ lb/ft}$$

$$= 61,632 \text{ lbs}$$

$$= 3,000 \text{ psi} \times 12.347 \times \pi / 4$$

$$2$$

F_{ptest}

$$= 359,198 \text{ lbs}$$

$$F_{\text{total}} = F_{\text{air}} - F_{\text{buoy}} + F_{\text{bend}} + F_{\text{ptest}}$$

$$= 702,000 - 168,425 + 61,632 + 359,198 \text{ lbs}$$

$$= 954,405 \text{ lbs}$$

5.5.2 Tensile Drilling and Production Loads

This stage involves assessing the suitability of the selected casing for withstanding the loads that could be imposed on the casing after the cement has set.

It is assumed that the casing is fixed at surface and fixed at the top of cement in the annulus.

The loads that need to be considered are as follows.

Landing Casing

Any additional tension applied to the casing after waiting on cement. This is normally restricted to jackup or platform operations where tension is applied prior to setting slips in the wellhead.

Buckling

The potential for buckling exists if any of the following occur:

- Internal mud density increases
- Internal surface pressure increases
- Annular fluid is removed or annulus mud density is reduced
- Casing is landed with less than full hanging weight
- Temperature of the casing increases

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If any of the above occurs this results in ballooning of the uncemented portion of the casing string, which could result in buckling.

Buckling itself does not mean that the casing has failed. However as buckling develops into a helical form this can promote increased dogleg severity and restrict tool drift length and can promote casing wear.

In general, for all wells deeper than 10,000 ft, casing designs must be evaluated for the impact of buckling.

The effects of buckling can be reduced by raising the top of cement (reducing the amount of uncemented pipe) or (if practical) increasing the amount of tension applied prior to setting the slips.

5.5.3 Confirming that the Selected Casing Meets the Tensile Loads

From the tensile loads generated by looking at the various installation, drilling and production load cases, the worst case tensile load that the casing will have to tolerate can be determined.

The tensile capacity of the casing is down rated by the Design Factor and if this exceeds the worst case tensile load then the selected casing is acceptable for service.

If the tensile capacity after down rating by the design factor is less than the worst case tensile load then the selected casing is not acceptable for service. The next weight and / or grade up that still meets the collapse and burst loads is then selected and the tensile loads re-calculated (changing the weight of the casing affects the internal diameter of the casing and these two properties affect all of the loads that make up the tensile load). This process is repeated until a suitable casing is selected.

Example

13-3/8" 72 lb/ft N80 Intermediate Casing set at 9,750 ft

From previous examples, this casing meets the collapse and burst worst case loadings

From previous calculations, worst case tensile load occurs at plug bump.

$F_{total} = 954,405 \text{ lbs}$

From casing tables the tensile rating of the pipe body and connections are obtained and downrated by the design factor of 1.3

Tensile Capacity Downrated Capacity

Pipe body 1,661,000 lbs 1,277,000 lbs

STC 1,040,000 lbs 800,000 lbs

Connection

BTC 1,693,000 lbs 1,302,000 lbs

Connection

From this it can be seen that the pipe body and the buttress (BTC) connection have an acceptable tensile rating and are suitable for this application. However the short round (STC) connection is not acceptable as it's downrated tensile capacity is below F_{total} .

It is important, when looking at tensile capacity that the weaker of the pipe body and connection is used.

5.6 Triaxial Design

The collapse, burst and tensile loads calculated so far have all assumed that the stresses are in a single or uniaxial direction. In practice, service loads generate triaxial stresses.

The three principal stresses for casing are axial (s_a), radial (s_r) and tangential (s_t) as shown below.

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σ_a

σ_r

σ_t

σ_t

σ_r

σ_a

$$\sigma_{VME} = \{0.5[\sigma_a - \sigma_t]^2 + (\sigma_t - \sigma_r)^2 + (\sigma_r - \sigma_a)^2\}^{1/2}$$

The recommended theory for calculating triaxial stress is known as the Von Mises theory. This theory consists of defining an equivalent stress (s_{vme}) and then relating that stress to the minimum specified yield stress (s_y) of the casing.

The calculation to determine triaxial stress is best conducted using a suitable casing design programme.

Triaxial design should be performed whenever any of the following conditions apply

- Expected pore pressure > 12,000 psi
- Temperature > 250°F
- H₂S service
- $OD/t < 15$

According to Von Mises theory, an axial tensile stress can increase the tangential stress capacity and vice versa. This is shown in the diagram below.

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Pressure

Internal

Triaxle ellipse

API Burst Line for $P_n = 0$

API Axial API Axial

Compression Line Tension Line

"Burst Region"

Axial Compression Axial Tension

"Collapse Region"

Triaxle ellipse API Collapse

for $P_i = 0$ Line

Pressure

External

Triaxial Load Capacity Diagram

The equivalent stress (s_{vme}) should be calculated at the top and bottom of each casing interval of weight and grade, at the top of cement, at a particular depth where there is a specific change in internal or external pressure or a specific hole geometry (DLS, washout, etc).

Specialist casing design software is required to perform a triaxial casing design.

6.0 Casing Wear

Having an understanding of the cause of casing wear will allow the well planner to optimise the well design and drillpipe specifications in order to reduce the occurrence to a minimum.

Casing wear takes the form of a wear groove generated by a rotating drill string that is forced into the casing wall. High sidewall forces and extended contact with a rotating drill string will wear the section down. Areas that are commonly identified with casing wear include kick off points and dogleg's. The implications of casing wear can be recognised as:

- Reduction in the pressure integrity due to wear groove(s), reducing the burst / collapse values
- Expensive repairs to drillpipe hardfacing.
- Friction (surface torque) can be high.
- Wear groove may act as a starting point for future corrosion.

The types of casing wear are as follows:

Adhesive Wear

The transfer of material from a low strength body to a high strength body by solid-phase welding. However, the weld bond is weak and often the material 'falls' off the tool joint and is incorporated into the drilling fluid system as a flake.

Abrasive Wear – Machining

Often stems from pieces of exposed tungsten carbide on tool joints removing material from the casing wall. Again debris is generated that is incorporated into the fluid system.

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Abrasive Wear – Grinding / Polishing

Hard particles produce a stand off between the casing wall and the tool joint. The casing is cyclically loaded by the particles due to the drill pipe rotation causing fatigue and the surface to become brittle.

These three types of casing wear are given wear factors to highlight their severity:

Wear Mechanism Debris Produced Wear Factor

Adhesive Cuttings 400-1800

Abrasive – Machining Flakes 20-50

Abrasive – Grinding Powder 0.1-10

6.1 Casing Wear Prediction

Casing wear mechanisms can be identified prior to the start of a well, and it is the control of the mechanism that reduces the amount of casing wear.

6.1.1 Contact Pressure and Load

Initially this is very high due to the small contact area resulting in high pressures and a large amount of wear. As wear continues it will expose a larger surface area and will help to distribute and reduce the wear pressure. Doglegs and build and drop off sections deserve special attention though. Transocean sets acceptable sidewall force limits as 2200 for water based mud systems and 2500 for oil base mud systems.

6.1.2 Well Design

Anticipate and consider ‘what if’ scenarios. Likelihood of additional drilling, stuck pipe and fishing operations for example.

6.1.3 Doglegs

Doglegs are unavoidable in many wells, and when designing a well it is important to understand their effect on casing wear. Whilst drilling the dogleg it is important to maintain drilling parameters as constant as possible to control the dogleg severity. Smoothing out the dogleg profile can also have an impact on wear reduction and is strongly advised. Implementing a deep kick off will lower the amount of sidewall forces exerted on the casing.

6.2 Control of Casing Wear

Certain areas have been identified that require to be considered during well planning to minimise casing wear. In the field an MFCT log can be run before and after the predicted wear to measure the extent of the damage.

6.2.1 Casing Material

It cannot be assumed that the higher the strength for resisting collapse and burst the higher the resistance to wear. Materials must be looked at closely if significant casing wear is expected.

6.2.2 Wear Bushing, Crossovers, Centralisers and Cementing

Regular inspection of the wear bushing is important based purely on the number of times it is exposed to tool joints travelling in and out of the hole.

When crossing over between two different weights and grades of casing this section needs to

be supported due to the different stiffness of the two sections. Support can either come from cement or adequate centralisation, otherwise a local dogleg might well develop. The internal wall of the crossover is shouldered and will experience wear from drill pipe tool joints, the shoulder should be chamfered as much as possible.

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It is prudent to locate the planned top of cement away from a zone of high sidewall forces as immediately above the cemented pipe a local dogleg is likely to develop.

Centralisers should be placed across the pipe body as opposed to locating them at the coupling. The additional stand-off provided by the centraliser may result in a local dogleg establishing itself above the coupling.

6.2.3 Drill Pipe Hardbanding

Today the vast majority of hardbanding is a 'wear resistant alloy', such as Armacor M and Arnco 200 XT, that exhibits a uniform surface hardness, as opposed to needing the addition of tungsten carbide. Nevertheless it is still important to check what is received to make sure it is 'user friendly'.

6.2.4 Drillpipe Protectors

These are placed on the body of the drill pipe a few feet above the tool joint to provide stand-off between the tool joint and the casing. They are an elastomer and are manufactured to be $\frac{1}{2}$ " larger than the OD of the tool joint, during their use they require regular calipering for wear and subsequent renewal. An alternative is to incorporate rotating sleeve subs within the drill string.

6.2.5 Mud Types

Water Based Mud's

Severe adhesive wear with friction. Caused by a lack of solid barriers in the mud.

Oil based Mud's

Reduced friction. Potential for casing wear dependant upon drill pipe hardbanding and sidewall contact pressures.

Unweighted / Weighted Mud's

Incorporating a weighting material will provide solids to the mud system that will act as particulate matter between the tool joint and the casing, providing a layer between the rolling surfaces. Particle size and hardness are relevant, a larger particle will provide greater stand-off, and a softer material will perform better. For example barite performs better than additions of haematite and quartz.

Sand / Silt

Sand particles are too large to 'roll' in the space between tool joint and casing. Unweighted systems experience so much adhesive wear as to make the abrasive influence of sand unnoticed. In a weighted system the effect of sand is again unnoticed due to the diluting effect of the weighting product.

Lubricants

Their effect is dependent on surface conditions of the casing / tool joint and the amount of solids in the system. Lubricants produce a film that lies on the casing and the tool joint providing a surface of low resistance. However as the solids content of a drilling system may increase these solids will penetrate the film to such a degree as to prevent any film / film contact, rendering the lubricant ineffective. If lubricants are included solids control will be a high priority.

7.0 Material Selection.

Two types of service need to be considered:

- Strings that are exposed to brines and mud's.
- Strings that are exposed to reservoir fluids.

7.1 Strings Exposed to Brines and Mud's.

Casing material is commonly composed of carbon or low-alloy steels; there is a wide range available as shown in API 5CT. Depending on geographical location the buyer may have access to non-API standard tubulars that claim to have a better performance.

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When dealing with completion brines and drilling muds, corrosion resistance is often not critical to the design. However if exposure to hydrogen sulphide is anticipated, casing that is manufactured for sour service will have to be selected.

API controlled grades of steel for high pressure sour wells can lead to very heavy wall casing designs. There is growing interest to use specialised high strength grades from non-API vendors.

7.2 Strings Exposed to Reservoir Fluids.

Primarily this section concerns itself with production tubing, however production casing / liners need to be considered for possible exposure due to a leak or tubing failure.

A rough guide exists for materials required, based on the partial pressures of carbon dioxide and hydrogen sulphide in the gas phase.

1. $\text{CO}_2 < 3 \text{ psia}$ and $\text{H}_2\text{S} < 0.05 \text{ psia}$
2. $\text{CO}_2 < 3 \text{ psia}$ and $\text{H}_2\text{S} > 0.05 \text{ psia}$
3. $\text{CO}_2 > 3 \text{ psia}$ and $\text{H}_2\text{S} < 0.05 \text{ psia}$
4. $\text{CO}_2 > 3 \text{ psia}$ and $\text{H}_2\text{S} > 0.05 \text{ psia}$

To determine the partial pressure in the gas phase multiply the mole fraction by the bottom hole pressure.

For example:

10,000 psi bottom hole pressure with 50 ppm H_2S = 0.5 psia
 $(50 / 1,000,000 \times 10,000)$

10,000 psi bottom hole pressure with 3% CO_2 = 300 psia
 $(3 / 100 \times 10,000)$

1. $\text{CO}_2 < 3 \text{ psia}$ and $\text{H}_2\text{S} < 0.05 \text{ psia}$

Carbon or low alloy steels. If carbon steels are selected a rigorous look at their service life and

corrosion potential is needed.

2. $\text{CO}_2 < 3$ psia and $\text{H}_2\text{S} > 0.05$ psia

Carbon or alloy steels. However the materials need to meet the standards of NACE MR0175-91. Grades that are for sour service are found in API 5CT and NACE MR-0175-91.

3. $\text{CO}_2 > 3$ psia and $\text{H}_2\text{S} < 0.05$ psia

It is not recommended to use carbon or low alloy steels because they will have insufficient corrosion resistance. Inhibitors and plastic lined pipe is one option however the job starts to get a little complicated.

A material that has been proven in the field is 13% chrome steel tubulars. Although a number of limitations do exist, above grade C-95 the material becomes brittle if in sour service and above 0

125 C there is an increased risk of pitting or general corrosion.

If the grade or temperature is affecting the design then a more exotic tubular is required like duplex stainless steel for example.

4. $\text{CO}_2 > 3$ psia and $\text{H}_2\text{S} > 0.05$ psia.

This environment is indicative of a corrosive well and a sour well. Corrosion resistant alloys and chloride / sulphide stress cracking resistant metals are needed for the design. 13% Chrome steel tubulars are not adequate and consideration can be given for 22% or 25% Cr, but with an H_2S partial pressure above 1.5 psia austenitic stainless steels need to be used.

As much information on the operating conditions is required in order to complete the final selection. Such as:

- Design Life
- Fluid types

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- Partial pressures of CO_2 and H_2S
- Operating and design pressures
- Operating temperature
- Flow rate / regime
- Possibility of sand production
- Water chemistry
- Any changes likely during the life of the field / tubulars
- Is gas lift likely?

7.3 Considerations for Corrosion Resistant Alloys

7.3.1 Chlorides and Bromides

Unexpectedly high levels for the casing design are likely to cause pitting and chloride stress cracking. An increase in temperature will heighten this. It may be necessary to restrict the temperature and / or chloride / bromide levels to complete the design.

7.3.2 Mechanical Properties

Tensile Properties

The yield strength can be anything up to 10% less in the transverse direction with materials that depend, for their strength, on cold working. If a specific application is required then the exact amount it is reduced by will need to be determined.

Yield strengths that are quoted in manuals are for an ambient temperature, hence for any increase in the temperature a decrease in the yield strength will occur. This is particularly important when selecting corrosion resistant alloys, and the amount of reduction for the designed temperature will need to be known.

The highest grade recommended when using duplex and highly alloyed austenitic steels is 125k psi. Above this and there is a chance for there to be a reduction in the corrosion resistance.

Toughness

This varies with temperature and tubulars can be ductile or brittle. Brittle behaviour needs to be avoided because it can be responsible for sudden failures. A control measure that can be used is the Charpy Impact Test, which is relatively inexpensive.

Erosion – Corrosion

The fluid velocity can advance tubular corrosion, and the design of the system needs to remain below the critical velocity. The critical velocity will naturally vary between the different alloys.

7.4 Corrosion Mechanisms

Carbon Dioxide

Carbon dioxide combines with water to form carbonic acid and will attack carbon steels where the iron carbonate layer is missing, hence localised pitting develops.

The rate of corrosion is controlled by a number of factors including CO₂ partial pressure, pH, temperature, fluid velocity and other chemicals.

Hydrogen Sulphide

Sources include: Well fluids

Bacterial activity

Breakdown products of other chemical additives

Hydrogen sulphide dissolves in water and can form a protective layer of iron sulphide scale, pitting occurs where the scale is not present.

Hydrogen is a natural by-product of water corrosion and the hydrogen molecules are too large to diffuse into the metal. In the presence of hydrogen sulphide the hydrogen atoms are prevented from combining into molecules, leaving the atoms to diffuse into the metal.

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The hydrogen atoms tend to concentrate in areas that are already under stress, their accumulation increases the stress levels and reduces the strength of the material. Stressed areas are stressed even further and cracking results which can occur rapidly.

Oxygen

Dissolved oxygen attacks iron converting it to an oxide and / or hydroxide. The rate of corrosion

is controlled by the ability of the oxygen to diffuse to the area to support the corrosion process. This can be a problem when using water based drilling mud and in a water injection system. Scavengers are added to the mud and degassers or stripping units are associated with water injection.

Halide Ions

Cause localised pitting and / or crevice corrosion on materials used for downhole equipment. Halide ions are also responsible for causing stress corrosion cracking, in particular materials

such as austenitic steels. Cracking will normally occur above 50 C and the result is brittle failure well below the materials yield strength.

Erosion – Corrosion

Damage is increased because of the velocity of a passing fluid. Velocity limits to minimise this form of erosion are related to API RP 14E.

Galvanic Corrosion

Corrosion that takes place when two metals are coupled in an erosive environment. One metal will act as an anode and corrode faster, the other the cathode. The controls on this sort of erosion are the conductivity of the corrosive medium, surface area of the two metals and the difference in equilibrium potentials of the two metals.

Localised Corrosion

Corrosion pitting where certain areas of the metal act as an anode. Process is enhanced by dissolved oxygen and is strongly influenced by temperature. Pitting is particularly damaging, as it is more penetrative than general corrosion.

Crevice corrosion is localised and confined to a gap between two materials. Within the gap the environment can be quite different to that outside enhancing the corrosive process, higher temperatures are conducive to this process.

8.0 Pore Pressure and Fracture Gradient Prediction

8.1 Pore Pressure Prediction

Pore pressure can be predicted before, during and after drilling by applying various methods, some directly and others indirectly or by correlation. The “before drilling” is important as a basis of design for the drilling programme. The “during drilling” determination will serve to confirm the design whilst the “after drilling” determination will provide data for future well designs in the area giving a greater level of confidence.

Before drilling estimation:

Lithology correlation – knowledge regarding abnormally pressured formations can be applied to occurrences of the same lithology if predicted along the proposed wellpath.

Shallow seismic (bright spots) – used to detect shallow gas and can infer seabed characteristics e.g. firm for jacking up.

Seismic depth correlation – used to identify salt structures and predict depths of rocks whose signatures were determined on previous offset wells.

Seismic interval velocity plots – if a shot point corresponds with the well location, can be used as a sonic log

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During drilling:

Corrected d exponent d_c – in over-pressured zones, compaction trends are reduced and formations are more porous/permeable than expected. Hence drilling rates increase as there is less resistance (less drill chip hold down force)

MWD data – similarly, as compaction trends decrease in over-pressured zones, the MWD will identify higher porosities than predicted against a normal trend.

Kicks – indicate pore pressure greater than hydrostatic

Drillstem tests – measure pore pressure directly

Changes in pore pressure indications:

Gas cut mud – indicates increasing pore pressure

Flowline temperature changes

Change in drilling rate

Chloride change – increase as pore pressure increases (influx of formation water)

Torque and drag – decrease as pore pressure increases

Volume, shape and size of cuttings – increase in pore pressure yields “pressure cavings” in increasing quantities

After drilling estimation:

Well logging – any logs that allow porosity or resistivity to be determined e.g. sonic, density.

Readings should be taken in shale and avoid washouts. Again deviations from trend lines of normally pressured formations will indicate the onset of an abnormally pressured zone.

Bottom hole pressure surveys

Note:

1. Trend lines and pressure correlation can vary both from area to area and even with formations of differing geological age within the same lithological column.
2. Analysis assumes consistent shale properties which is unlikely.
3. In hard rock areas, (carbonate sequences), compaction trends are interrupted since carbonates do not compact in the same manner as clays.

8.2 Fracture Gradient Prediction

Fracture gradient prediction is now largely based on the work carried out by Hubbert and Willis in 1957. This work predicted fracture pressure for vertical fractures based on a relationship between the overburden pressure gradient, pore pressure gradient and the matrix stress coefficient (ratio of horizontal to vertical stress). They assumed that fracture pressure was equal to pore pressure plus the horizontal component of the vertical stress in the rock.

Later workers have developed this basic relationship into the more usual forms today.

Mathews and Kelly 1967: $F = P/D + K_i(\sigma/D)$

Where F = fracture gradient, psi/ft

P = formation pore pressure, psi

σ = matrix stress at point of interest, psi

D = depth at point of interest, ft tvd

K_i = matrix stress coefficient, no units

Eaton 1976: $F = ((S/D - P_f/D) \cdot (y/(1-y)) + (P_f/D)$

Where P_f = formation pressure, psi

D = depth at point of interest, ft tvd

S = overburden gradient, psi

Y = Poisson's ratio (shear strain/axial strain)

Offset well data may yield leak off test data that can be plotted against vertical depth and lithology and correlated against the proposed well plan to give additional fracture gradient reference points.

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7 DRILLING PROBLEM

7.1 Introduction

The main objectives of every drilling operation are :

1. High drilling efficiency through the achievement of

- Optimum penetration rate
- Minimum downtime

2. Minimum operating cost

3. Effective safety

4. Environmental Impact

There are however a number of problems encountered in the hole which mitigate against the achievement of the above key objectives. The problems which constitute a hazard to drilling are:

1. Lost Circulation

2. Differential Sticking

3. Washouts

4. Kick / Blowout

5. Formation Damage

The design of drilling fluids and drilling techniques adopted play a major role in the propagation or prevention of these problems. This note therefore covers

- The functions/properties of drilling fluids
- Causes, prevention and control of the drilling problems.

7.2.1 Lost Circulation

Lost Circulation can be described as the loss of the substantial quantity of

whole mud to “thief” formations otherwise known as lost circulation zones.

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7.2.1 Causes of Lost Circulation

The problem can be caused by natural or induced factors.

Natural Causes

They are basically caused by the presence of natural fractures, high permeability formations, cracks, vugs and fissures.

Induced Causes

- a. High Mud Density: When the mud density is so high as to make the hydrostatic or bottom hole pressure greater than the formation fracture gradient resulting in a breakdown of the formation. Fractures can be generated which will result in mud loss
- b. Increase in Annular Pressure: Increase in annular pressure beyond the maximum allowable annular surface pressure (MAASP) can result in the generation of fracture - a potential thief zone.
- c. Pressure Surge: During trips, running-in the drillstring at a very high velocity can result in localised pressure surges that can lead to the breakdown of weak formations.
- d. Penetration of depleted or subnormal formations

7.2.2 Implication of Lost circulation.

1. Formation Damage: Loss of whole mud results in loss of filtrate and mud solids which will result in plugging and permeability reduction.
2. Kick: Loss of whole mud will lead to loss of mud column height. This can make the hydrostatic pressure of mud column to be less than the formation pressure. The net result is an influx of formation fluid into the wellbore. This is known as kick.
3. Massive increase in mud cost.

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7.2.3. Prevention of Lost Circulation

1. Use of Loss Control Materials (LCM) as a regular mud additive. Examples of LCMs are nut plug, kwikseal, mica, corn fibres, calcium carbonate, sized salt, Oil soluble resin (OSR)
2. Prevention of pressure surge and maintenance of annular pressure less than MAASP.
3. Good estimate of fracture gradient.

7.2.4 Control of Lost Circulation

The first step involved in the control of lost circulation is the identification of the thief zone. The techniques used include

1. Spinner Survey
2. Temperature survey
3. Radioactive tracer survey
4. Pressure transducer's survey

These techniques are now normally part of regular MWD techniques.

Control of lost circulation is achieved through the use of loss control materials.

1. For High Permeability formations: Calcium Carbonate and sized salts may be used
2. For cracks, fissures, vugs, etc, mica, nut plugs, fibres, etc may be used.
2. For severe cases, spotting of 'gunk' - mixture of diesel/barite or cement may help.
3. Total isolation with intermediate casing cemented in place may help.
4. "Drilling Blind" - a process in which the total return mud including cuttings is allowed to flow into the thief zone and plug whilst being practised in

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some special problem cases, is not a good technique. Where the lost circulation is severe, it is useful to displace "gunk" rather than drilling blind

7.3. Pipe Sticking

7.3.1 Causes of Pipe Sticking

Drill string get stuck due to a variety of reasons

1. Junk inside the hole
2. Keyseating
3. Sloughing/Heaving Shales
4. Bit/Drillcollar balling
5. Drill cuttings above the bit or drill collar due to improper solids control
6. Pressure Differential sticking

(a) Differential sticking

Differential sticking occurs when the drillstring becomes motionless against a permeable formation due to excessive hole-to-formation differential pressure (excessive overbalance) acting on an isolated area. The differential pressure exerts a considerable force against the drillpipe and literally glues it to the borehole wall.

The drillstring is also amenable to seeking when the filter cake is too thick and

if the string is left static for extended period of time in the hole.

7.3.2. Prevention of Pipe Sticking

Pipe sticking can be prevented by adopting the following strategies:

1. Minimising the margin of overbalance and maintaining efficient control of mud properties
2. Minimising the drillpipe-to-borehole wall contact area by:
 - Minimising filter cake thickness through efficient solids control and using low fluid loss mud
 - Using stabilisers on drillcollars
 - Using grooved or non-circular/external upset drillcollars

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- Minimising non-rotating time
3. Minimising friction factor by using possibly emulsion mud.

7.3.3 Control of Stuck pipe

Once a pipe is stuck, it has become a fish. The only way to control stuck pipe is by initiating fishing operation.

The first step in fishing operation is to evaluate the depth at which the string is stuck. This is known as the free point. Two methods are in common use for evaluating the free point:

1. Pipe stretch method using surface observations (Pg62)
2. Free point indicators - Logging while drilling(LWD) techniques which employ the use of specialised strain gauge and subsurface probes.

Pipe Stretch Method

This simple method requiring measurement from the rig floor involve the following procedure:

1. Pull drillstring using normal hook load designated here as F1 Mark a reference point X1 on the pipe.

Pull additional load between 20klbs and 40klbs, F2 less than yield point of drillpipe. This will cause the free portion of pipe to stretch to a new position X2.

3. Calculate the free point using the following equation:

$$Lst = 735294(X2 - X1)WdP/F$$

Lst= Depth of stuck pipe, ft

WdP = Weight of drillpipe, lb/ft

F = Additional pull, lbs

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Once the free point is evaluated, the pipe can be freed using the following methods:

1. Spotting Organic Fluids

A mixture of diesel oil and surfactant is usually spotted across the stuck point to reduce the mud cake thickness, thus reducing the friction factor. The pipe is worked continuously during the spotting. Fluid has to be spotted against the correct zone and given sufficient time to dissolve and destroy the cake. The procedure involves pumping the oil/surfactant into the drillpipe and the gradually pumping small volumes into the annulus until the stuck pipe is adequately soaked.

2. Hydrostatic pressure reduction

This method involves creating a pressure differential inside the drillpipe and outside the drillpipe in the annulus. The pressure differential is achieved by either pumping a new lower density mud, or by pumping a small volume of lower specific gravity fluid such as diesel oil into the drillpipe and then bleeding it from the drillpipe. This will drop the mud level in the annulus thus reducing the pressure of the mud at the stuck point. The maximum volume of oil or water to be pumped into the drillstring and the maximum drillpipe gauge pressure are given by the following equations:

$$L_{st} V_{dp} = P_p \times V_{an} \times \rho_0$$

$$V_0 = \frac{P_p}{\rho_0 - \rho_m} + \frac{P_p}{\rho_m - \rho_0}$$

$$\rho_m - \rho_0 = 0.052 \frac{L_{st} V_{dp}}{m}$$

$$\rho_m - \rho_0 = 0.052 \frac{L_{st} V_{dp}}{m}$$

$$\rho_m - \rho_0 = 0.052 \frac{L_{st} V_{dp}}{m}$$

$$\rho_m - \rho_0 = 0.052 \frac{L_{st} V_{dp}}{m}$$

$$0.052 V_0 (\rho_m - \rho_0)$$

$$(P)$$

$$dp_{max} =$$

$$V_{dp}$$

The fluid level drop in the annulus when the mud and formation pressure are equal is:

$$\Delta FL = L_{st} - [P_p / 0.052 \rho_m]$$

The mud weight equivalent at depth during the bleedoff is

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$$(P) - P$$

$$\rho_e = \rho_m -$$

$$dp_{max} dp$$

$$\rho_m V_{dp} \rho_0$$

$0.052 \text{ Lst} \frac{1}{1000} + \frac{an}{1000}$

$\frac{1}{1000} - \rho \frac{1}{1000}$

0

$\frac{1}{1000} \frac{1}{1000}$

$\frac{1}{1000} \text{ Vdp} \frac{1}{1000} \text{ m} \frac{1}{1000}$

Vdp= Drillpipe capacity, bbis/ft;

Pp = pore pressure, psi.

(Pdp)max = max. drillpipe pressure,

Van.= annular capacity, bbis/ft

Pdp = Drillpipe pressure at surface

3. Backoff Operations

When attempts to free the stuck pipe fails, a backoff operation is the last resort. This technique involves parting the drillstring at or above the stuck point and recovering the drillstring. The remaining string can be freed using a combination of:

1. Hydraulic jars
2. Overshot
3. Hook, etc.

7.4 Kicks and Blowout

7.4. 1. Introduction

A kick is the term used to describe the inflow of formation fluid into the wellbore during drilling operation. This mainly arises due to the formation pressure being greater than the hydrostatic pressure imposed by the mud column. This can happen in a normal or abnormal formation especially geopressed formation.

A kick may be primarily composed of salt water, oil, natural gas or a combination of the fluids. The influx of formation fluid arises for a variety of reasons. These include:

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1. Insufficient mud weight
2. Low level of fluid column in the annulus
3. Excessive Swabbing
4. Excessive drilling rate in gas bearing formation
5. Diffusion of gas through mud cake.

Kicks generally occur during trips with influx occurring from a combination of swabbing and failure to keep the hole full. Swabbing is the result of the piston

effect of the drillpipe due to excessive trip speed resulting in instantaneous vacuum being created below the bit. Kicks occurring during drilling operations most often occur due to insufficient mud weight.

1. Insufficient Mud Weight

Insufficient mud weight may occur due to:

1. Penetration of a geopressured zone
2. Accidental dilution of the mud by fluid addition at surface
3. Dilution of mud by influx from an aquifer exposed to open hole
4. Gradual mud density reduction due to gas cut and failure to degas the mud at surface
5. Improper mud mixing and poor quality control in measurement

However, for strategic operational reasons, a borehole may be drilled underbalanced with controlled kick taking place.

2. Low level of drilling fluid column

Two conditions may lower the fluid column in the annulus. These are:

- Failure to fill the hole during a trip
- Lost circulation during drilling

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In pulling out the drillstring, the hole driller must refill the hole with mud of equivalent volume. Monitoring of the volume of mud filling the hole is done with the use of the Trip tank or Possom Belly Tank.

The trip tank is a small mud pit (Approx. 50 bbls) separate from the active pits. This trip tank is fully equipped with a volume gauge that can be monitored from rig floor or mud loggers console. In monitoring the trip, the driller must calculate the theoretical mud volume displaced by one or five stands of drillpipe. During the trip, the driller then measures the actual mud volume pumped into the hole for each stand of pipe pulled.

Lost Circulation is the loss of substantial quantity or whole mud into the formation. It may occur as a result of:

- ◆ mud loss to cavernous or vugular formation
- ◆ penetration of depleted or subnormal zones
- ◆ mud loss to fractures opened by excessive annulus pressure. This may be due to annular blockage, surge pressures, excessive hydraulics, pressures imposed to break mud gel strength.

Lost circulation will result in insufficient mud column and thus reduction in bottom hole pressure.

Kick due to lost circulation can be a major well control problem. If the well cannot be shut in, an underground blow out can occur.

3 Excessive Swabbing

Swabbing is the reduction of bottom hole pressure caused by upward pipe movement. As the pipe is being pulled out during trips, mud flows down through the annulus to fill the space in the hole. The downward mud flow creates a frictional pressure loss in the direction of the surface. The result is reduced bottom hole pressure. High trip speed can lead to instantaneous vacuum below the bit. The swab pressure depends on :

- Pipe velocity

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- clearance between pipe and hole - The smaller the annular cross section, the greater the swabbing action.
- mud rheology - The higher the viscosity, the greater the swabbing action
- total measured depth - The longer the section, the greater the swabbing action.

Excessive drilling rate in gas-bearing formations

The densities of formation fluids(gas, oil or water) are much lower than that of conventional drilling fluids. Any influx of formation fluid therefore reduces the mud density in the annulus. This influx otherwise known as mud cut is measured using sensors that measure mud density and conductivity at injection and at flow line

Mud cut may occur from:

- kick due to inadequate mud weight or swabbing
- diffusion of fluid into the hole due to negative differential pressure
- air entrained in mud from the so called Kelly air during trips
- expansion of drilled gas as it reaches the surface.

Majority of kick problems arise from the reduction of mud weight due to the expansion of drilled gas especially in large diameter holes such as surface holes drilled at high rates. For such holes drilled without a blowout preventer in place; a small reduction in mud weight can lead to a violent kick and with the loss of primary control, lead to blowout.

Kick Detection

A kick can occur at any time during the drilling of a well. To prevent any major catastrophe therefore, early detection of a kick is essential to primary well control.

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As a primary precaution, it is important for all personnel including the mud logger to always be in a state of readiness. Certain pre-kick information must be collected and available for use in case a kick occurs. These include volumes and pressures data. A kick control Work Sheet must be maintained for this purpose.

The pre-kick data requirements include

1. Maximum Working Casing Pressure

The casing and blowout preventer(BOP) are designed for the different expected pressures during each drilling phase . This is determined during well planning and is generally the maximum pressure rating of the BOP and its outlets or 80% of the burst pressure of the last casing. The 80% consideration is a safety factor for the casing string.

2. Maximum Allowable Annular Surface Pressure(MAASP)

This is the annular pressure at the surface that corresponds to the pressure at the weakest point of the hole. Fracture gradient at the last casing shoe defines the weakest point. However for some boreholes, the MAASP can be the casing burst pressure or BOP rupture pressure if they are less than the fracture gradient. The MAASP represents the maximum annulus pressure permitted while controlling a kick.

The MAASP is estimated from the following equation:

$$\text{MAASP} = 0.052 * (\text{FRAC} - \rho) * D_w$$

where

FRAC = Fracture pressure at shoe given as equivalent mud weight in ppg.

ρ = Current mud density, ppg

D_w = True Vertical Depth (TVD) of weakest point, ft

In Metric Units

$$\text{MAASP} = (\text{FRAC} - \rho) * D_w / 10$$

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$$[\text{FRAC}] = \text{Kg/cm}^2$$

$$[\rho] = \text{Kg/cm}^2$$

$$[D_w] = \text{metres}$$

3. Rig Capacity for Weighting Mud

Although this data does not appear in the worksheet, it is important as it defines the number of circulation cycles necessary to regain primary control with given change of mud. The rig capacity depends on total reserves of weighting materials (Barites, etc) and the maximum rate of addition to system.

The maximum rate of addition is measured in lb/min or Kg/min. Backup supply MUST always be available on rig.

4. System Pressure Losses at Slow Pump Rate

Circulation at pump rate for drilling may produce surface pressures greater than the MAASP. Therefore a slow pump rate for kick control must be determined at the beginning of the well. This is usually 1/3 to 1/2 of normal rate. This slow rate must be taken

- at the start of each tour
- if the mud weight changes
- if bit arrangement changes
- after bit changes

If a kick occurs, the influx will be circulated out at the maximum pump rate that does not exceed the MAASP. This is called the kill rate.

The pressure losses must always include the choke line pressure losses especially offshore where this could be substantial.

5. Pump, String, and Hole Configuration

This includes the data used for the determination of the lag time and contains

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- Pump capacity
- Hole size
- Drill string capacity (internal volume)
- Annular capacity (bit to casing shoe and bit to BOP)
- Casing data (size; weight; burst pressure; shoe depth)
- The worksheet includes the space to fill in lag time in strokes or minutes for the proposed kill rate.

This estimation is usually computerised nowadays and continuously logged.

7.4.3 Kick Detection Techniques

There are a number of indicators that provide early warning of kick occurrence. These indicators include:

The positive kick indicators are :

1. Mud Pit Level/Flow Increase
2. Incorrect hole fillup during trip
1. Decrease in Standpipe pressure/increase in pump rate
2. Increase/Decrease in drillstring weight

The potential indicators are

1. Increase in Penetration Rate
2. Lost Circulation
3. Changes in gas levels, mud density and conductivity

1. Mud Pit Level/Flow Increase

These parameters are major positive indicators. As the influx of formation fluid displaces the mud in the annulus, the pit level would increase in the active

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mud tanks. There also be evidence of increase in return flow rate if logged.

The effectiveness of pit gain/flow increase depend on

- Initial size and type of kick
- Mud fluid base and rheology
- Minimum sensitivity of the measuring systems

Gas kicks are more difficult to detect than oil/water due to its compressibility. It tends to go into solution in the drilling fluid. However, a small gas kick can expand multiple times its original volume by the time it reaches the surface.

Return flow is more sensitive to gas kick than pit level monitoring overall. The return flow sensors may therefore provide the first warning of a gas kick.

Sensor resolution determines the minimum size of kick detected. Therefore, the mud logger should normally set alarms on the pit volume parameters close to the actual pit value. Report should be sent to the driller no matter how small the pit gain is. Daily checks on the alarms and routine maintenance is essential to effective monitoring.

2. Incorrect Fillup during trips

When pulling out of the hole, if the volume of mud pumped to keep the hole full is less than that normally required, then there is an evidence of influx. The mud volume required should be equal to or slightly greater than the volume of drillpipe (normally five stands) already pulled. Mud loggers MUST follow trips even though an automatic trip monitor is usually used.

Conversely during a trip into the hole, the downward movement of the drillpipe expels fluid from the annulus to the trip tank or active mud system. This return flow should cease a few seconds after pipe movement stops. If flow continues, then there is a kick.

3. Decrease in Standpipe pressure/Increase in Pump rate

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Influx of fluid into the annulus creates an imbalance resulting in a decrease in hydrostatic pressure in the annulus. In such an unbalanced system, gravity

helps move drilling fluid down the hole, requiring less energy from the pump. This will result in a decrease in the standpipe pressure.

4. Increase/Decrease in Drillstring Weight

Any influx into the wellbore from the formation reduces the buoyancy of the annular drilling fluid. A sensitive weight indicator will register this change in weight as an increase in drill string weight. For very large kicks, fluid may enter the annulus with enough force to cause a decrease in indicated string weight.

5. Increase in Penetration rate (Drilling Break)

A marked increase in rate of penetration (ROP) may indicate either changes in the type of formation being drilled or a reduction in the differential between the mud pressure and pore pressure.

Generally, the following parameters affect the ROP:

- rock type
- formation bulk density/porosity
- differential pressure
- bit type/wear
- hydraulics
- weight on bit
- rotary speed
- personnel/equipment

Drilling breaks are generally evidence of porosity change. Drilling rate tends to decrease with depth. Thus when a drilling break occurs, it may be an

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evidence of transition to an abnormal zone. It is crucial at this point to stop drilling and check for flows.

6. Lost Circulation

Loss of substantial quantity of mud into the formation will result in reduction in hydrostatic column height. If not checked, this can result in a kick.

7. Changes in gas levels/mud density/conductivity

Gas extracted from the mud comes from one or more of four sources

i Liberated gas which is the measured gas from the return mud flow, released from the pore spaces of the drilled cuttings. It is the so-called "background gas" during circulation. If there is overbalance, and the ROP is constant with flow rate, this liberated gas.

ii Produced Gas This gas enters the wellbore from adjacent permeable

formations when underbalanced exists. A background gas increase when ROP is constant is evidence of produced gas.

iii Recycled gas is the gas re-circulated into the hole. It appears on detection equipment as an increase in background levels.

iv Contamination Gas results from chemical breakdown of mud additives

v Connection gas and trip gas are short duration gas peaks caused by swabbing action

Depending on sensitivity level, surface monitors should detect a relatively steady level of gas extracted from mud during normal drilling. This background gas level may show occasional variations depending on penetration rate, mud pumping rate, hydrocarbon content of section drilled. Under normal conditions, the background gas should remain within about 50% of local average. It is crucial that all gas values must be reported whether they are significant or not. Thorough inspection of gas monitoring systems and calibration as part of routine maintenance exercise is essential to preventing potential disasters.

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Connection and trip gas are most common while drilling. The connection gas peaks are a clear sign that pressures are near balance, making for optimum ROP. Swabbing is the main cause of connection/trip gas as it creates the set of negative differential pressure. .

Effects of gas expansion at surface are varied. Mud loggers and drilling crew can see evidence of:

- Rapid fall in flow line mud density
- Increase in return flow
- Mud pit level increase
- Rapid increase in total gas or hydrocarbon levels:

7.5 Well Control Procedure

Control procedures must be initiated immediately after a kick has been detected. Decisions concerning control operation rest on the Company man (The so-called client rep.) as well as the on-site Tool Pusher. The mud loggers are responsible for monitoring and recording circulation pressures, gas and pump strokes.

The primary and secondary well controls involve the prevention and having occurred, the removal of influx without damage to hole, personnel and equipment as well as prevention of further influx.

Kick control procedures involve the following:

1. Shut in the well - Follow a predefined procedure

2. Observe shut-in pressures and kick volume

3. Make kill calculations for:

- Formation pressure
- BHP to maintain while circulating
- Kill mud density
- Initial and final circulation pressures
- Drillpipe pressure schedules

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- Weighting material volume required

4. Define weighing up and circulating procedure using either of the following methods:

- Drillers method
- Wait and Weight method (so-called Engineers method)
- Concurrent Method
- Top kill method

5. Make interpretation of the influx including

- height and density of influx
- annulus pressure behaviour while circulating

6. Proceed with circulation using the following guide rules

a At all times during circulation, the bhp must be high enough to prevent further influx.

b For the safety of the rig and personnel, surface pressure must not at any time exceed the predetermined MAASP value. Maintenance of bhp at proper level is through the application of backpressure via the adjustable choke.

7.6 Surface Well Control Equipment

The control equipment is laid out as shown in Fig 7.1 and is made up of:

1. Blow out preventers - Usually mounted in a stack including Annular and ram type preventers(Fig. 7.2)
2. Drilling Spool (Fig 7.3) for attachment of high pressure choke and kill lines, for circulation with the BOP closed.
3. Casing head - Ibis is welded to the first surface casing and it provides support and pressure seal for the BOP stack and future casing strings.

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Fig. 7.1

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Fig 7.2: Blowout Preventer Stacks

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Fig. 7.3 BOP Components with Drilling Spool

4. Choke Manifold (Fig. 7.4) - To control flow of produced fluids and route them to separator, flare or holding tank or pit. A typical choke manifold includes:

a One hydraulically operated BOP outlet valve for positive closure

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b Three adjustable chokes (variable-opening valves)

c Several manual valves

Fig. 7.4:

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At least one adjustable choke is manually operated and permits precise control of return flow rate and backpressure. Operation of other chokes is from the choke control console (Fig. 7.5), which is also linked to the Mud loggers cabin by remote sensors for automatic logging.

Fig. 7.5: Control Console

5 Kelly Cock, Float Valve or Inside BOP (Fig 7.6) - To prevent back flow via the drillpipe.

6 Mud Gas separators and Degassers - Equipment for removing gas from mud

For subsea well control equipment, the layout and configuration are rather different. They may include (Fig 7.7)

- Templates - Temporary and permanent guide bases
- Subsea BOP stack

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- Marine riser, etc

Fig. 7.6: Drillstring Well Control System

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Fig. 7.7: Subsea Well Control Equipment

7.7 Kill Calculations

To make kill calculations, the following initial data must be available:

1. SIDPP 2. SICP 3. Volume of influx 4. Initial mud weight

Calculations required are:

1. Formation or pore pressure
2. Bottom hole pressure to maintain well during killing operation
3. Kill mud density

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4. Initial circulation pressure
5. Final circulation pressure
6. Volume of weighting material needed.

There are a variety of software that can be used on-site to carry out these calculations. Nevertheless, many company men will feel comfortable with hand calculated values. Recordings must also be made on the Kick control worksheet.

Formation Pressure Computation

$$PF = PH + SIDPP$$

$$PH = \text{Hydrostatic pressure } 0.052 \cdot \rho_m \cdot D$$

ρ_m = Mud density, ppg; D = Depth, ft

Bottom Hole Pressure to maintain While Killing the well

This must be in excess of the formation pressure to prevent influx. This should include a safety factor (SF) called trip margin to compensate for calculation errors and swabbing. SF can be about 0.3 to 0.5 ppg equivalent. A better practice involves calculation based on yield point of mud in use.

$$SF = YP/6(D_h - D_p)$$

D_h = Hole diameter

D_p = Drillpipe

YP = Yield point.

SF = Safety factor, ppg

$$BHP = PF + SF$$

Kill Mud Weight

This is the mud weight required to prevent further influx while circulating out the kick and during later operations.

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$$\rho_{kill} = \text{SIDPP} / (0.052 * D) + \rho_m + SF$$

Initial Circulation Pressure

During circulation of influx, the choke operator controls bhp by varying the choke opening or casing pressure. The choke operator monitors the bhp with the drillpipe pressure gauge. The operator obtains the Initial Circulation Pressure (ICP) by opening the choke and waiting for the driller to bring the pump to kill rate. Operator then adjusts the choke to obtain the desired pressure on the drillpipe pressure gauge.

$$\text{ICP} = \text{SIDPP} + P_c + SF$$

P_c is pressure loss for initial mud weight at the kill rate,.

When the driller pumps higher mud weight into the hole, the hydrostatic pressure of the mud increases in the drillpipe. This reduces differential pressure. As the new mud travels down the pipe, the circulation pressure declines gradually until new mud reaches the bit. At kill mud weight, the differential pressure should decline to zero. Thus the circulation pressure should equal the kill rate pressure loss corrected for kill mud. This is then the final circulation pressure.

Final Circulation Pressure (FCP)

$$\text{FCP} = P_c \times \rho_K / \rho_m$$

Weighting Material Volume Estimation

The Company man determines the well control procedure and the number of circulation necessary to bring the mud up to the kill weight. If the kick can be controlled in one circulation cycle, then the control equipment spends less time under pressure. If the amount of weighting material needed is greater than rig capacity to mix mud, well control will require weighting up stages. These considerations require calculation of weighting material volume requirement.

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Total Circulating system volume, $T_v = \text{Annular vol.} + \text{Drillstring I \> D volume} + \text{Active pits volume.}$

$$\text{Barite requirement, Weight} = T_w = 0.736 * T_v * (\rho_{kill} - \rho_m) / (\rho_w - \rho_{kill})$$

In metric units

$$T_w = 4.2 * T_v * (\rho_{kill} - \rho_m) / (\rho_w - \rho_{kill})$$

ρ_w = Density of weighting material

Next is the calculation of volume requirement.

$$V_w = T_w / \rho_w$$

Finally, rate of barite or weighting material addition is calculated.

$$\text{Rate} = 1472 * q_r * (\rho_{\text{kill}} - \rho_m) / 42 * (\rho_w - \rho_{\text{kill}})$$

Interpretation of Influx

A first step towards the control of influx after shut-in and data gathering is the identification of the type and size of the influx which will eventually guide the choice of the appropriate control measures.

The influx may be oil, water or gas or a combination of the three. Water and oil do not show significant compressibility effects under typical wellbore conditions but gas can be extremely difficult to handle because of its high compressibility. Typical characteristics of a gas kick include the following:

1. A gas kick will expand as it rises in the hole because of the reduction in hydrostatic pressure especially close to the surface. This will greatly affect the pit volume gain to be handled.

Gasses have much lower densities than most drilling fluids. They therefore tend to rise up the annulus in the mud. If well remains shut in during migration, a gas influx will move up as a slug and will retain its volume and pressure.

Circulating out the influx through an adjustable choke normally restricts gas expansion. This leads to surface pressure increase as the gas moves up the annulus and it is highest when the gas influx reaches the surface. As the gas

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bubble rises, the hydrostatic pressure of mud column pressing down the gas becomes less, thus increasing the surface casing pressure. Thus, the greater the migration the higher will be the surface pressure. It is imperative therefore that once there is a kick especially if it is gas, circulation should start as soon as possible after the shut-in.

If a weighted mud reaches the hole bottom before the top of a gas kick migrate up the annulus and reaches the casing shoe, the pressure at the shoe will be less by a height of the weighted mud in the annulus and the differences between the mud weights.

Effective quality control is essential to a successful killing process. The simpler the control procedure the lower the chance of error. If time permits, it may usually be simpler to circulate the kill mud and kill the influx at source.

A second quality assurance is that the chosen circulation method should not exceed the rig capacity to weight up mud.

Computation of Kick Height, L_k

The general assumptions in the calculations are:

- The influx behaves as a slug and does not mix with the wellbore fluid.

- The Pit gain is the kick volume
- The hole diameter is to gaug
- Hole is vertical.

Assume the annular capacities in the regions of the drillpipe and casing are v_{adp} (barrels per foot of gallons per foot of hole length) and v_{adc} respectively

Given that the pit gain or kick volume = V_{Kick}

The annular geometry is normally not uniform due to bottom hole assembly(BHA) diameter being greater than drillpipe diameter. The first step in the calculation of kick height therefore is to find out if the volume of kick is greater than the annular volume in the region of the BHA.

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The two conditions for influx height calculation are:

1 If $V_{Kick} < V_{adc}$

where $V_{adc} = V_{adc} \times L_{dc}$.

L_{dc} = Length of drillcollar

Then

$L_{Kick} = V_{Kick} / v_{adc}$

2 If $V_{Kick} > V_{adc}$,

$L_{Kick} = L_{dc} + [V_{Kick} - L_{dc} \times v_{adc}] / v_{adp}$

3. Calculation of Kick Density, ρ_{Kick}

$\rho_{Kick} = \rho_m - [(SICP - SIDPP) / 0.052 L_{Kick}]$

If $\rho_{Kick} > 8.3$, then influx is water

If $7.6 \text{ppg} < \rho_{Kick} < 8.33 \text{ppg}$, then influx is oil

If $< 2.3 \text{ppg}$, then influx is gas

If $2.3 < \rho_{Kick} < 7.6 \text{ppg}$ then influx is condensate, gas-cut oil or gas-cut water.

Different service companies have 'Kick' software for computing the kick densities and can create so-called "bubble plot" that indicate annular pressure changes as influx is circulated up the hole.

4. Calculation of Annular Pressure Profile.

Strictly speaking, Well control procedures do not require the calculation of annular pressure profile for a kick. But this may be useful especially for gas kick.

The P_{max} is calculated as :

$P_{max} = 0.052 \rho_{Kick} \times (1029.2 \times V_{Kick} \times D) / A_s$

A_s = Annular suffice area at wellhead, in²

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7.8 Kill Methods

There are three major kill methods usually adopted by industry for the control of influx into the wellbore. These are:

1. The Driller's Method
2. Wait and Weight Method (Also called the Engineers Method)
3. Concurrent Method

There are also two quasi-control methods, which do not require circulation through the choke. These are:

1. Top Kill Method
2. Volumetric Method

Driller's Method

In this method, the well is originally shut-in after the kick has occurred with the appropriate shut-in procedures followed. Thereafter, the influx is circulated out using the original mud in the hole. Once the kick is out of the hole, the hole is subsequently filled with the heavier kill mud.

This control method therefore requires two circulation cycles. Circulation 1 removes the influx and circulation 2 circulates the kill mud. Operators tend to use the Driller's method when they are certain the kick contains little or no gas.

Merits of Driller's Method

1. Minimum circulation
2. Minimum idle time. No waiting
3. Minimum well information required

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Disadvantages of Driller's Method

1. High annular pressures if kick contains gas
2. Maximum time with well under pressure
3. Maximum time spent controlling choke!

Wait and Weight Method

In this method, otherwise known as the Engineers Method, the influx is circulated out by the kill mud in one cycle. This requires the manipulation of the choke to maintain constant bottom hole pressure as the new mud moves from surface to bottom. The kill mud may require several circulation cycles

with mud density raised in stages.

Advantages of the Engineers Method

1. Lowest wellbore and surface pressure
2. Minimum time spent "on-choke"

Disadvantages

1. Idle time weighting up with gas migration possible
2. Possibility of several circulation cycles if large mud weight increase is necessary

Concurrent Method

This method mixes the features of the Driller's and Engineer's methods. It can be applied in cases where the surface pressures need to be kept as low as possible and shut-in time to a minimum.

Circulation starts immediately after observing the shut-in pressures using the current mud weight. Then the mud weight is increased in stages. As each stage of increased mud weight reaches the bit, the driller starts pumping the next stage. When the new mud reaches the kill mud weight, circulation continues until new mud reaches the surface.

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This method requires manipulation of the of the choke to maintain constant BHP just as in the engineers method. However the mud weight increases determine the pipe pressure schedule and the pump stroke from surface to bit. The pipe pressure schedule covers a longer interval than the other two methods.

Advantages of the Concurrent Method

1. Minimum idle time. Control starts immediately
2. Continuous, smooth adjustment of mud density.

Main Disadvantages

1. Slightly more complex calculations than the other two methods
2. Maximum time spent circulating under pressure

7.9 Well Control Without Circulation

If influx is gas, and the drillstring is plugged or completely out of the hole, then there is no means of circulation. Without circulation, gas will rise in the annulus leading to casing and bottom hole pressure increase. Excessive increase may cause damage at surface or at weak point of casing shoe.

In such cases the driller may use the Volumetric Method to bleed off mud volume, allowing gas to expand as it migrates upward.

In the real sense of the word, Volumetric Method is not a kill method. Rather

it involves the bleefoff of the influx or kick pressure. The well control team must still eventually circulate the kill mud to kill the well dead. This method is most effective when the kick occurs due to swabbing. In most cases under this condition, the mud weight is probably sufficient to prevent further influx before the circulation of weighted mud. However, if the cause of kick is insufficient mud weight, then there is likely to be further influx when the kick bleeds off at the surface.

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7.9.1 Control Procedure for Volumetric Method

The procedure for the Volumetric Method of control is as follows:

1. Shut in the well according to defined procedure in well plan.
2. Monitor and record SICP. By definition, use of the SIDPP does not exist since there can be no circulation through the drillpipe.
3. Calculate the bottom hole pressure to maintain while killing the well
4. If the SICP gradually rises after 10-20 minutes of shut-in, the kick contains gas and the well control team must apply the volumetric method.
5. Allow the SICP to rise by a convenient increment. About 5bar(75psi) or maximum 100psi is recommended. Make sure the rise does not exceed the MAASP.
6. Slowly open the manual choke and bleed off mud volume until the casing pressure drops by the predefined increment. This allows the gas influx to expand slightly in the annulus.
7. Close the manual choke and note the new SICP. After bleed-off, the new SICP will be higher than the original SICP.
8. Again allow the SICP to rise to predefined increment level.
9. Repeat steps 6-8 as needed until the kick reaches the surface. The bleed-off mud volume required will increase as the gas expands, and the bleed-off volume may become very large when gas nears surface.
10. As the gas reaches surface and bleeds off through the choke, slowly pump mud into the hole via the kill line, to replace the lost fluid volume.
11. The control team can then "strip or snub" the drill string into the hole to complete the kill.

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7.9.2 Stripping and Snubbing

Normally, kick control using balanced bottom hole pressure requires that the

drillstring be at or near bottom. If the drillstring is not on bottom at the time of shut-in, the driller must run in hole under pressure. Stripping and snubbing are the two optional processes normally adopted for running-in of drillpipe under pressure.

The primary objective of these processes is to maintain a constant bottom hole pressure while Running-in the drillstring. This requires careful manipulation of a manually operated adjustable choke to bleed off mud volume displaced by the pipe.

1. Stripping-in uses the travelling block to lower the pipe into the hole, through a partially Closed blowout preventer(BOP). The method is most practical when the pipe weight is sufficient to overcome the upward force generated by well pressure on the pipe cross section.
2. Snubbing uses hydraulic or mechanical equipment to apply downward force to the pipe. This way the snubbing process overcomes the excess upward force generated by well pressure on the pipe cross-section.

For either process, the BOP must be closed against the pipe wall to maintain at least a partial seal. The driller may choose to run the pipe in through

1. the annular presenter only or
2. Combination of annular or pipe ram preventers.

7.9.3 The Top Kill Method

If the drillstring is off bottom, or out of the hole, and running in appears to be very risky, then the top kill control technique can be adopted. Top kill methods include:

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1. Bullheading - This is a process involving the injection of mud into the closed annulus to force invading fluids back into formation. It is a popular method adopted for cleaning up the wellbore after a drill stem test especially in cased hole. It is rare in kick control except in unusual circumstances where everything else is not possible. The main disadvantage is that it risks failure of the casing shoe and in a shallow hole may result in creation of fissures to the surface behind the casing. Bullheading works mainly If the producing formation is isolated from the rest of the wellbore..
2. Setting barite or cement plugs A barite plug is a slurry of barite and fresh water or alternatively barite and diesel, spotted above the producing formation to bridge off the hole. Merits of this method are:
 - The barite is fine enough to pump through the bit nozzles

- Setting a barite plug offers a reasonable chance of recovering the drillstring.
- The plug can be drilled out if necessary.

A major disadvantage however is that the barite may plug the drillstring if not fully displaced into the annulus.

Examples of Barite plug requirement are-

1. Possible slurry densities: 1.76 - 2.58SG
2. Barite in fresh water: 485 - 15301bslbbl(This may require thinning with phosphates or lignosulfonates.

Phosphate mixing requirements: 2.81bislbbl

Lignosulfonate mixing requirement: 0.41bislbbl. i

3. Barite in diesel : 570 - 1660 lbs/bbl

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The recommended volume of barite slug is about 10barrels.

Mixing/displacement must be prompt to prevent barite sag and possible plugging of drillstring. Use of cement pump may improve matters.

The use of a cement plug however should be regarded as a last option as this is tantamount to well abandonment including loss of much of drillstring.

7.10. Hole Enlargement

7.10.1 Causes

Hole enlargement otherwise known as washout can be the result of drilling through salt section or result of sloughing/heaving shales

7.10.2 Prevention/Control

1. For salt section - use saturated salt-water mud
1. For Heaving/sloughing Shales : Use KCL/Polymer mud
2. Maintenance of good equivalent circulation density.

7.11 Formation Damage

Formation damage can be defined as a measure of the reduction in the production capacity or injectivity of a given reservoir formation . This can be due to the combined effect of :

- reduction in the formation absolute permeability
- alteration in the relative permeability of the formation to a specific fluid
- alteration in the viscosity of the mobile fluid

It is mainly a consequence of a number of operations conducted in a well from the initial drilling, through the production and workover phases to final abandonment.

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From the radial flow theory, the flow through a composite reservoir with damage can be viewed as a combined flow through two reservoir sections in series. Thus, the total pressure drop in the reservoir can be defined as:

$$\Delta P_{TOTAL} = \Delta P_{SKIN} + \Delta P_{CLEAN RESERVOIR}$$

7.1.1 Causes of Formation Damage

Formation damage can be considered to be caused by a physical or chemical interaction between the following:

1. Invading fluid liquid phase and the reservoir fluids
2. Invading liquid phase and the reservoir rock constituents
3. The solid constituents of the invading fluid and the reservoir rock.

The damage mechanisms do vary but these can be listed as:

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1. Formation damage resulting in reduction of absolute permeability due to

◆ Particle plugging due to solids in drilling and completion fluids invading the reservoir rock pore spaces.

◆ Clay problems: In-situ clays are found in majority of reservoir rocks especially sandstone. Majority are very sensitive to fluids invasion and may swell thus blocking the pores

◆ Fines migration: A typical sand stone reservoir rock consists of insitu quartzitic or siliceous fines generally held within the insitu connate film surrounding the sandstone grains. These fines can become mobilised either due to high production rate or due to increase in wetting phase saturation. Once mobilised, fines are transported and can result in particle plugging.

2. Inorganic scale deposit resulting from reaction of incompatible invading and reservoir fluids. Changes in pressure and temperature also contribute to the precipitation of solids forming scales

3. Plugging by organic deposits such as wax and asphaltenes.

4. Damage resulting from reduction in relative permeability - Invasion by wellbore fluids cause hydrocarbon saturation to decrease around the wellbore. This lowers the relative permeability to oil and , especially during early stages of production, a significant pressure drop is created around the wellbore.

5. Damage due to emulsion blocking - Result of formation of emulsion between oil and water. Bacterial action may also form emulsion blockage.

6. Gas Breakthrough and Water Coning - Increase in gas and water saturation reduces the relative permeability to oil.

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7. Stress-induced permeability change - Changes in stress distribution will result in changes in compaction status and thus pore pressure change. This will affect the net permeability of the reservoir.

8. Sand production from unconsolidated reservoir rocks. These are known to be production rate sensitive. Thus high production rate and water encroachment among other factors can lead to the migration of in-situ sand grains and massive sand intrusion.

7.11.2 Prevention and Control Measures

One important aspect of formation damage is the location of its occurrence. it can occur close to the wellbore and this will have the greatest impact on well productivity since it will affect fluid flow in the region of highest velocity convergent flow for radial production and divergent flow for radial injection in injection wells. Occurrence in the proximity of the wellbore makes it accessible for removal or bypassing. it is usually the most severe and therefore needs to be handled with care. On the other hand damage which occurs deep inside the reservoir is unlikely to be severe since the effects of pressure drop, velocity and concentration are not as high. Further since the in-situ velocity is lower, the impact of damage is reduced and can be easily bypassed by the

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large number of alternative flow channels. therefore the removal of very deep damage is not generally technically feasible.

Prevention

The best control technique is prevention. Preventive measures for formation damage are limited to the ways of ameliorating the three potential causes. In all cases, the following is required:

1. An appreciation of the causes of formation damage and factors governing the severity and location
2. An analysis of well operation to assess the potential risk for damage
3. The modification of operation to avoid or minimise damage
4. The evaluation and implementation of techniques to remove or bypass

the effects of damage.

The potential preventive measures are as follows:

1. Interaction of invading fluid liquid phase and reservoir fluid.

Combating reaction between the fluids that can lead to precipitation, emulsion or water blockage, etc requires the selection of appropriate drilling and completion fluids that are compatible with the reservoir fluids at the downhole temperature and pressure. Likewise, the reduction of fluid loss through the use of appropriate control agents is essential. Total prevention of lost circulation is also very important.

2. Interaction of invading fluid liquid phase and the formation

This problem leads mainly to potential clay swelling, wettability change and potential water blocking. To minimise the problem, use of appropriate clay inhibitor/stabiliser completion fluids would be desirable to prevent clay swelling.

Most formations are usually water-wet but this may change due to completion operations. For such cases, the use of appropriate

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surfactant to alter wettability will be desirable. For example, for improved oil relative permeability, the formation needs to be water wet. Thus the water wettability of the formation can be restored through appropriate surfactant treatment.

3. Interaction of invading fluid solids and formation.

This generally leads to particle plugging or result of fines migration. Prevention generally involves effective solids control for drilling fluids and the use of solids free brines as completion fluids. Effective well preparation before completion operation is carried out is very essential. Thus, the drilling fluid can be displaced by solids free completion fluids.

Specific cases of Preventive measures during well completion operations are discussed below.

1. Damage during drilling and cementing

Damage during drilling can be due to

- Fluid loss or lost circulation
- mud solids

Therefore it is essential to select compatible drilling fluids. Avoidance of unnecessary downtime with no circulation is essential. Minimisation of pressure surge and effective solids control are a key to minimising damage.

During cementing, casing should be run in the hole with minimum delay but smoothly to avoid excessive pressure surge. Cement formulation must be with adequate fluid loss control. Compatibility and conditioning of mud to avoid unnecessary contamination is also essential.

2. Formation damage during well completion operation

Completion operation is perhaps the most susceptible to damage. The completion involves the preparation of the open hole or cased wellbore, installation of completion string and placement of completion and packer fluids.

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The principal sources of damage are

1. Fluid leakoff and particle plugging
2. Fluid interaction with reservoir rock and fluids.

In these cases therefore it is essential to

i Prepare the well for completion by removing the mud and carry out effective wellbore cleanup free of debris. Casing, tanks and lineworks must be cleaned

ii Use solids free completion fluid to avoid plugging. However, loss control materials such as calcium carbonate, oil-soluble resin or sized salts that are easily removable during back production are added to minimise fluid loss. They are also removable with acid wash.

iii Exposure time especially for open hole completion should be minimised.

iv Use of minimal overbalance

v Perforation should be done with minimal overbalance if overbalance perforating is used. Preferably underbalanced perforating is best to minimise damage. In any case, perforations should be cleaned of debris either by backsurgings or perforation washing.

Increasing perforation shot density and penetration can also minimise damage.

3. Damage during production operations

The main causes of damage during production are

- (1) Formation of inorganic scales
- (ii) Deposition of wax/asphaltenes
- (iii) Migration of fines towards wellbore
- (iv) Sand production from unconsolidated reservoirs

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Scales.

Scales are formed especially by the mixing of incompatible waters and changes in the wellbore conditions such as salt concentration, declining temperature and pressure. Use of scale inhibitors where the problem is suspected is therefore essential.

Organic Deposits

Deposition of wax/asphaltenes depends on production conditions and produced fluid composition. It may not be possible to modify drawdown to avoid asphaltene deposition but the effect can be reduced by increasing, for example, the perforation density, phasing and penetration.

Fines Migration

This is prompted by increased water cut and excessive drawdown, therefore should involve the minimisation of these causes.

Sand Production/Control

Completion of unconsolidated reservoirs should involve the use of pack sands(Gravelpacks) and other screen systems in the attempt to:

- ◆ maximise productivity
- ◆ minimise impairment through the prevention of plugging

If not controlled the migration/production of solids which may erode downhole and surface facilities such as valves, pumps, tubing and flow line components besides the loss of reserves.

Formation Damage during injection operation

These include water or gas injection. For water injection(water flooding), the principal causes of damage are:

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- (a) Solids plugging; (b) bacteria plugging (c) inorganic scale deposition;
(d) Clay swelling and fines migration

it is therefore essential to:

- filter sea water to remove solids over 2-10microns
- remove rust and use oxygen scavengers to minimise corrosion
- use bactericide
- use scale inhibitors

Control of Formation damage

The best control measure is the prevention as discussed above. However, when these fail, then it is essential to stimulate the reservoir by using the

appropriate Stimulation techniques.

Before stimulation, it is essential to analyse the damage through well testing in order- to quantify the magnitude. Details of well test and damage quantification otherwise known as SKIN or skin effect will be dealt with later

A positive skin represented by the symbol 's' implies damage while a negative skin represents enhancement.

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7 DRILLING PROBLEM

7.1 Introduction

The main objectives of every drilling operation are :

1. High drilling efficiency through the achievement of
 - Optimum penetration rate
 - Minimum downtime
2. Minimum operating cost
3. Effective safety
4. Environmental Impact

There are however a number of problems encountered in the hole which mitigate against the achievement of the above key objectives. The problems which constitute a hazard to drilling are:

1. Lost Circulation
2. Differential Sticking
3. Washouts
4. Kick / Blowout
5. Formation Damage

The design of drilling fluids and drilling techniques adopted play a major role in the propagation or prevention of these problems. This note therefore covers

- The functions/properties of drilling fluids
- Causes, prevention and control of the drilling problems.

7.2.1 Lost Circulation

Lost Circulation can be described as the loss of the substantial quantity of whole mud to “thief” formations otherwise known as lost circulation zones.

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7.2.1 Causes of Lost Circulation

The problem can be caused by natural or induced factors.

Natural Causes

They are basically caused by the presence of natural fractures, high permeability formations, cracks, vugs and fissures.

Induced Causes

- a. High Mud Density: When the mud density is so high as to make the hydrostatic or bottom hole pressure greater than the formation fracture gradient resulting in a breakdown of the formation. Fractures can be generated which will result in mud loss
- b. Increase in Annular Pressure: Increase in annular pressure beyond the maximum allowable annular surface pressure (MAASP) can result in the generation of fracture - a potential thief zone.
- c. Pressure Surge: During trips, running-in the drillstring at a very high velocity can result in localised pressure surges that can lead to the breakdown of weak formations.
- d. Penetration of depleted or subnormal formations

7.2.2 Implication of Lost circulation.

1. Formation Damage: Loss of whole mud results in loss of filtrate and mud solids which will result in plugging and permeability reduction.
2. Kick: Loss of whole mud will lead to loss of mud column height. This can make the hydrostatic pressure of mud column to be less than the formation pressure. The net result is an influx of formation fluid into the wellbore. This is known as kick.
3. Massive increase in mud cost.

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7.2.3. Prevention of Lost Circulation

1. Use of Loss Control Materials (LCM) as a regular mud additive. Examples of LCMs are nut plug, kwikseal, mica, corn fibres, calcium carbonate, sized salt, Oil soluble resin (OSR)
2. Prevention of pressure surge and maintenance of annular pressure less than MAASP.
3. Good estimate of fracture gradient.

7.2.4 Control of Lost Circulation

The first step involved in the control of lost circulation is the identification of the thief zone. The techniques used include

1. Spinner Survey

2. Temperature survey
3. Radioactive tracer survey
4. Pressure transducer's survey

These techniques are now normally part of regular MWD techniques.

Control of lost circulation is achieved through the use of loss control materials.

1. For High Permeability formations: Calcium Carbonate and sized salts may be used
 2. For cracks, fissures, vugs, etc, mica, nut plugs, fibres, etc may be used.

2. For severe cases, spotting of 'gunk' - mixture of diesel/barite or cement may help.

3. Total isolation with intermediate casing cemented in place may help.

4. "Drilling Blind" - a process in which the total return mud including cuttings is allowed to flow into the thief zone and plug whilst being practised in

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some special problem cases, is not a good technique. Where the lost circulation is severe, it is useful to displace "gunk" rather than drilling blind

7.3. Pipe Sticking

7.3.1 Causes of Pipe Sticking

Drill string get stuck due to a variety of reasons

1. Junk inside the hole
2. Keyseating
3. Sloughing/Heaving Shales
4. Bit/Drillcollar balling
5. Drill cuttings above the bit or drill collar due to improper solids control
6. Pressure Differential sticking

(a) Differential sticking

Differential sticking occurs when the drillstring becomes motionless against a permeable formation due to excessive hole-to-formation differential pressure (excessive overbalance) acting on an isolated area. The differential pressure exerts a considerable force against the drillpipe and literally glues it to the borehole wall.

The drillstring is also amenable to seeking when the filter cake is too thick and if the string is left static for extended period of time in the hole.

7.3.2. Prevention of Pipe Sticking

Pipe sticking can be prevented by adopting the following strategies:

1. Minimising the margin of overbalance and maintaining efficient control of

mud properties

2. Minimising the drillpipe-to-borehole wall contact area by:

- Minimising filter cake thickness through efficient solids control and using low fluid loss mud
- Using stabilisers on drillcollars
- Using grooved or non-circular/external upset drillcollars

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- Minimising non-rotating time

3. Minimising friction factor by using possibly emulsion mud.

7.3.3 Control of Stuck pipe

Once a pipe is stuck, it has become a fish. The only way to control stuck pipe is by initiating fishing operation.

The first step in fishing operation is to evaluate the depth at which the string is stuck. This is known as the free point. Two methods are in common use for evaluating the free point:

1. Pipe stretch method using surface observations (Pg62)
2. Free point indicators - Logging while drilling(LWD) techniques which employ the use of specialised strain gauge and subsurface probes.

Pipe Stretch Method

This simple method requiring measurement from the rig floor involve the following procedure:

1. Pull drillstring using normal hook load designated here as F1 Mark a reference point X1 on the pipe.

Pull additional load between 20klbs and 40klbs, F2 less than yield point of drillpipe. This will cause the free portion of pipe to stretch to a new position X2.

3. Calculate the free point using the following equation:

$$Lst = 735294(X2 - X1)WdP/F$$

Lst= Depth of stuck pipe, ft

WdP = Weight of drillpipe, lb/ft

F = Additional pull, lbs

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Once the free point is evaluated, the pipe can be freed using the following methods:

1. Spotting Organic Fluids

A mixture of diesel oil and surfactant is usually spotted across the stuck point to reduce the mud cake thickness, thus reducing the friction factor. The pipe is worked continuously during the spotting. Fluid has to be spotted against the correct zone and given sufficient time to dissolve and destroy the cake. The procedure involves pumping the oil/surfactant into the drillpipe and the gradually pumping small volumes into the annulus until the stuck pipe is adequately soaked.

2. Hydrostatic pressure reduction

This method involves creating a pressure differential inside the drillpipe and outside the drillpipe in the annulus. The pressure differential is achieved by either pumping a new lower density mud, or by pumping a small volume of lower specific gravity fluid such as diesel oil into the drillpipe and then bleeding it from the drillpipe. This will drop the mud level in the annulus thus reducing the pressure of the mud at the stuck point. The maximum volume of oil or water to be pumped into the drillstring and the maximum drillpipe gauge pressure are given by the following equations:

$$LstVdp = Pp - Van - \rho_0$$

$$V_0 = \frac{Pp - P_1}{\rho_1 - \rho_0}$$

$$\rho_1 - \rho_0$$

$$\rho_m - \rho_0 = 0.052 Lst Vdp - \rho_m$$

$$m$$

$$\frac{Pp - P_1}{\rho_1 - \rho_0}$$

$$0.052V_0 (\rho_m - \rho_0)$$

$$(P)$$

$$dp_{max} =$$

$$Vdp$$

The fluid level drop in the annulus when the mud and formation pressure are equal is:

$$\Delta FL = Lst - [Pp/0.052\rho_m]$$

The mud weight equivalent at depth during the bleedoff is

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$$(P) - P$$

$$pe = \rho_m -$$

$$dp_{max} dp$$

$$\frac{Pp - P_1}{\rho_1 - \rho_0}$$

$$0.052 Lst \frac{Pp - P_1}{\rho_1 - \rho_0} + an$$

$$\rho_1 - \rho_0$$

$$0$$

□□ □

□□ Vdp □□ m □□

Vdp= Drillpipe capacity, bbis/ft;

Pp = pore pressure, psi.

(Pdp)max = max. drillpipe pressure,

Van.= annular capacity, bbis/ft

Pdp = Drillpipe pressure at surface

3. Backoff Operations

When attempts to free the stuck pipe fails, a backoff operation is the last resort. This technique involves parting the drillstring at or above the stuck point and recovering the drillstring. The remaining string can be freed using a combination of:

1. Hydraulic jars
2. Overshot
3. Hook, etc.

7.4 Kicks and Blowout

7.4. 1. Introduction

A kick is the term used to describe the inflow of formation fluid into the wellbore during drilling operation. This mainly arises due to the formation pressure being greater than the hydrostatic pressure imposed by the mud column. This can happen in a normal or abnormal formation especially geopressured formation.

A kick may be primarily composed of salt water, oil, natural gas or a combination of the fluids. The influx of formation fluid arises for a variety of reasons. These include:

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1. Insufficient mud weight
2. Low level of fluid column in the annulus
3. Excessive Swabbing
4. Excessive drilling rate in gas bearing formation
5. Diffusion of gas through mud cake.

Kicks generally occur during trips with influx occurring from a combination of swabbing and failure to keep the hole full. Swabbing is the result of the piston effect of the drillpipe due to excessive trip speed resulting in instantaneous vacuum being created below the bit. Kicks occurring during drilling operations most often occur due to insufficient mud weight.

1. Insufficient Mud Weight

Insufficient mud weight may occur due to:

1. Penetration of a geopressured zone
2. Accidental dilution of the mud by fluid addition at surface
3. Dilution of mud by influx from an aquifer exposed to open hole
4. Gradual mud density reduction due to gas cut and failure to degas the mud at surface
5. Improper mud mixing and poor quality control in measurement

However, for strategic operational reasons, a borehole may be drilled underbalanced with controlled kick taking place.

2. Low level of drilling fluid column

Two conditions may lower the fluid column in the annulus. These are:

- Failure to fill the hole during a trip
- Lost circulation during drilling

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In pulling out the drillstring, the hole driller must refill the hole with mud of equivalent volume. Monitoring of the volume of mud filling the hole is done with the use of the Trip tank or Possum Belly Tank.

The trip tank is a small mud pit (Approx. 50 bbls) separate from the active pits. This trip tank is fully equipped with a volume gauge that can be monitored from rig floor or mud loggers console. In monitoring the trip, the driller must calculate the theoretical mud volume displaced by one or five stands of drillpipe. During the trip, the driller then measures the actual mud volume pumped into the hole for each stand of pipe pulled.

Lost Circulation is the loss of substantial quantity or whole mud into the formation. It may occur as a result of:

- ◆ mud loss to cavernous or vugular formation
- ◆ penetration of depleted or subnormal zones
- ◆ mud loss to fractures opened by excessive annulus pressure. This may be due to annular blockage, surge pressures, excessive hydraulics, pressures imposed to break mud gel strength.

Lost circulation will result in insufficient mud column and thus reduction in bottom hole pressure.

Kick due to lost circulation can be a major well control problem. If the well cannot be shut in, an underground blow out can occur.

3 Excessive Swabbing

Swabbing is the reduction of bottom hole pressure caused by upward pipe movement. As the pipe is being pulled out during trips, mud flows down through the annulus to fill the space in the hole. The downward mud flow creates a frictional pressure loss in the direction of the surface. The result is reduced bottom hole pressure. High trip speed can lead to instantaneous vacuum below the bit. The swab pressure depends on :

- Pipe velocity

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- clearance between pipe and hole - The smaller the annular cross section, the greater the swabbing action.
- mud rheology - The higher the viscosity, the greater the swabbing action
- total measured depth - The longer the section, the greater the swabbing action.

Excessive drilling rate in gas-bearing formations

The densities of formation fluids(gas, oil or water) are much lower than that of conventional drilling fluids. Any influx of formation fluid therefore reduces the mud density in the annulus. This influx otherwise known as mud cut is measured using sensors that measure mud density and conductivity at injection and at flow line

Mud cut may occur from:

- kick due to inadequate mud weight or swabbing
- diffusion of fluid into the hole due to negative differential pressure
- air entrained in mud from the so called Kelly air during trips
- expansion of drilled gas as it reaches the surface.

Majority of kick problems arise from the reduction of mud weight due to the expansion of drilled gas especially in large diameter holes such as surface holes drilled at high rates. For such holes drilled without a blowout preventer in place; a small reduction in mud weight can lead to a violent kick and with the loss of primary control, lead to blowout.

Kick Detection

A kick can occur at any time during the drilling of a well. To prevent any major catastrophe therefore, early detection of a kick is essential to primary well control.

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As a primary precaution, it is important for all personnel including the mud logger to always be in a state of readiness. Certain pre-kick information must

be collected and available for use in case a kick occurs. These include volumes and pressures data. A kick control Work Sheet must be maintained for this purpose.

The pre-kick data requirements include

1. Maximum Working Casing Pressure

The casing and blowout preventer(BOP) are designed for the different expected pressures during each drilling phase . This is determined during well planning and is generally the maximum pressure rating of the BOP and its outlets or 80% of the burst pressure of the last casing. The 80% consideration is a safety factor for the casing string.

2. Maximum Allowable Annular Surface Pressure(MAASP)

This is the annular pressure at the surface that corresponds to the pressure at the weakest point of the hole. Fracture gradient at the last casing shoe defines the weakest point. However for some boreholes, the MAASP can be the casing burst pressure or BOP rupture pressure if they are less than the fracture gradient. The MAASP represents the maximum annulus pressure permitted while controlling a kick.

The MAASP is estimated from the following equation:

$$\text{MAASP} = 0.052 * (\text{FRAC} - \rho) * \text{Dw}$$

where

FRAC = Fracture pressure at shoe given as equivalent mud weight in ppg.

ρ = Current mud density, ppg

Dw = True Vertical Depth (TVD) of weakest point, ft

In Metric Units

$$\text{MAASP} = (\text{FRAC} - \rho) * \text{Dw} / 10$$

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[FRAC] = Kg/cm²

[ρ] = Kg/cm²

[Dw] = metres

3. Rig Capacity for Weighting Mud

Although this data does not appear in the worksheet, it is important as it defines the number of circulation cycles necessary to regain primary control with given change of mud. The rig capacity depends on total reserves of weighting materials (Barites, etc) and the maximum rate of addition to system. The maximum rate of addition is measured in lb/min or Kg/min. Backup supply MUST always be available on rig.

4. System Pressure Losses at Slow Pump Rate

Circulation at pump rate for drilling may produce surface pressures greater than the MAASP. Therefore a slow pump rate for kick control must be determined at the beginning of the well. This is usually 1/3 to 1/2 of normal rate. This slow rate must be taken

- at the start of each tour
- if the mud weight changes
- if bit arrangement changes
- after bit changes

If a kick occurs, the influx will be circulated out at the maximum pump rate that does not exceed the MAASP. This is called the kill rate.

The pressure losses must always include the choke line pressure losses especially offshore where this could be substantial.

5. Pump, String, and Hole Configuration

This includes the data used for the determination of the lag time and contains

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- Pump capacity
- Hole size
- Drill string capacity (internal volume)
- Annular capacity (bit to casing shoe and bit to BOP)
- Casing data (size; weight; burst pressure; shoe depth)
- The worksheet includes the space to fill in lag time in strokes or minutes for the proposed kill rate.

This estimation is usually computerised nowadays and continuously logged.

7.4.3 Kick Detection Techniques

There are a number of indicators that provide early warning of kick occurrence. These indicators include:

The positive kick indicators are :

1. Mud Pit Level/Flow Increase
2. Incorrect hole fillup during trip
1. Decrease in Standpipe pressure/increase in pump rate
2. Increase/Decrease in drillstring weight

The potential indicators are

1. Increase in Penetration Rate
2. Lost Circulation
3. Changes in gas levels, mud density and conductivity

1. Mud Pit Level/Flow Increase

These parameters are major positive indicators. As the influx of formation fluid displaces the mud in the annulus, the pit level would increase in the active

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mud tanks. There also be evidence of increase in return flow rate if logged.

The effectiveness of pit gain/flow increase depend on

- Initial size and type of kick
- Mud fluid base and rheology
- Minimum sensitivity of the measuring systems

Gas kicks are more difficult to detect than oil/water due to its compressibility. It tends to go into solution in the drilling fluid. However, a small gas kick can expand multiple times its original volume by the time it reaches the surface.

Return flow is more sensitive to gas kick than pit level monitoring overall. The return flow sensors may therefore provide the first warning of a gas kick.

Sensor resolution determines the minimum size of kick detected. Therefore, the mud logger should normally set alarms on the pit volume parameters close to the actual pit value. Report should be sent to the driller no matter how small the pit gain is. Daily checks on the alarms and routine maintenance is essential to effective monitoring.

2. Incorrect Fillup during trips

When pulling out of the hole, if the volume of mud pumped to keep the hole full is less than that normally required, then there is an evidence of influx. The mud volume required should be equal to or slightly greater than the volume of drillpipe (normally five stands) already pulled. Mud loggers MUST follow trips even though an automatic trip monitor is usually used.

Conversely during a trip into the hole, the downward movement of the drillpipe expels fluid from the annulus to the trip tank or active mud system. This return flow should cease a few seconds after pipe movement stops. If flow continues, then there is a kick.

3. Decrease in Standpipe pressure/Increase in Pump rate

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Influx of fluid into the annulus creates an imbalance resulting in a decrease in hydrostatic pressure in the annulus. In such an unbalanced system, gravity helps move drilling fluid down the hole, requiring less energy from the pump. This will result in a decrease in the standpipe pressure.

4. Increase/Decrease in Drillstring Weight

Any influx into the wellbore from the formation reduces the buoyancy of the annular drilling fluid. A sensitive weight indicator will register this change in weight as an increase in drill string weight. For very large kicks, fluid may enter the annulus with enough force to cause a decrease in indicated string weight.

5. Increase in Penetration rate (Drilling Break)

A marked increase in rate of penetration (ROP) may indicate either changes in the type of formation being drilled or a reduction in the differential between the mud pressure and pore pressure.

Generally, the following parameters affect the ROP:

- rock type
- formation bulk density/porosity
- differential pressure
- bit type/wear
- hydraulics
- weight on bit
- rotary speed
- personnel/equipment

Drilling breaks are generally evidence of porosity change. Drilling rate tends to decrease with depth. Thus when a drilling break occurs, it may be an

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evidence of transition to an abnormal zone. It is crucial at this point to stop drilling and check for flows.

6. Lost Circulation

Loss of substantial quantity of mud into the formation will result in reduction in hydrostatic column height. If not checked, this can result in a kick.

7. Changes in gas levels/mud density/conductivity

Gas extracted from the mud comes from one or more of four sources

i Liberated gas which is the measured gas from the return mud flow, released from the pore spaces of the drilled cuttings. It is the so-called "background gas" during circulation. If there is overbalance, and the ROP is constant with flow rate, this liberated gas.

ii Produced Gas This gas enters the wellbore from adjacent permeable formations when underbalanced exists. A background gas increase when ROP is constant is evidence of produced gas.

iii Recycled gas is the gas re-circulated into the hole. It appears on

detection equipment as an increase in background levels.

iv Contamination Gas results from chemical breakdown of mud additives

v Connection gas and trip gas are short duration gas peaks caused by swabbing action

Depending on sensitivity level, surface monitors should detect a relatively steady level of gas extracted from mud during normal drilling. This background gas level may show occasional variations depending on penetration rate, mud pumping rate, hydrocarbon content of section drilled. Under normal conditions, the background gas should remain within about 50% of local average. It is crucial that all gas values must be reported whether they are significant or not. Thorough inspection of gas monitoring systems and calibration as part of routine maintenance exercise is essential to preventing potential disasters.

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Connection and trip gas are most common while drilling. The connection gas peaks are a clear sign that pressures are near balance, making for optimum ROP. Swabbing is the main cause of connection/trip gas as it creates the set of negative differential pressure. .

Effects of gas expansion at surface are varied. Mud loggers and drilling crew can see evidence of:

- Rapid fall in flow line mud density
- Increase in return flow
- Mud pit level increase
- Rapid increase in total gas or hydrocarbon levels:

7.5 Well Control Procedure

Control procedures must be initiated immediately after a kick has been detected. Decisions concerning control operation rest on the Company man (The so-called client rep.) as well as the on-site Tool Pusher. The mud loggers are responsible for monitoring and recording circulation pressures, gas and pump strokes.

The primary and secondary well controls involve the prevention and having occurred, the removal of influx without damage to hole, personnel and equipment as well as prevention of further influx.

Kick control procedures involve the following:

1. Shut in the well - Follow a predefined procedure
2. Observe shut-in pressures and kick volume
3. Make kill calculations for:

- Formation pressure
- BHP to maintain while circulating
- Kill mud density
- Initial and final circulation pressures
- Drillpipe pressure schedules

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- Weighting material volume required

4. Define weighing up and circulating procedure using either of the following methods:

- Drillers method
- Wait and Weight method (so-called Engineers method)
- Concurrent Method
- Top kill method

5. Make interpretation of the influx including

- height and density of influx
- annulus pressure behaviour while circulating

6. Proceed with circulation using the following guide rules

a At all times during circulation, the bhp must be high enough to prevent further influx.

b For the safety of the rig and personnel, surface pressure must not at any time exceed the predetermined MAASP value. Maintenance of bhp at proper level is through the application of backpressure via the adjustable choke.

7.6 Surface Well Control Equipment

The control equipment is laid out as shown in Fig 7.1 and is made up of:

1. Blow out preventers - Usually mounted in a stack including Annular and ram type preventers(Fig. 7.2)
2. Drilling Spool (Fig 7.3) for attachment of high pressure choke and kill lines, for circulation with the BOP closed.
3. Casing head - Ibis is welded to the first surface casing and it provides support and pressure seal for the BOP stack and future casing strings.

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Fig. 7.1

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Fig 7.2: Blowout Preventer Stacks

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Fig. 7.3 BOP Components with Drilling Spool

4. Choke Manifold (Fig. 7.4) - To control flow of produced fluids and route them to separator, flare or holding tank or pit. A typical choke manifold includes:

a One hydraulically operated BOP outlet valve for positive closure

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b Three adjustable chokes (variable-opening valves)

c Several manual valves

Fig. 7.4:

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At least one adjustable choke is manually operated and permits precise control of return flow rate and backpressure. Operation of other chokes is from the choke control console (Fig. 7.5), which is also linked to the Mud loggers cabin by remote sensors for automatic logging.

Fig. 7.5: Control Console

5 Kelly Cock, Float Valve or Inside BOP (Fig 7.6) - To prevent back flow via the drillpipe.

6 Mud Gas separators and Degassers - Equipment for removing gas from mud

For subsea well control equipment, the layout and configuration are rather different. They may include (Fig 7.7)

- Templates - Temporary and permanent guide bases
- Subsea BOP stack

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- Marine riser, etc

Fig. 7.6: Drillstring Well Control System

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Fig. 7.7: Subsea Well Control Equipment

7.7 Kill Calculations

To make kill calculations, the following initial data must be available:

1. SIDPP 2. SICP 3. Volume of influx 4. Initial mud weight

Calculations required are:

1. Formation or pore pressure
2. Bottom hole pressure to maintain well during killing operation
3. Kill mud density

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4. Initial circulation pressure
5. Final circulation pressure
6. Volume of weighting material needed.

There are a variety of software that can be used on-site to carry out these calculations. Nevertheless, many company men will feel comfortable with hand calculated values. Recordings must also be made on the Kick control worksheet.

Formation Pressure Computation

$$PF = PH + SIDPP$$

$$PH = \text{Hydrostatic pressure } 0.052 \times \rho_m \times D$$

ρ_m = Mud density, ppg; D = Depth, ft

Bottom Hole Pressure to maintain While Killing the well

This must be in excess of the formation pressure to prevent influx. This should include a safety factor (SF) called trip margin to compensate for calculation errors and swabbing. SF can be about 0.3 to 0.5 ppg equivalent. A better practice involves calculation based on yield point of mud in use.

$$SF = YP / 6(D_h - D_p)$$

D_h = Hole diameter

D_p = Drillpipe

YP = Yield point.

SF = Safety factor, ppge

$$BHP = PF + SF$$

Kill Mud Weight

This is the mud weight required to prevent further influx while circulating out the kick and during later operations.

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$$\rho_{kill} = SIDPP / (0.052 \times D) + \rho_m + SF$$

Initial Circulation Pressure

During circulation of influx, the choke operator controls bhp by varying the choke opening or casing pressure. The choke operator monitors the bhp with the drillpipe pressure gauge. The operator obtains the Initial Circulation Pressure (ICP) by opening the choke and waiting for the driller to bring the pump to kill rate. Operator then adjusts the choke to obtain the desired pressure on the drillpipe pressure gauge.

$$ICP = SIDPP + P_c + SF$$

P_c is pressure loss for initial mud weight at the kill rate,.

When the driller pumps higher mud weight into the hole, the hydrostatic pressure of the mud increases in the drillpipe. This reduces differential pressure. As the new mud travels down the pipe, the circulation pressure declines gradually until new mud reaches the bit. At kill mud weight, the differential pressure should decline to zero. Thus the circulation pressure should equal the kill rate pressure loss corrected for kill mud. This is then the final circulation pressure.

Final Circulation Pressure (FCP)

$$FCP = P_c \times \rho_K / \rho_m$$

Weighting Material Volume Estimation

The Company man determines the well control procedure and the number of circulation necessary to bring the mud up to the kill weight. If the kick can be controlled in one circulation cycle, then the control equipment spends less time under pressure. If the amount of weighting material needed is greater than rig capacity to mix mud, well control will require weighting up stages. These considerations require calculation of weighting material volume requirement.

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Total Circulating system volume, $T_v = \text{Annular vol.} + \text{Drillstring I\>D volume} + \text{Active pits volume.}$

$$\text{Barite requirement, Weight} = T_w = 0.736 * T_v * (\rho_{\text{Kill}} - \rho_m) / (\rho_w - \rho_{\text{kill}})$$

In metric units

$$T_w = 4.2 * T_v * (\rho_{\text{Kill}} - \rho_m) / (\rho_w - \rho_{\text{kill}})$$

ρ_w = Density of weighting material

Next is the calculation of volume requirement.

$$V_w = T_w / \rho_w$$

Finally, rate of barite or weighting material addition is calculated.

$$\text{Rate} = 1472 * q_r * (\rho_{\text{Kill}} - \rho_m) / 42 * (\rho_w - \rho_{\text{kill}})$$

Interpretation of Influx

A first step towards the control of influx after shut-in and data gathering is the identification of the type and size of the influx which will eventually guide the choice of the appropriate control measures.

The influx may be oil, water or gas or a combination of the three. Water and oil do not show significant compressibility effects under typical wellbore conditions but gas can be extremely difficult to handle because of its high compressibility. Typical characteristics of a gas kick include the following:

1. A gas kick will expand as it rises in the hole because of the reduction in hydrostatic pressure especially close to the surface. This will greatly affect the pit volume gain to be handled.

Gasses have much lower densities than most drilling fluids . They therefore tend to rise up the annulus in the mud. If well remains shut in during migration, a gas influx will move up as a slug and will retain its volume and pressure.

Circulating out the influx through an adjustable choke normally restricts gas expansion. This leads to surface pressure increase as the gas moves up the annulus and it is highest when the gas influx reaches the surface. As the gas

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bubble rises, the hydrostatic pressure of mud column pressing down the gas becomes less, thus increasing the surface casing pressure. Thus, the greater the migration the higher will be the surface pressure, It is imperative therefore that once there is a kick especially if it is gas, circulation should start as soon as possible after the shut-in.

If a weighted mud reaches the hole bottom before the top of a gas kick migrate up the annulus and reaches the casing shoe, the pressure at the shoe will be less by a height of the weighted mud in the annulus and the differences between the mud weights.

Effective quality control is essential to a successful killing process. The simpler the control procedure the lower the chance of error. If time permits, it may usually be simpler to circulate the kill mud and kill the influx at source.

A second quality assurance is that the chosen circulation method should not exceed the rig capacity to weight up mud.

Computation of Kick Height, Lk

The general assumptions in the calculations are:

- The influx behaves as a slug and does not mix with the wellbore fluid.
- The Pit gain is the kick volume
- The hole diameter is to gaug

- Hole is vertical.

Assume the annular capacities in the regions of the drillpipe and casing are v_{adp} (barrels per foot of gallons per foot of hole length) and v_{adc} respectively

Given that the pit gain or kick volume = V_{Kick}

The annular geometry is normally not uniform due to bottom hole assembly(BHA) diameter being greater than drillpipe diameter. The first step in the calculation of kick height therefore is to find out if the volume of kick is greater than the annular volume in the region of the BHA.

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The two conditions for influx height calculation are:

1 If $V_{Kick} < V_{adc}$

where $V_{adc} = V_{adc} \times L_{dc}$.

L_{dc} = Length of drillcollar

Then

$L_{Kick} = V_{Kick} / v_{adc}$

2 If $V_{Kick} > V_{adc}$,

$L_{Kick} = L_{dc} + [V_{Kick} - L_{dc} \times v_{adc}] / v_{adp}$

3. Calculation of Kick Density, ρ_{Kick}

$\rho_{Kick} = \rho_m - [(SICP - SIDPP) / 0.052 L_{Kick}]$

If $\rho_{Kick} > 8.3$, then influx is water

If $7.6 \text{ppg} < \rho_{Kick} < 8.33 \text{ppg}$, then influx is oil

If $< 7.6 \text{ppg}$, then influx is gas

If $2.3 < \rho_{Kick} < 7.6 \text{ppg}$ then influx is condensate, gas-cut oil or gas-cut water.

Different service companies have 'Kick' software for computing the kick densities and can create so-called "bubble plot" that indicate annular pressure changes as influx is circulated up the hole.

4. Calculation of Annular Pressure Profile.

Strictly speaking, Well control procedures do not require the calculation of annular pressure profile for a kick. But this may be useful especially for gas kick.

The P_{max} is calculated as :

$P_{max} = 0.052 \rho_{Kick} (1029.2 \times V_{Kick} \times D) / A_s$

A_s = Annular surface area at wellhead, in²

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7.8 Kill Methods

There are three major kill methods usually adopted by industry for the control of influx into the wellbore. These are:

1. The Driller's Method
2. Wait and Weight Method (Also called the Engineers Method)
3. Concurrent Method

There are also two quasi-control methods, which do not require circulation through the choke. These are:

1. Top Kill Method
2. Volumetric Method

Driller's Method

In this method, the well is originally shut-in after the kick has occurred with the appropriate shut-in procedures followed. Thereafter, the influx is circulated out using the original mud in the hole. Once the kick is out of the hole, the hole is subsequently filled with the heavier kill mud.

This control method therefore requires two circulation cycles. Circulation 1 removes the influx and circulation 2 circulates the kill mud. Operators tend to use the Driller's method when they are certain the kick contains little or no gas.

Merits of Driller's Method

1. Minimum circulation
2. Minimum idle time. No waiting
3. Minimum well information required

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Disadvantages of Driller's Method

1. High annular pressures if kick contains gas
2. Maximum time with well under pressure
3. Maximum time spent controlling choke!

Wait and Weight Method

In this method, otherwise known as the Engineers Method, the influx is circulated out by the kill mud in one cycle. This requires the manipulation of the choke to maintain constant bottom hole pressure as the new mud moves from surface to bottom. The kill mud may require several circulation cycles with mud density raised in stages.

Advantages of the Engineers Method

1. Lowest wellbore and surface pressure

2. Minimum time spent "on-choke"

Disadvantages

1. Idle time weighting up with gas migration possible
2. Possibility of several circulation cycles if large mud weight increase is necessary

Concurrent Method

This method mixes the features of the Driller's and Engineer's methods. It can be applied in cases where the surface pressures need to be kept as low as possible and shut-in time to a minimum.

Circulation starts immediately after observing the shut-in pressures using the current mud weight. Then the mud weight is increased in stages. As each stage of increased mud weight reaches the bit, the driller starts pumping the next stage. When the new mud reaches the kill mud weight, circulation continues until new mud reaches the surface.

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This method requires manipulation of the of the choke to maintain constant BHP just as in the engineers method. However the mud weight increases determine the pipe pressure schedule and the pump stroke from surface to bit. The pipe pressure schedule covers a longer interval than the other two methods.

Advantages of the Concurrent Method

1. Minimum idle time. Control starts immediately
2. Continuous, smooth adjustment of mud density.

Main Disadvantages

1. Slightly more complex calculations than the other two methods
2. Maximum time spent circulating under pressure

7.9 Well Control Without Circulation

If influx is gas, and the drillstring is plugged or completely out of the hole, then there is no means of circulation. Without circulation, gas will rise in the annulus leading to casing and bottom hole pressure increase. Excessive increase may cause damage at surface or at weak point of casing shoe.

In such cases the driller may use the Volumetric Method to bleed off mud volume, allowing gas to expand as it migrates upward.

In the real sense of the word, Volumetric Method is not a kill method. Rather it involves the bleefoff of the influx or kick pressure. The well control team must still eventually circulate the kill mud to kill the well dead. This method is most effective when the kick occurs due to swabbing. In most cases under

this condition, the mud weight is probably sufficient to prevent further influx before the circulation of weighted mud. However, if the cause of kick is insufficient mud weight, then there is likely to be further influx when the kick bleeds off at the surface.

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7.9.1 Control Procedure for Volumetric Method

The procedure for the Volumetric Method of control is as follows:

1. Shut in the well according to defined procedure in well plan.
2. Monitor and record SICP. By definition, use of the SIDPP does not exist since there can be no circulation through the drillpipe.
3. Calculate the bottom hole pressure to maintain while killing the well
4. If the SICP gradually rises after 10-20 minutes of shut-in, the kick contains gas and the well control team must apply the volumetric method.
5. Allow the SICP to rise by a convenient increment. About 5bar(75psi) or maximum 100psi is recommended. Make sure the rise does not exceed the MAASP.
6. Slowly open the manual choke and bleed off mud volume until the casing pressure drops by the predefined increment. This allows the gas influx to expand slightly in the annulus.
7. Close the manual choke and note the new SICP. After bleed-off, the new SICP will be higher than the original SICP.
8. Again allow the SICP to rise to predefined increment level.
9. Repeat steps 6-8 as needed until the kick reaches the surface. The bleed-off mud volume required will increase as the gas expands, and the bleed-off volume may become very large when gas nears surface.
10. As the gas reaches surface and bleeds off through the choke, slowly pump mud into the hole via the kill line, to replace the lost fluid volume.
11. The control team can then "strip or snub" the drill string into the hole to complete the kill.

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7.9.2 Stripping and Snubbing

Normally, kick control using balanced bottom hole pressure requires that the drillstring be at or near bottom. If the drillstring is not on bottom at the time of shut-in, the driller must run in hole under pressure. Stripping and snubbing are the two optional processes normally adopted for running-in of drillpipe under

pressure.

The primary objective of these processes is to maintain a constant bottom hole pressure while Running-in the drillstring. This requires careful manipulation of a manually operated adjustable choke to bleed off mud volume displaced by the pipe.

1. Stripping-in uses the travelling block to lower the pipe into the hole, through a partially Closed blowout preventer(BOP). The method is most practical when the pipe weight is sufficient to overcome the upward force generated by well pressure on the pipe cross section.
2. Snubbing uses hydraulic or mechanical equipment to apply downward force to the pipe. This way the snubbing process overcomes the excess upward force generated by well pressure on the pipe cross-section.

For either process, the BOP must be closed against the pipe wall to maintain at least a partial seal. The driller may choose to run the pipe in through

1. the annular presenter only or
2. Combination of annular or pipe ram preventers.

7.9.3 The Top Kill Method

If the drillstring is off bottom, or out of the hole, and running in appears to be very risky, then the top kill control technique can be adopted. Top kill methods include:

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1. Bullheading - This is a process involving the injection of mud into the closed annulus to force invading fluids back into formation. It is a popular method adopted for cleaning up the wellbore after a drill stem test especially in cased hole. It is rare in kick control except in unusual circumstances where everything else is not possible. The main disadvantage is that it risks failure of the casing shoe and in a shallow hole may result in creation of fissures to the surface behind the casing. Bullheading works mainly If the producing formation is isolated from the rest of the wellbore..
2. Setting barite or cement plugs A barite plug is a slurry of barite and fresh water or alternatively barite and diesel, spotted above the producing formation to bridge off the hole. Merits of this method are:
 - The barite is fine enough to pump through the bit nozzles
 - Setting a barite plug offers a reasonable chance of recovering the drillstring.
 - The plug can be drilled out if necessary.

A major disadvantage however is that the barite may plug the drillstring if not fully displaced into the annulus.

Examples of Barite plug requirement are-

1. Possible slurry densities: 1.76 - 2.58SG
2. Barite in fresh water: 485 - 1530 lbs/bbl (This may require thinning with phosphates or lignosulfonates.

Phosphate mixing requirements: 2.81 lbs/bbl

Lignosulfonate mixing requirement: 0.41 lbs/bbl

3. Barite in diesel : 570 - 1660 lbs/bbl

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The recommended volume of barite slug is about 10 barrels.

Mixing/displacement must be prompt to prevent barite sag and possible plugging of drillstring. Use of cement pump may improve matters.

The use of a cement plug however should be regarded as a last option as this is tantamount to well abandonment including loss of much of drillstring.

7.10. Hole Enlargement

7.10.1 Causes

Hole enlargement otherwise known as washout can be the result of drilling through salt section or result of sloughing/heaving shales

7.10.2 Prevention/Control

1. For salt section - use saturated salt-water mud
1. For Heaving/sloughing Shales : Use KCL/Polymer mud
2. Maintenance of good equivalent circulation density.

7.11 Formation Damage

Formation damage can be defined as a measure of the reduction in the production capacity or injectivity of a given reservoir formation . This can be due to the combined effect of :

- reduction in the formation absolute permeability
- alteration in the relative permeability of the formation to a specific fluid
- alteration in the viscosity of the mobile fluid

It is mainly a consequence of a number of operations conducted in a well from the initial drilling, through the production and workover phases to final abandonment.

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From the radial flow theory, the flow through a composite reservoir with damage can be viewed as a combined flow through two reservoir sections in series. Thus, the total pressure drop in the reservoir can be defined as:

$$\Delta P_{TOTAL} = \Delta P_{SKIN} + \Delta P_{CLEAN\ RESERVOIR}$$

7.1.1 Causes of Formation Damage

Formation damage can be considered to be caused by a physical or chemical interaction between the following:

1. Invading fluid liquid phase and the reservoir fluids
2. Invading liquid phase and the reservoir rock constituents
3. The solid constituents of the invading fluid and the reservoir rock.

The damage mechanisms do vary but these can be listed as:

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1. Formation damage resulting in reduction of absolute permeability due to

- ◆ Particle plugging due to solids in drilling and completion fluids invading the reservoir rock pore spaces.

- ◆ Clay problems: In-situ clays are found in majority of reservoir rocks especially sandstone. Majority are very sensitive to fluids invasion and may swell thus blocking the pores

- ◆ Fines migration: A typical sand stone reservoir rock consists of insitu quartzitic or siliceous fines generally held within the insitu connate film surrounding the sandstone grains. These fines can become mobilised either due to high production rate or due to increase in wetting phase saturation. Once mobilised, fines are transported and can result in particle plugging.

2. Inorganic scale deposit resulting from reaction of incompatible invading and reservoir fluids. Changes in pressure and temperature also contribute to the precipitation of solids forming scales

3. Plugging by organic deposits such as wax and asphaltenes.

4. Damage resulting from reduction in relative permeability - Invasion by wellbore fluids cause hydrocarbon saturation to decrease around the wellbore. This lowers the relative permeability to oil and, especially during early stages of production, a significant pressure drop is created around the wellbore.

5. Damage due to emulsion blocking - Result of formation of emulsion between oil and water. Bacterial action may also form emulsion blockage.

6. Gas Breakthrough and Water Coning - Increase in gas and water saturation reduces the relative permeability to oil.

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7. Stress-induced permeability change - Changes in stress distribution will result in changes in compaction status and thus pore pressure change. This will affect the net permeability of the reservoir.

8. Sand production from unconsolidated reservoir rocks. These are known to be production rate sensitive. Thus high production rate and water encroachment among other factors can lead to the migration of in-situ sand grains and massive sand intrusion.

7.11.2 Prevention and Control Measures

One important aspect of formation damage is the location of its occurrence. it can occur close to the wellbore and this will have the greatest impact on well productivity since it will affect fluid flow in the region of highest velocity convergent flow for radial production and divergent flow for radial injection in injection wells. Occurrence in the proximity of the wellbore makes it accessible for removal or bypassing. it is usually the most severe and therefore needs to be handled with care. On the other hand damage which occurs deep inside the reservoir is unlikely to be severe since the effects of pressure drop, velocity and concentration are not as high. Further since the in-situ velocity is lower, the impact of damage is reduced and can be easily bypassed by the

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large number of alternative flow channels. therefore the removal of very deep damage is not generally technically feasible.

Prevention

The best control technique is prevention. Preventive measures for formation damage are limited to the ways of ameliorating the three potential causes. In all cases, the following is required:

1. An appreciation of the causes of formation damage and factors governing the severity and location
2. An analysis of well operation to assess the potential risk for damage
3. The modification of operation to avoid or minimise damage
4. The evaluation and implementation of techniques to remove or bypass the effects of damage.

The potential preventive measures are as follows:

1. Interaction of invading fluid liquid phase and reservoir fluid.

Combating reaction between the fluids that can lead to precipitation, emulsion or water blockage, etc requires the selection of appropriate drilling and completion fluids that are compatible with the reservoir fluids at the downhole temperature and pressure. Likewise, the reduction of fluid loss through the use of appropriate control agents is essential. Total prevention of lost circulation is also very important.

2. Interaction of invading fluid liquid phase and the formation
This problem leads mainly to potential clay swelling, wettability change and potential water blocking. To minimise the problem, use of appropriate clay inhibitor/stabiliser completion fluids would be desirable to prevent clay swelling.

Most formations are usually water-wet but this may change due to completion operations. For such cases, the use of appropriate

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surfactant to alter wettability will be desirable. For example, for improved oil relative permeability, the formation needs to be water wet. Thus the water wettability of the formation can be restored through appropriate surfactant treatment.

3. Interaction of invading fluid solids and formation.

This generally leads to particle plugging or result of fines migration. Prevention generally involves effective solids control for drilling fluids and the use of solids free brines as completion fluids . Effective well preparation before completion operation is carried out is very essential. Thus, the drilling fluid can be displaced by solids free completion fluids.

Specific cases of Preventive measures during well completion operations are discussed below.

1. Damage during drilling and cementing

Damage during drilling can be due to

- Fluid loss or lost circulation
- mud solids

Therefore it is essential to select compatible drilling fluids. Avoidance of unnecessary downtime with no circulation is essential. Minimisation of pressure surge and effective solids control are a key to minimising damage.

During cementing, casing should be run in the hole with minimum delay but smoothly to avoid excessive pressure surge. Cement formulation must be with adequate fluid loss control. Compatibility and conditioning of mud to avoid

unnecessary contamination is also essential.

2. Formation damage during well completion operation

Completion operation is perhaps the most susceptible to damage. The completion involves the preparation of the open hole or cased wellbore, installation of completion string and placement of completion and packer fluids.

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The principal sources of damage are

1. Fluid leakoff and particle plugging
2. Fluid interaction with reservoir rock and fluids.

In these cases therefore it is essential to

i Prepare the well for completion by removing the mud and carry out effective wellbore cleanup free of debris. Casing, tanks and lineworks must be cleaned

ii Use solids free completion fluid to avoid plugging. However, loss control materials such as calcium carbonate, oil-soluble resin or sized salts that are easily removable during back production are added to minimise fluid loss. They are also removable with acid wash.

iii Exposure time especially for open hole completion should be minimised.

iv Use of minimal overbalance

v Perforation should be done with minimal overbalance if overbalance perforating is used. Preferably underbalanced perforating is best to minimise damage. In any case, perforations should be cleaned of debris either by backsurgings or perforation washing.

Increasing perforation shot density and penetration can also minimise damage.

3. Damage during production operations

The main causes of damage during production are

- (1) Formation of inorganic scales
- (ii) Deposition of wax/asphaltenes
- (iii) Migration of fines towards wellbore
- (iv) Sand production from unconsolidated reservoirs

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Scales.

Scales are formed especially by the mixing of incompatible waters and changes in the wellbore conditions such as salt concentration, declining temperature and pressure. Use of scale inhibitors where the problem is suspected is therefore essential.

Organic Deposits

Deposition of wax/asphaltenes depends on production conditions and produced fluid composition. It may not be possible to modify drawdown to avoid asphaltene deposition but the effect can be reduced by increasing, for example, the perforation density, phasing and penetration.

Fines Migration

This is prompted by increased water cut and excessive drawdown, therefore should involve the minimisation of these causes.

Sand Production/Control

Completion of unconsolidated reservoirs should involve the use of pack sands(Gravelpacks) and other screen systems in the attempt to:

- ◆ maximise productivity
- ◆ minimise impairment through the prevention of plugging

If not controlled the migration/production of solids which may erode downhole and surface facilities such as valves, pumps, tubing and flow line components besides the loss of reserves.

Formation Damage during injection operation

These include water or gas injection. For water injection(water flooding), the principal causes of damage are:

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- (a) Solids plugging; (b) bacteria plugging (c) inorganic scale deposition;
(d) Clay swelling and fines migration

it is therefore essential to:

- filter sea water to remove solids over 2-10microns
- remove rust and use oxygen scavengers to minimise corrosion
- use bactericide
- use scale inhibitors

Control of Formation damage

The best control measure is the prevention as discussed above. However, when these fail, then it is essential to stimulate the reservoir by using the appropriate Stimulation techniques.

Before stimulation, it is essential to analyse the damage through well testing in order- to quantify the magnitude. Details of well test and damage

quantification otherwise known as SKIN or skin effect will be dealt with later

A positive skin represented by the symbol 's' implies damage while a negative skin represents enhancement.

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NORTHERN TERRITORY GEOLOGICAL SURVEY

DRILLING MANUAL

JN DUNSTER

Record 2004-007

Northern Territory Government

Department of Business, Industry & Resource Development

NORTHERN TERRITORY DEPARTMENT OF BUSINESS,
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INTRODUCTION

NTGS has an obligation to provide geological information in areas of poor outcrop and inadequate drillhole coverage. Despite

the best geophysical data, this can only be reliably addressed by stratigraphic drilling, where such drillholes provide lithological

and stratigraphic information and allow collection of pristine samples for biostratigraphic, geochemical, geochronological and

mineralisation studies. These data can then be used to build regional geological models and encourage exploration companies

into new areas. Considering the paucity of outcrop and the limited coverage of existing drillholes over much of the Northern

Territory, there is an ongoing need for the NTGS to undertake drilling programs. In addition, drilling undertaken by NTGS

should be documented in a way that allows explorers to optimise their own drilling programs.

Objective and content of this document

This in-house manual is designed to assist NTGS staff in organising stratigraphic drilling programs and supervising on site. It

is directed to those with little or no previous experience of drilling. It gives in-house guidelines and does not purport to

reiterate government policy concerning drilling, unless this is explicitly acknowledged. It is not a textbook on drilling nor is it

intended to be as comprehensive as company manuals. For comparison, the 1996 Santos onshore drilling operations manual is

over 400 pages in length, including 58 pages of forms. In the present manual, much of the drillers' jargon is explained in

references listed in Appendix 1, and a glossary of common abbreviations provided in Appendix 2. Australian Standards referred

to throughout the document are shown in the format AS1234. The symbol % is used to indicate guidelines, procedures and

checks for the NTGS site Representative. The symbol □ is used to distinguish worked examples of calculations. An index

provides quick reference to specific topics.

It is anticipated that this document will be revised regularly, and any corrections or comments should be directed to the

author or the Editorial Geologist.

History of NTGS drilling

The various governments responsible for what is now the Northern Territory have been involved in drilling for almost a century. No specific drilling reports were published prior to 1908 when the Territory was administered from South Australia, but summaries were included in earlier annual reports at least as far back as 1905. The first internal review of drilling activities by the NT government was undertaken in 1912 by HI Jensen, the then Director of Mines. He rationalised the 250 rapidly decaying boxes of core stored in the Government Stables, 'put an end to the old system of boring at random' and 'introduced business methods into the work of the drills' (Jensen 1915). He reported that during 1912-1913 eight holes totalling 3626' (1105 m) were drilled using steam-powered diamond rigs at known prospects. The average cost was £1 15s 2½d per foot. The Mines Branch, as it was called, continued to drill prospects at the behest of leaseholders until its demise in the late 1970s. From NT self-government and the birth of NTGS in 1978, the emphasis shifted to stratigraphic drilling as a complement to geological mapping. Another government division is responsible for waterbore drilling but their samples are stored in NTGS core libraries. As of July 2003, 644 NT government water and stratigraphic drillholes totalling over 57 000 m had been registered in our COREDAT database. The average depth is 88.8 m; 155 holes are ≥ 100 m and 19 of the holes exceed 500 m, the deepest to date being 845.9 m.

Types of drilling

The various types of drilling may be classified according to the hole-making action and the hole cleaning method. Those of relevance to NTGS are described below.

Open hole rotary drilling is widely used in petroleum and minerals exploration. It uses a slowly rotating (typically <100 rpm) drillstring and bits which cut by scraping (blade bit), point pressure chipping (roller bit with three cones of steel or tungsten carbide teeth) or percussion (rotary downhole hammer with rounded tungsten carbide buttons; 10-40 rpm). Figure 1 shows a selection of bits used in open hole drilling. Penetration rates while actually drilling are in excess of 8 m/hr, sometimes as high as 30 m/hr. The flushing of cuttings may be by water, mud or air (rotary air blast – RAB). The circulation can be normal (cuttings up the annulus) or reverse circulation (RC - in which cuttings come up the inside of the drillstring). There are numerous combinations of the various open hole rotary drilling techniques and hole clearing techniques. For example, blade bit and tricone rotary drilling can be done using mud or air. Hammer drilling uses air and specific hammers are designed for

normal and reverse circulation. Mist and foam lift cuttings employ a combination of air and mud and can be used in each of these examples.

Coring, in which an annular bit cuts a solid cylinder of rock, is the technique most used in NTGS stratigraphic drilling. It is discussed at length in CORE DRILLING.

Alternatively, individual holes or drilling programs may use a combination of different drilling techniques. For example,

the overburden may be drilled quickly and cheaply (termed a precollar) using a hammer before the target is diamond cored.

Geochemical sampling may involve grid drilling using RAB for reconnaissance and the more expensive RC for more reliable samples.

1

Types of drill rigs

Petroleum rigs range from heli-transportable to huge offshore platforms. On most petroleum rigs, the drillstring consists of

drill pipe with upsets (thicker walls at the joins) and heavy weight collars at the base which provide the weight on the bit. The

drillstring is supported above the ground in a derrick and spun using a Kelly bushing in a rotary table on the drill floor. The

string is said to be tripped in and out of the hole. Changing the bit necessitates a round trip.

Mineral rigs are typically truck, trailer or track mounted. They have a one-piece mast rather than a derrick and use a top

drive. The rig itself provides the pulldown on the bit. The drillstring is usually thin-walled slick (no upsets) drill rods. Moving

the drillstring in and out of the hole is referred to as running in and pulling out.

Drill rigs were originally designed specifically for a particular style of drilling and many remain so. However, the need for

several different styles of drilling in the same hole or drilling program led to the concept of multipurpose rigs. These rigs were

first developed in Australia and the United States during the 1970s-1980s. Mineral rigs, such as the heli-transportable Bourne

4000S (capable of 2500 m in 4.35" hole) were extensively modified to take blowout preventers (BOPs) for both open hole

drilling and continuous coring of petroleum wells. These hybrid rigs use a top drive and either a heavy duty (CHD series) slick

string for coring or small diameter drill collars and heavy weight drillpipe for open hole rotary drilling. Such light-weight

petroleum rigs are only used for specialist exploration work, often in remote or inaccessible locations. These rigs are also

referred to as slimhole, meaning a substantial portion of the drillhole is <100 mm diameter.

Multipurpose rigs were also

developed for mineral exploration, where they have become almost ubiquitous. These universal rigs come in a range of sizes and are capable of rotary, downhole percussion and diamond coring of both vertical and angled holes (Figure 2).

a b

TUNGSTEN CARBIDE INSERTS

CONE

END ROLLER BEARING

BALL BEARING

ROLLER BEARING

BALL PLUG

GUAGE PROTECTIVE INSERTS

SHIRT TAIL PROTECTION

LEG PROTECTIVE INSERTS

BACK OF LEG WITH SURFACING

FOR PROTECTION

LEG

AIR PASSAGE

AIR JET NOZZLE

BACKFLOW VALVE

c d

THREADED PIN CONNECTION

m04-068.dgn

Figure 1. Types of drill bits most commonly used in open hole, non-core drilling. (a) Chevron wing blade bit with replaceable cutters,

commonly used for poorly consolidated near-surface formations. (b) Tricone roller bits; these are among the oldest and most versatile types

of bit. The bit on the left has steel teeth; longer, more pointed teeth are used for softer formations. The bit on the right has tungsten carbide

buttons; buttons come in a range of shapes from chisel-shaped for moderately hard formations to dome-shaped for extremely hard formations.

Both bits are designed for mud drilling; mud is jetted through three nozzles (arrow) and differently sized jets can be fitted into the nozzles

to optimise the hydraulics; the bearings are sealed. (c) Small-diameter hammer bit with tungsten carbide buttons, for percussion drilling.

(d) Partial cross-section through a roller bit designed for air drilling. Air roller bits have different plumbing and bearings to those used with

mud, eg the bearings are air cooled and there is a single air jet nozzle with a backflow valve inside the bit. Images courtesy of Baker

Hughes, Kay Rock Bit and UralMBT.

2

a b

c

d e

f g

i

h j

m04-069.dgn

Figure 2. Multipurpose rigs. (a) UDR 650 MKII, a medium-sized multipurpose rig, reverse circulation (RC) drilling an angled hole; note

the flowline connection that enables the drilling fluid to be circulated down the annulus, bringing the cuttings up the inside of the drillstring.

This rig is capable of RC hammer to 280 m with 3.5" rods. (b) The same rig as in (a), diamond coring in NQ diameter. It is capable of coring

HQ to 575 m and NQ to 875 m. (c) Carrier-mounted UDR 5000 jacked up as a platform. This is one of the largest UDR multipurpose rigs,

rivalling some conventional petroleum rigs in depth capacity, capable of downhole hammering to 1830 m with 3.5" rods and of coring

CHD101 to 2960 m. (d) Bourne THD25, a medium-sized rig designed and built in Queensland, used for waterbore drilling and shallow

mineral exploration. It would normally be truck mounted; the hydraulic top head drive is visible at the bottom of its travel and the driller's

platform and controls are to the left. (e) UDR 1000 on a platform in Papua-New Guinea. A standard UDR 1000 is capable of RC hammering

to 415 m with 3.5" rods and of coring HQ to 1000 m and NQ to 1500 m. (f) Drillcorp rig 53, a UDR 1000 mounted on an 8x8 Man truck,

showing the rod rack on the right side of the rig. (g) Boart Longyear UDR 1000, similar to the previous, showing the left side of the rig with

pumps, compressor and driller's platform. (h) Truck-mounted UDR 1000 drilling an angled hole. (i) Truck-mounted UDR 1200, capable of

RC hammering to 550 m with 4.5" rods and of coring PQ to 820 m, HQ to 1100 m and NQ to 1650 m. (j) UDR 1500 shown jacked onto its

platform with the mast set to drill an angled hole; the caterpillar tracks used to transport it are visible to the left. A UDR 1500 MKIII can

hammer to almost 1000 m and core CHD101 to 1680 m and NQ to 2780 m. Images courtesy of Bournedrill, Drillcorp and UDR.

3

ADMINISTRATION

This section presents checklists and brief summaries to assist in the planning, execution and supervision of a drilling program.

Safety and duty of care are discussed.

PLANNING AND MANAGING THE PROGRAM

% Stage 1 Planning

- Define drillhole(s) objectives
- Organise access with Traditional Owners and other stakeholders
- Prioritise order of drilling
- Prepare Drillhole Proposal(s)
- Prepare Authorisation for Expenditure (AFE –see AUTHORISATION FOR EXPENDITURE)
- Purchase long lead time consumables
- Finalise Invitations to Tender
- Organise earthworks (build roads, dams, dig pits etc)
- Assure secondary water supply
- Prepare draft work flowchart
- Evaluate tenders
- Collect intelligence
- Model costs
- Inspect equipment
- Interview key contractor personnel

% Stage 2 Contracts management

- Award contracts
- Organise site inspection by representative of drilling company
- Refine work flowchart
- Obtain any necessary insurance

% Stage 3 Site supervision

- Coordinate mobilisation and inter-drillhole moves
- Undertake site duties (see ONSITE DUTIES)

% Stage 4 Closeout

- Site restoration
- Depermit all stakeholders
- Appraise contractors and NTGS performance
- Audit project
- Prepare Technical Note and Drillhole Completion Report

AUTHORISATION FOR EXPENDITURE

A drilling budget should include:

- Permitting and depermitting
- Water supply (waterbore drilling and/or haulage)
- Roadworks
- Roadworks supervision
- Site preparation
- Site preparation supervision
- Rig mobilisation/moves/demobilisation
- Rig day rates

- Camp
- Fuel/lubricants and delivery
- Casing
- Bits and drilling consumables

4

- Mud supply and engineering
- Wireline logging and vertical seismic profiling
- Travel and accommodation
- Freight
- Field supplies and equipment
- Analyses
- Drilling Proposal/Technical Note/Drilling Completion Report
- Insurance
- Communications
- Site restoration
- Road repairs
- Contingency

RISK ANALYSIS

Anticipating and assessing the risks or potential problems is part of any project plan. In drilling, the emphasis in risk analysis

will vary from one proposal to another. The checklist below gives examples of common risks.

Quantitative comparisons are

possible if the seriousness of the various facets of the program and their likelihood are each given a rating from one to ten with

one being the least serious and least likely. The weighted average then ranks the relative risk of this particular proposal.

Once identified and prioritised, potential problems need to be matched with preventative actions, contingent plans or

adequate solutions.

Planning and access problems

- Delays in Aboriginal clearance
- Problems with access logistics for mobilisation/demobilisation/interhole moves
- Difficulties with crew change
- Unable to get fuel to site

Technical problems while drilling

- Insufficient/unreliable water supply
- Bad weather delaying operations
- Lost circulation
- Fractured formation
- Overpressured formation
- Unsafe hydrocarbon intersection

- Intersecting faults or inclined bedding
- Intersecting evaporites
- No backup and replacement equipment has to be mobilised

Incorrect stratigraphic prognosis

- Stratigraphic prognosis too shallow, hole deeper than anticipated
- Original target beyond rig capacity

ONSITE DUTIES

% NTGS must maintain a technical presence on site from the arrival of the rig to mast down. The NTGS Geologist and

Technical Assistants are responsible for the following duties:

- Survey the drillhole location with GPS and ensure that the rig is set up appropriately (especially for an angled hole)
- Ensure that all work is carried out in accordance with relevant legislation and under the terms of contracts
- Check and sign contractors' daily operations reports (sometimes called DORs, DDRs, plods or tour sheets). They must be completed and signed daily and not allowed to accumulate. The driller will probably also have separate time sheets and

safety checks to sign

- Brief all site personnel on NTGS guidelines concerning safety and confidentiality
- Organise and chair weekly safety meetings
- Recommend the sample interval during rotary drilling
- Supervise collection of all samples
- Ensure core and other samples are handled, marked and labelled according to best practice

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- Provide a detailed geological description
- Detect and describe all potentially economically important aspects (eg oil shows, ore minerals)
- Provide an NTGS Daily Operations Report in a standard format (Appendix 3)
- Monitor costs relative to budget
- Photograph core
- Arrange transport of core and other materials
- Advise of termination depth (see Terminating the hole)
- Witness and QC wireline logging operations
- Supervise completion or abandonment of the hole in accordance with relevant legislation
- Provide photographic evidence and sign-off on environmental impact closeout

Terminating the hole

% Under normal circumstances, the NTGS Geologist on site will recommend when to terminate the hole. On a deep stratigraphic

hole, a cutoff will have been clearly defined as part of the proposal (eg into top of igneous basement). If the hole is to be

geophysically logged, remember to allow sufficient sump to enable meaningful readings of the lowermost geological unit (in this example, basement). Some of the large combo suites used in petroleum exploration need a 30 m sump; whereas 3 m might be adequate with other tools.

DATA HANDLING AND REPORT WRITING

NTGS drillhole naming convention

The NTGS drillhole naming convention is:

MapSheet code (two letters)_Year (last two digits)_Hole type (DD, RC, AC, RB) _Hole number (two digits)

eg VR03DD01 is the first diamond hole drilled in VICTORIA RIVER DOWNS in 2003. Drillhole names are best determined

early, during preparation of the Drillhole Proposal(s) or Invitations to Tender. The list of MapSheet codes to be used with the

convention is to be found at:

G:\Geological Survey\Administration\Standards & Procedures\GIS\File naming\NTGS_File_Names.xls

NTGS Technical Note, Drillhole Completion Report and data management

% All relevant background data from the drilling will be compiled into an NTGS Technical Note in a standard format (specified

in separate guidelines). These data will be simplified and accompanied by a comprehensive interpretative report in the final

Drillhole Completion Report. The geologist responsible for the project is also to provide all necessary metadata in standard

NTGS format.

SAFETY AND EMERGENCY PROCEDURES

The following are basic checklists relating to safety and emergency procedures for drilling operations.

Site access by trained personnel only

- Only trained personnel on site
- Barriers to prevent access by unauthorised personnel
- Contractors are not to grant ingress to any third party without the consent of the NTGS Representative

Work hours

- Shift duration
- Tour of duty (duration of field period)
- Adequate light
- Working in extreme conditions (avoiding heat stroke)

Site layout and housekeeping

- Good site layout, ground stability, drainage, flood and fire risk, wind direction to camp
- Access and turning circles for support trucks and service vehicles
- Remove any obstructions (loose rocks, tree stumps) from site

- All rigs should ideally be fitted with elevated walkways (AS1657 compliant) to create a uniform work platform, irrespective of local site conditions

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- No slippery walkways
- Check for underground cables and pipes
- Good housekeeping; site clean and tidy and free of tripping hazards
- Tubular items stacked in safe manner
- Safety signs must be displayed (AS1319)
- Any requirement for quarantine and disease control, for example steps to avoid spreading noxious weeds

Fire safety

- Fire breaks and fire fighting equipment; fire bans include campfires
- Fire extinguishers (AS1841, AS1845-48, AS1851 Part 1, AS2444)
- If rig is fitted with an automatic fire suppression system, include familiarisation in safety induction
- Fire prevention during welding (AS1674)
- Oxygen (from oxyacetylene) is not to contact hydrocarbons (eg grease or petrol) as this is potentially explosive

Fuel safety

- Trayback NTGS vehicles are limited in the number of 200 L fuel drums they may carry on a gazetted road without a permit (refer to NTGS Field Manual)
- Fuel stored away from rig and camp in accordance with regulations (eg tanks may require a bund wall)
- Spills or leakage of fuel for the use of Contractor are their responsibility but NTGS will monitor clean-up

Hazardous substances

- Appropriate signage in place
- Materials Safety Data Sheets for all potentially toxic or hazardous drilling additives
- Safe disposal of all potentially toxic wastes
- Spills or leakage of hazardous substances for the use of Contractor are their responsibility, clean-up is under NTGS supervision

Safety audit and safety meetings

- Site safety audit before spud
- % NTGS should organise weekly safety meetings of each shift, to include both NTGS and all contract personnel. Such meetings must be recorded in the Daily Operations Report

Personal protective equipment (PPE)

- Head: hard hats (AS1800, 1801, 2210) must be worn within 30 m of the rig. Note that metal hard hats are not permitted;

allowable accessories include sun brim, visor-type face shield, earmuff attachments, lampholder. Long hair must be restrained, even when a hard hat is worn

- Eye: safety glasses (AS1336:1982, AS1337), tinted or otherwise, must have the appropriate Australian Standard logo; welding

shields (AS1338): a full-face shield is to be worn when cutting core. Filters in fluoroscopes and UV boxes (AS1338 Part 2)

- Hearing: hearing protection device shall provide protection to a level not exceeding 85 dB (AS1270). This can be earmuffs,

disposable ear plugs or both, such that they do not compromise other safety equipment

- Respiratory: respiratory protection against dust (AS1715, AS1716). Breathing apparatus may be carried on some rigs and its use

requires formal training

- Skin: sunscreen and insect repellent will be supplied by the employer

- Hand: general work gloves (AS2161), welding gloves (AS1558)

- Foot: safety boots (AS2210) with a steel toe cap must be worn by all personnel within 30 m of an operating drill rig; boots must

have the Australian Standard logo

- Clothing: safe and adequate clothing, no loose clothing, a UPF (UV) rating of 50+. Some companies stipulate that long-sleeved

shirts and long trousers be worn on drill rigs; welding apron, raincoats

- Harness: all personnel aloft must have a safety belt or safety harness (AS1891, AS2626). Note that these standards forbid the use

of harnesses made from leather or natural fibre webbing. No tools to be hand carried into the mast

Personal health and hygiene

- Any medical condition that may affect Contractor performance must be reported to the NTGS Representative

- Be aware of high-risk individuals (eg asthmatics, diabetics, epileptics, angina sufferers)

- Prohibition of drugs; control of alcohol

- Camp conditions, especially food preparation areas and ablutions are to be clean and hygienic

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- Adequate rubbish and sanitary disposal facilities

- Firearms, bows or similar weapons are prohibited

- Domestic animals are prohibited

- Occupational driving (AS2299)

Use of radioactive sources

Use of a radioactive source in downhole logging requires a licence under the Radiation Health Act.

Any radioactive sources for use

in downhole logging must be transported, stored and handled in accordance with the Commonwealth NHMR Code of practice for the

safe transport of radioactive substances and Code of practice for the safe use of sealed radioactive sources in borehole logging.

Requirements for signage on vehicles carrying sources must be enforced.

Pressure vessels and associated equipment

- Must comply with AS3788, AS3873
- All high-pressure hoses must be restrained to prevent whipping in the event of breakage or connection failure. Particular attention should be paid to the adequacy and placement of the restraining line between the sample hose and the cyclone on an air rig (failure has caused fatalities and serious injuries in Australia)
- Pressure relief valves (mandatory on air compressors, triplex and water injection pumps) should be function tested where possible. No shutoff valve is permitted between the pressure relief valve and the pressure chamber
- Compressed air handtools such as button bit grinders must only be operated through a pressure regulator
- Take care when pumping out inner tubes with drilling fluid or water. Use of compressed air is forbidden; a bumper must be in place and no one within 5 m

Guards on rig

- Hydraulic rod spinners should be fitted to minimise rod handling and ideally, safety cages installed to protect rig personnel while drilling is in progress. These cages should be fitted with a hydraulic interlock which immediately stops rod rotation when the cage is opened
- Check guards on belts, chains and gears

Gas hazards

Petroleum gas is the most common hazardous gas encountered while drilling. CO₂ and H₂S are also potential problems and can

be associated with petroleum gas (up to 28% of natural gas could be H₂S) or gas-charged groundwater. Both petroleum gas and

H₂S are explosive. These and CO₂ can all be fatal if breathed in sufficient concentrations. Ensure that all personnel are informed

if gas is encountered. The driller will decide how to treat the problem, what fire prevention strategy to adopt, and if rig and/or

camp evacuation needs to be considered.

H₂S (rotten egg gas) is not widespread in Australia, but several of the major aquifers contain local pockets of detectable H₂S. For example,

the artesian groundwater at Winton, Queensland has a noticeable odour and there is anecdotal evidence of H₂S being encountered in mineral

exploration holes in the Georgina Basin. Many petroleum rigs (and some mineral rigs) have H₂S detectors as standard equipment and carry

breathing apparatus near the driller in the event that H₂S is encountered. It is extremely toxic, flammable at 4% concentration and makes steel

go brittle. One of the main reasons that it has caused fatalities is that it deadens the sense of smell and gives the false sense that it has dissipated.

It is colourless and heavier than air and will accumulate in tanks and confined and low-lying areas. Because it is soluble in water and hydrocarbons it is readily transported in the mud system. The driller must be informed at the first indication of H₂S (Table 1). Do not attempt to rescue anyone who has been overcome by H₂S unless you are wearing breathing apparatus. The OSHA Permissible Exposure Limit for a ceiling concentration is 20 ppm hydrogen sulfide, a level which may not ever be exceeded. The acceptable maximum peak, for 10 minutes only, once during an 8 hour day if there is no other measurable exposure, is 50 ppm.

Concentration of H₂S Effects

10 ppb First noticeable odour

5-10 ppm Obvious odour of rotten eggs

10-20 ppm Very unpleasant but safe for up to 8 hours exposure per day

20-100 ppm Headache, dizziness, nausea and vomiting may develop, together with irritation of the eyes and respiratory tract;

odour becomes sweet then disappears as it kills the sense of smell; continuous exposure to 100 ppm for several hours

may result in death within the next 48 hours

100-500 ppm Breathing difficulties; continued exposure fatal

500-700 ppm Unconsciousness, immediate brain damage; continued exposure fatal

700-1000 ppm Certain death within 4 minutes

Table 1. Effects of H₂S exposure.

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DUTY OF CARE

Reporting systems

All contractors must have their own internal accident reporting system. The current reporting systems used to record OHS data within

NT Government are:

- Hazard Report
- Incident Report: to record all incidents including near misses
- Incident Investigation: to record investigation details and outcomes

% Managers/Supervisors must ensure that all relevant forms are easily accessible and that they are completed and processed in

accordance with the requirements of the OHS management system. They must also ensure that all employees are familiar with the

forms and their responsibilities to use them. First Aid Kit - Contents List form ensures that kits have adequate materials and records

use of first aid materials. Forms are available at:

http://uluru.nt.gov.au/dbird/ohs/pages/management_system.htm#ohsmgtsystem

All completed forms are to be returned to the supervisor.

An external NT Fleet Motor Vehicle Accident form is raised when any NT Fleet vehicle has been damaged. As well, an external

DCIS Accident Report is raised when there is an injury to an employee and any of the following conditions apply:

- Employee(s) involved will be absent from work for one full day/shift or more (lost time)
- Hospitalisation is required
- The attention of a medical practitioner (this includes community nurses) is required
- Workers Compensation will be, or has the potential to be claimed (medical, pharmacy, physiotherapy, rehabilitation, etc)

Hazard Report

Any hazard that has the potential to cause injury or illness to an employee, visitor, contractor or member of the public

must be reported and assessed using the prescribed Hazard Report form.

Incident Report

All incidents and any workplace-related illness must be reported on the Incident Report form. If the person involved is

unable to do so, the supervisor or witness will complete the form as soon as possible. Any other witnesses should

independently complete forms.

Statutory accident reporting

The HR Manager is required to report all prescribed accidents to the Work Health Authority within 24 hours, followed by

a written report within 7 days. An external two-page DCIS Accident Report Form is used for this.

The standard NT Government definition of a prescribed accident is when:

- A death occurs
- The accident is likely to result in more than 5 days lost time
- A worker suffers an electrical shock
- Exposure to a hazardous substance results in admission to hospital
- The accident results in injury to a person other than a worker
- Overturning, collapse or failure of a lift, crane, hoist, lifting gear or scaffolding occurs (this would include the rig mast)
- Failure of pressure equipment is involved
- A height of more than 1.5 m of an excavation or shoring collapses
- Part of a building or structure collapses
- Fire or explosion results in normal work being stopped for more than 24 hours (see added comment below)
- An accident involves plant coming into contact with live electrical conductors, or
- Personal protective equipment fails affecting the health and safety of a person.

In the case of drilling we need to add:

- Any drillhole flows uncontrolled
- Any combustion of material in or flowing from a drillhole occurs

The scene of any such accident must be rendered safe, but otherwise disturbed as little as possible. Photographs may be advantageous. A safety inspector will probably need to visit the site.

Vehicle accidents

It is a legal requirement that all motor vehicle accidents resulting in bodily injury or damage are reported to the police as soon as possible. If an NT Fleet vehicle is damaged, an external NT Fleet Motor Vehicle Accident form must be completed and forwarded to the Office Services Manager. This form is in addition to the internal DBIRD Incident Report and Incident Investigation forms, which still have to be completed.

Qualified first aiders

The drilling Contractor's crew should include a designated first aid officer who has a minimum Senior First Aid qualification.

This should be documented in the contract and records kept on site. All NTGS personnel on the rig site should have current similar qualifications.

First Aid Kit - Contents List and first aid reporting

The First Aid Kit - Contents List ensures that kits have adequate materials and records use of first aid materials. Managers and supervisors are to inform all employees of the location of the nearest first aid kit and the nominated First Aider. Where applicable, they are also to include a copy of the procedures to be followed with all first aid equipment. The Manager of the work area will ensure that a procedure is implemented to restock the First Aid kits on a regular basis. If a situation arises whereby an employee requires first aid treatment, the Manager should record the materials used from the first aid kit on the form provided inside the kit. The nominated First Aider will collect and forward these forms to the HR Manager for collation and review of first aid usage. This procedure does not replace the requirement to notify all incidents to your Supervisor as soon as possible.

Casualty evacuation and general rig evacuation

% As part of the NTGS Field Plan, casualty evacuation and general rig evacuation procedures must be formulated in consultation with the drilling Contractor and submitted to the NTGS Field Supervisor before drilling commences. A Royal Flying Doctor Service-approved airstrip has to be nominated for evacuation. Ideally, there should be all-weather access to and from the airstrip. If not, contingencies for helicopter evacuation should be put in place. Remember that not all helicopters can carry a stretcher.

Safety induction

% All rig-site personnel should receive a safety induction specific to the rig and location before their first shift. This includes

all NTGS, contract and subcontract personnel. This should be documented in writing.

Emergency communications

% In addition to the satellite phone/fax on site, the NTGS Field Plan should include at least one form of standby communications

that can be relied upon in an emergency. This may be VHF/UHF radio or the drillers' satellite phone. Note that if vehicle-

mounted satellite phones are used there must be at least one such NTGS vehicle on site at all times.

CORE DRILLING

INTRODUCTION

Coring uses a rapidly rotating (350-1000+ rpm) thin-walled drillstring and an annular bit to cut a solid sample. The volume of

the annulus in a cored hole is only one tenth that of a conventional rotary-drilled open hole of the same diameter.

In conventional coring, as done on a petroleum rig and for some bottom hole cores (eg waterbores), the core is retrieved by

pulling out the entire drillstring to get the core barrel at the end. This technique was pioneered in engineering and mining and

first introduced to the oilfield in the early 1920s, where it remains prohibitively expensive and is used sparingly, mainly for

reservoir evaluation. Petroleum core barrels come in 30', 60' and 90' lengths.

Continuous coring uses an overshot on a wireline run inside the drillstring to retrieve the core in an inner tube and has

become the standard of the minerals exploration industry. There have been over 200 different core diameters used in the

minerals industry and for slimhole petroleum coring. Many sizes are obsolete. Some are specific to a country, for example,

South Africa or Canada, and rarely seen in Australia. There are as many as ten in routine use in any one place at any one time.

The core and hole diameters commonly used in Australia over the past 40 years are shown in Table 2. Hartley (1994) gives a

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more comprehensive listing of 150 Australian types. In the old DCDMA system, the first letter refers to range of hole size. The

second letters X and W are group letters indicating that all similar letter equipment (eg HW and NW) are complementary for

decreasing hole diameter. DCDMA 'X' is now referred to as 'WG'. Q indicates wireline, J and other letters are thread types and

the remaining letters and numbers are barrel type, bit type, rod thickness or metric hole diameter. For example, in HQ3, H

refers to a hole diameter of about 60 mm, Q indicates that it is wireline and 3 denotes triple tube. Note that different manufacturers

claim different specifications for the now largely obsolete AWL core. For example, Bradley's is 27.0 mm (others AQ), Hartley

(1994) gives 29.59 mm and Longyear's is 30.1 mm (others AX or AWM).

The core bit may be face set with diamonds, or diamonds can be impregnated throughout a matrix that is designed to

progressively wear away. Other core bits have cutters or disc-shaped insets made of PDC. This polycrystalline diamond compact

is synthetic diamond powder bonded into tungsten carbide. Some specialist core bits use tungsten carbide chips. A selection of

such bits is shown in Figure 3. Figure 4 shows a typical diamond bit in cross-section with the components labelled. Traditional

diamond bits cut using a combination of crushing and scraping; most of the other types have a greater shearing action. The

circulating fluid is usually a polymer-based liquid. Diamond coring is normally at least three times more expensive than any

other method and 5 m/hr is a realistic overall average penetration rate for holes <1000 m. Figure 5 shows the components of

a core barrel and associated equipment. Core barrels come in a variety of lengths. The most common contain 1 m, 3 m or 6 m

of core. The more efficient 6 m barrels are routine in deep holes where there are few drilling problems. The smaller barrels are

used when only a spot core or bottom-hole core is required, or when coring is difficult. Normal practice in wireline coring is to

have at least two inner tubes on site so that one can be dropped and drilling resumed while the other tube is being emptied. The

interval drilled to fill the tube is called a run.

Air coring is a combination of RC and diamond coring in which a special bit (similar to that shown in Figure 3c) cuts a

small-diameter core which is air-lifted up the inside of the drillstring. This technique is suited to sticky clays and semi-consolidated

rocks with hard bands. Overall penetration rates are typically about 15 m/hr.

Size Core diameter (mm) Hole diameter (mm)

AQ, AQ-U 27.0 48.0

ATW 30.3 48.0

AWG, AWM, AWL, AX, AXM 27.0, 29.6 or 30.1 48.0

BTW 42.0 60.0

BQ, BQ-U, BQWL 36.5 60.0

BQ2.32 38.6 58.9

BQ3 33.5 59.9 or 60.0

BW, BWG, BWM, BWL, BX, BXM 42.0 59.9

CHD101 63.5 101.3

CHD134 85.0 134.3

CHD76 43.5 76.3
 EW, EWG, EWM, EWL, EX, EXM 21.5 37.7
 HQ, HQWL 63.5 96.0 or 96.1
 HQ3 61.1 96.0 or 96.1
 HQ3.18 66.2 93.5
 HW, HWG, HX 76.2 99.2
 LTK46 35.6 46.2
 LTK56 45.2 56.3
 NQ, NQ-U, NQWL 47.6 75.7 or 75.8
 NQ2 50.7 75.7
 NQ3 45.0 or 45.1 75.7
 NTW 56.0 75.7
 NW, NWG, NWM, NWL, NX, NXM 54.7 75.7
 PQ, PQWL 85.0 122.6
 PQ3 83.0 or 83.1 122.6
 RWG 18.7 29.8
 2¾" x 3" 68.3 98.4
 4" x 5½" 100.8 139.6
 6" x 7¾" 151.6 196.9

Table 2. Common diamond drillcore and hole diameters.

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a b c d

e f g h

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Figure 3. There is a wide variety of different types of core bits using diamonds, tungsten carbide or PDC, and the correct bit must be chosen

to optimise penetration and bit life for specific formations and drilling conditions. (a) Impregnated flat-crown diamond bit, probably the most

versatile and commonly used type in mineral exploration. More diamond fragments are exposed as the matrix is worn away, so that these bits

have a long life when drilling homogeneous material; these bits are disposable. (b) Surface-set, step-faced diamond bit for hard sedimentary

formations. This is an older design and the bit has to be pulled once the exposed diamonds are worn away, but such bits are then reset and

reused: an eight-step bit would typically make at least 450 m at >2.5 m/hr in average Proterozoic or Cambrian carbonate and fine siliciclastic

rocks; 600 m at 3.0-5.0 m/hr is achievable in the best drilling conditions. (c) Carbide chip bit can be used to core sedimentary rocks, but is

more usually used to clean metal junk out of a hole. A similar design is used for air core. (d) Large tungsten carbide saw teeth on this bit are

used in coring very soft rocks in geotechnical investigations. (e) Ridge-set, round-crown natural diamond bit for hard, dense, moderately

abrasive rocks. (f) Diamond bit for general purpose coring in medium to hard rocks such as limestone. (g) Small cutter PDC for medium to hard rocks. (h) Medium-set PDC with large fluid courses internally and externally, designed for soft to medium rocks. Images courtesy of Baker Hughes Christensen and Dimatec.

A.P.I.

Shank Bore Connection
(Upper Section)

Pin Chamfer

Bit Breaker

slots

(Hairpin)

(Wraparound)

Junk Slot Alignment

Threads

Weld

Gauge

Chamber

Gauge Gauge

Broaches Steel section

Blank

Diamond Pad Gauge

with Diamonds Point

Crowfoot O.D.R.

Low Pressure

Collectors Matrix

Taper

Cone

Figure 4. Partial section through a typical diamond bit, with component

High Pressure

Feeders features labelled.

Cone Angle Nose m04-066.dgn

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Figure 5. Components of a core barrel. Image courtesy of Longyear.

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Ordering core trays

% At present, either NTGS or the drillers can supply the core trays. Normally, unused core trays provided by the driller are non-

returnable. Trays should be ordered, with the assistance of the DBIRD Core Library Manager, well in advance of mobilisation. Note

that the DBIRD core libraries use 400 mm x 1000 mm trays and that this size may not be standard in other states. Also remember to obtain core trays appropriate to the diameter of the core; for example, NQ2 core is too tight in NQ trays. At the very least, lids will be required for trays that will be stacked as the top layers of a pallet. However, it is considered standard practice for every tray to have its own lid. The nominal capacity of trays is given in Table 3. Allow about 5% extra for drillers' blocks and gaps between core pieces.

Core size Nominal metres per tray

BQ 9

NQ 7

HQ 5

PQ 3

Table 3. Nominal capacity of core trays for different core sizes; allow 5% extra when ordering.

DRILLING PROBLEMS WHILE CORING

Dropped core

Unconsolidated, brittle or fissile rocks can all cause problems with dropped core. Such core falls from the core lifter in the base of the inner tube as the tube is being recovered. The core falls to the base of the hole and has to be overdrilled during the next run. It is difficult to redrill because the bit tends to spin it and there is a tendency for this piece of core to lodge within the inner tube causing a wedge off. In the early days of coring, dropped core had to be crushed up with a rotary bit before coring could resume. This often necessitated dozens of round trips to change back and forth between tricone and diamond core bits. Even today's core bits will wear prematurely if repeatedly used to overdrill dropped core. If there is no indication that the formation is the cause of dropped core, equipment malfunction should be suspected. Core will be dropped if the inner tube fails to seat, if an inappropriate type of core lifter is used, or if the core lifter is bent or worn.

The degree of wear can be easily checked at the surface using a piece of in-gauge core. It is often difficult to snap hard, competent rock such as massive limestone at the end of a run. Core of such rocks commonly has chatter marks left by the teeth on the core lifter skidding up the core. These are a sign that the lifter is under stress and may bend or wear prematurely.

Wedging off in the inner tube

Faulted, fractured or overpressured formation and overdrilled core commonly become wedged inside the inner tube as a run is being drilled. This is termed 'wedging off' and necessitates a short run. Split liners and triple tubes are designed to help overcome this problem.

Poor quality of core

Overdrilled core occurs where the base of the cut core falls from the core lifter and gets redrilled at the start of the next run. A

sudden change in core diameter will occur when a worn out bit is replaced or where there was a sudden change in pressure on

the bit. Corkscrewed core of variable diameter is a sign of excessive bit pressure. Breaking in a new face-set bit too quickly will

also result in inferior core.

CORE HANDLING PROCEDURES

Preparing core trays

% To prepare trays for drillsite use, paint one long side and, as viewed facing this painted long side, the top left lip and bottom

right lip, using two-pack white epoxy paint (Dulux Durebild STE or equivalent). Personal protection equipment must be used

in accordance with the Materials Safety Data Sheet(s) for the particular brand of paint being used.

When the paint is dry, write

'Start' on the painted top left lip of the core tray using a black paint marker (Artline 440XF or equivalent). The tray is now

ready to receive core (see also Labelling core trays).

Lining core trays

% The galvanising on metal core trays as used in NTGS core libraries is prone to abrasion by core during transit. Some

sulfides (Figure 6) and evaporites, if present in the core, break down to produce acid and salt which attack the tray. Aside from

the damage to the trays, this can result in significant contamination of the core, particularly by zinc. To minimise this, trays

containing suspect core should be lined with plastic sheeting.

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Figure 6. The good, the bad and the ugly. (a) Core interval from an intersection of sulfide ore that contains reactive pyrite, photographed

immediately after drilling in 1988. (b) The same core interval after slabbing and storage, photographed in 1996. (c) Another example showing

extreme breakdown of sulfides. The white material is gypsum, formed by the reaction of calcite with remobilised sulfate; this reaction also

generates acid, which attacks the metal core tray, liberating Zn from the galvanised coating. The Zn is scavenged by the gypsum and the

altered rock, so that geochemical analysis of this material is misleading.

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Core catching

There are several methods whereby the driller can lay out the

recovered core. In the petroleum industry, where every Safety note conventional core costs tens of thousands of dollars, each core On no account is anyone to hold their hand under the run is laid out on a suitable rack (two lengths of pipe tied together end of the inner tube or put their fingers up the end of the over two 200 L drums are adequate). Normally the core is tube at any time. Safety glasses must be worn if the core is recovered in a disposable liner which is cut away. There is no being broken. possibility of getting pieces of core out of order or upside down. The inner tube of a triple tube system should be pumped Different practices are used during continuous coring in the out using liquid, never air. An appropriate bumper must be minerals industry. If a spilt inner tube is used, the splits function placed at the bottom of the tube to prevent core flying out similarly to a liner and provide a platform for the core to be under pressure and noone is to stand within 5 m of the tube examined and marked up. The core can then be slid directly while pumping is in progress. into the core tray in the correct orientation. Otherwise, when using a conventional barrel, normal practice is for the driller's assistant to deposit the core directly into the trays while the tube is still suspended by the wireline. This is fraught with problems and is not recommended. Firstly, the core comes out of the tube bottom first, so the assistant has to judge how far into the empty tray to begin depositing the core. It is normally necessary to hammer the core lifter with a rubber mallet to free the lowermost piece of core protruding from the tube. This invariably flies off and has to be reoriented. Once freed, careful coordination is necessary to prevent the remaining core falling uncontrolled from the tube. The driller has to feed out the wireline from the winch and the assistant move the tube along the segments of the trays while keeping the lower end of the tube close enough to the tray to prevent core spilling out. If the core is recovered in long solid sticks, the assistant has to juggle the tube while using a hammer to break the core at the start of the next core box segment. Not only does this risk misorienting core but it is potentially dangerous (see Safety note). Most drillers have a tendency to overfill each segment of the tray, making it difficult to get the core out again. % Preferably, when using a conventional barrel, onsite NTGS staff should ensure that core is deposited onto a suitable rack for marking up and is then transferred from there to the trays by a trained NTGS Technical Assistant. If – and only if – this is not feasible, core may be dumped directly into trays. Drillers should be closely supervised during this procedure for

each and every run. (This is one of the reasons for NTGS to maintain a technically competent presence on site during drilling).

Edge matching and marking-up of the trayed core will then need to be done, metre by metre.

Irrespective of barrel type, NTGS practice is to load core trays from the top left ('Start'), so that contained core runs

continuously from shallowest at top left to deepest at bottom right when viewed facing the painted long side of the tray.

Edge matching, washing and core orientation lines

A standard convention of marking the core longitudinally:

- Enables broken edges to be easily rematched (between and within runs), ensuring accurate recovery measurements
- Provides a datum for relative measurement of sedimentary features and metamorphic fabrics (eg relative directions of crossbeds, orientation of cleavage)

- Means that large sections of the core are already oriented when an absolute determination of orientation is undertaken (see

Determining absolute core orientation and measuring true dip and strike)

- Ensures that composite sampling of core is consistent relative to a longitudinal frame of reference
- Ensures that core can never be returned to the tray upside down

% The last piece of marked core from the previous run should be left on the core rack or near where the core is being examined

in the boxes. Immediately after removal from the inner tube the new core should be laid out starting with the bottom piece from

the previous run, so that all the ends are matched. The new core is then washed. A nylon brush may be used but cloth is to be

avoided since lint fluoresces under UV light. Once dry, the core can be marked up. If there is no apparent fit between adjacent

pieces and it is certain that pieces of core have been laid out in the correct order, two arrow-up chevrons are marked on either

side of the cores that do not match. All core is to be marked with permanent continuous longitudinal orientation lines using

the AAPG standard of black line on the right, red line on the left when core is viewed in correct orientation (but see Sealing

oil-soaked core). Two marking pens taped together are used (Figure 7b). The tips of the marking pens will wear rapidly, so be

sure to have adequate supplies. Chisel-tipped pens last longest.

Labelling driller's blocks

% The drilling Contractor should label driller's blocks with end-of-run metreage (Figure 7c), depth of change in core diameter,

absolute orientation points and end-of-hole metreage. Be sure that it is clear in the contract as to who is supplying the blocks.

Wooden blocks or plastic markers are acceptable. The latter have the disadvantages that they must be the exact width for the

core tray and are easily dislodged by hosing the core or if open trays are left out in high winds. An approved permanent marking

pen is to be used. Alternatively, inscribed aluminium tags can be stapled to the wooden block. All labels are to be oriented to

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read such that the top of the label is toward the top of the hole. Labels must be legible from eye height when the core tray is on the ground.

Dealing with lost core

% In the event of less than full recovery, drillers assume that core loss occurs at the top of the run and will put in blocks

to indicate that. In reality, the lithology and edge matching will usually locate the actual interval of loss. If this is not

possible, a better convention is to assume, at least initially, that any unrecovered core comes from the base of the interval

cored. This is because with continuous coring, the core usually breaks some centimetres above the total depth to which it

was cut. This will appear as core loss in one run, but the next run may recover it (ie the second run recovers more core

than the actual length of the run drilled). If these few centimetres not recovered accumulate over successive runs and the

accumulated total is then successfully picked up, it may completely fill the inner tube before the length of the inner tube

has been drilled. After this, the apparent core losses can be rationalised. Be sure to check that driller's depths on core

blocks are not being 'fudged' to account for discrepancies in recovery between runs. The graphical geological log (see

Describing core) should show the core loss as interpreted by the Geologist not the driller.

Measuring and depth labelling

% The full length of each core recovered is measured and recorded in the appropriate column on the drilling log. Depths

are written on the core, between the red and black lines, every full metre. Labels may face so as to read uphole or from

top right to bottom left when the labelled side of the core tray is facing the viewer (as in Figure 7c). Every half metre

may be marked by either writing the depth or putting a tick mark between the red and black lines. Depth marks for 100%

recovery should approximately line up along a line from upper left to lower right in a tray (Figure 7c). Top and bottom

of intervals of substantial core loss should also be so labelled.

Core sample depths

% Sample depths are almost always given as driller's depth to correspond to labels on core. This applies even if there is

a discrepancy with wireline depths or if the original depth labels are subsequently proved wrong. Sidewall core points, which are picked from wireline logs, are the only exception.

In most onshore petroleum wells and larger mineral rigs on a platform, the Kelly bushing or slips are used as datum

and depths may need to be so specified or corrected. For NTGS drilling, the depth datum will normally be the ground

surface. The driller should be advised of this prior to commencement of drilling.

Arrow up

% Arrows on samples always indicate stratigraphic-up. In the case of overturned bedding this is not correct with respect

to the present frame of reference but any ambiguity is resolved by adding an overturned dip and strike symbol. Arrows on

core from underground drilling are not necessarily up-hole since such holes can drill up, rather than down, stratigraphy.

In contrast, drillers tend to use arrows to indicate a downhole direction on the start of core trays and driller's blocks.

Labelling core trays

% NTGS uses a standard layout of labels in its core libraries and this is to be followed on site (Figure 7). Labelling of

pre-painted core trays (see Preparing core trays) must be done using black paint markers: Artline 400XF or equivalent

for sides of trays, and finer Artline 440XF or equivalent for lips of trays. Neat, legible labelling is essential.

Once a tray is filled and its contained core measured and marked up, label the painted top left lip ('Start') with the

interval start depth and the diagonally opposite, painted bottom right lip with the interval finish depth, so that these are

visible when the tray is photographed. The white-painted side of the tray is then permanently labelled with (from left to

right) tray (box) number, location, drillhole name, and start and finish depth. Remember to leave a 75 mm blank space

on the far left for rack location numbers to be added later in the Core Library, and to leave a gap along the top so that the

labels will not be obscured if a lid is added to the tray. Example:

[75 mm blank space] 1 (tray number) Golden Grove (location) NTGS DD04GG01 (hole name)
100.0-105.5 m

(interval)

Determining absolute core orientation and measuring true dip and strike

% Even if the azimuth and declination of the hole are known (see ENGINEERING), and the long axis of the core

marked with continuous red and black lines, the core is still unoriented relative to the hole. Rarely, a penetrative fabric in

the core (such as cleavage) can be matched to nearby outcrop. More usually, this has to be done by using a special core

lifter that scribes a line along the core and various downhole devices that use gravity to orient the core. These instruments

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only work in holes with declinations of less than -75° (ie greater than 15° from vertical). The recommended device is the

simplest. After the core tube has been pulled, a heavy metal spear is lowered down the hole on a wireline. The tip of the spear

makes a percussion mark on the lowest point of the rock yet to be drilled (Figure 8). When the percussion-marked core is

recovered in the next run, it can be oriented with the mark at the lowest point of that end of the core. In very hard rocks, a wax

pencil can be added to the tip of the spear. Spear marks will not work if the core has broken in a jagged manner. Pencil and

percussion marks can be easily destroyed if the ends of core rub together during transport. Ensure that an appropriately

labelled driller's block separates them and a permanent ink triangle should be marked on the outside of the core opposite the

orientation mark. Repeated trials by the author have demonstrated that in a hole declined -70° (ie 20° from vertical), 90% of the

spear marks fall within 5° either side of vertical from the core axis.

When both hole and core orientations are known, features within the core can be accurately oriented. This can be done

mathematically using trigonometry, a Wulff stereonet, specialist software or by physically orienting a section of the core in

special jig or a sandbox. True dip and strike should be calculated this way on site as part of routine core logging.

b

a

c

98.0 m

99.0

100.0

101.0

102.0

103.0

BOX 9 NTGS VR04DD01 96.5 - 103.01m

m04-073.dgn

Figure 7. Correct labelling of core and core trays. (a) The start of the tray should be labelled in advance, with start depth recorded on white-

painted corner of lip as indicated. (b) Core is marked with continuous red and black lines using two marking pens taped together; black line

is always on the right when the core is stratigraphically oriented, and full and half metres are labelled between the red and black lines. (c)

Depth labels on core in a tray with full core recovery should approximately line up diagonally from top left to bottom right (in the illustrated

orientation). Note that the driller's block reads from the same side as the core depth labels; the block and all other labels should be legible from

eye height when the core tray is on the ground. A specimen label for the white-painted long side of the core tray is shown at bottom.

18

BO

TT

O

M

DR

IL Percussion Mark

HE

L

AV

Y

HO

CO

Wire Line attachment

LE

ST

RE

EE

L

SP

EA

R

Rock stub is top

of next Core run

END VIEW OF CORE

Core Centre

Trace of Vertical Plane

Percussion Mark m04-062.dgn

Figure 8. A percussion spear is used to make an orientation mark on core. Modified from unpublished work by RW Marjoribanks.

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Recording maximum dip in core

% In situations other than those described above, it is important to routinely record the maximum dip in core. Note that some mineral companies term this the acute bedding to core angle. This is done by removing a section of bedded core from the tray and rotating it such that the maximum dip can be measured (ie the strike is at right angles to the line of sight) (Figure 9).

Measurements should be made nominally every 5-6 m (ie once in each core tray) and at every significant change. In most

cases, the maximum dip can be used as a proxy for true dip even if the strike is unknown. The maximum dip is used to estimate

true vertical thickness of a formation.

Calculating true stratigraphic thickness and Stratigraphic Efficiency

% Ideally, a stratigraphic drillhole should penetrate bedding at right angles and the bedding thickness in core will be a true

stratigraphic thickness. In reality, hole deviation, faults and folds complicate the situation. The true dip (or maximum dip as a

proxy) and hole deviation are used to calculate true stratigraphic thickness. Stratigraphic Efficiency is the ratio of equivalent

true stratigraphic thickness to metres drilled, expressed as a percentage. There are several freeware software packages that will

calculate these. A graphic representation and manual calculation are shown below.

□ Example 1

A vertical hole that has a true dip of 60° in the core will actually drill double the true stratigraphic thickness (Figure 10). This

is a stratigraphic efficiency of only 50%.

Hole deviation (see ENGINEERING) introduces an extra complication. In this case, the true stratigraphic thickness of a

formation is calculated as follows:

$$T = AB (\sin a \times \cos b - \cos a \times \sin b \times -\cos c)$$

where: T = true thickness of formation

a = declination of hole expressed as the angle from vertical

b = true dip of formation

c = angle between direction of formation dip and direction of hole

AB = thickness of intersection in hole

□ Example 2

Target formation True dip from horizontal = 60°

Dip direction = 130°

Hole Declination (deviation from vertical) = 10°

Dip direction (azimuth) = 110°

Thickness intersected in hole = 100.00 m

$$c = 130^\circ - 110^\circ = 20^\circ$$

$$T = 100 (\sin 10 \times \cos 60 - \cos 10 \times \sin 60 \times -\cos 20)$$

$$= 100 (0.1736 \times 0.50 - 0.985 \times 0.866 \times -0.940)$$

$$= 100 (0.0868 + 0.8016)$$

$$= 88.84 \text{ m}$$

Stratigraphic Efficiency is 88.84 m of true thickness for 100 m drilled, so 89%.

Photographing core

% All core must be photographed on site. This is recommended practice under formal state and federal exploration guidelines

where such legislation exists. Such photographs provide a useful record of the pristine core before transport and deterioration

due to mineral breakdown, natural fracturing and clay expansion (Figure 6). The photographs are invaluable should a core tray

be accidentally dropped and have to be repacked. It also prevents deliberate tampering. Usual practice is to photograph core

trays in lots of two with a blackboard or similar showing the hole name and depths. The core is normally photographed wet in

natural light, avoiding any distracting shadows. A standard colour scale is to be included. To enable the trays to be framed with

minimum distortion by parallax, a zoom lens is recommended and the photographer should stand on a sturdy support. For best

results, a frame can be constructed to support the camera at a constant focal length and an infill flash and cable release

incorporated.

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m04-072.dgn

Figure 9. Measuring maximum dip or acute core-to-bedding angle in core. Rotate the core so that the strike is at right angles to the viewer and

use a protractor to measure the maximum dip angle.

Describing core

% Core logging is normally done on site by filling in a pro-forma core description sheet (Appendix 4) by hand at a vertical

scale appropriate to the level of detail (normally 1:100 in stratigraphic holes). This form can be modified to include details

specific to the project. This will probably be the most comprehensive record of the entire core ever undertaken by NTGS. It has

to be legible for inclusion in the NTGS Technical Note and dark enough to photocopy. It will be simplified for later entry into

geological software for use in the final Drillhole Completion Report.

Be objective when logging core; it is easy to allow personal preference for a particular geological discipline to bias the

description. Petrology, structural geology, economic geology and palaeontology are all important. Remember, we are logging facts, and where interpretation is given, it should be so noted. Core is normally described in natural light and wet. However, the internal features in some carbonates and shales become less obvious when the core is wet. There are fewer tricks and pitfalls when describing core as opposed to cuttings, but many a geologist has logged 'conglomerate' or 'massive limestone' when a cement plug, with or without cavings, has been drilled. Phenolphthalein stains cement purple. For examination, individual core pieces should be removed from the tray and all sides examined. Look at the ends of core! The author knows of a supposedly Proterozoic type section core which had been logged in centimetre detail on site and examined by dozens of other geologists in the Core Library, all of whom missed trilobite fossils only visible on the ends of the core. Core will break along natural fractures and veins. These surfaces should be checked for epigenetic mineralisation.

Packing and palletising core trays and their weight

% NTGS will provide special framed steel pallets designed to carry core trays. Once packed with trays and fully assembled, these should be strapped at least twice in both directions. Where these special pallets are not available, Figure 11 shows how to pack core trays on a pallet. Wooden pallets should normally be used in this case. Again, packed pallets should be strapped at least twice in both directions. Note that the labels are visible and that there are only two trays (of HQ size) per layer. In either case, it is considered standard practice for every tray to have its own lid affixed. Use Table 4 to estimate weights of core. If the 31 trays shown in Figure 11 were full of HQ sandstone core, they would weigh 1.2 tonnes. About 220 m of NQ dolostone core, equating to 32 trays, weighs 1 tonne; 123 m of HQ weighs 1 tonne. A single 3 m tray of PQ lead-zinc ore is almost 60 kg, with only 17 trays per 1 tonne pallet.

Slabbing core and composite core sampling

% Core sampled for multielement geochemistry should be quartered lengthways using a diamond saw, not mechanically split or edge ground. Properly done, this is not a trivial exercise. Samples have to be consistently oriented with respect to the red and black core marks such that all the quarter cores should be able to join end to end. Ensure that core with depth labels and longitudinal lines is returned to tray. Composite sample intervals should not span intervals of lost core and intervals should be adjusted according to core recovery. For example, including intervals of 100% and 70% recovery in the same composite is contaminating good sample with bad.

HYDROCARBON DESCRIPTION

There is an old saying that 'oil is where you find it'. % All stratigraphic holes should therefore be monitored for the possibility

of hydrocarbons, irrespective of whether they are in a known petroleum basin or not. There are several mineral exploration

21

m

60

50

T=

TS

100m

200m

drilled thickness =

100m

m04-061.dgn

Figure 10. A vertical hole that intersects bedding with a true dip Figure 11. NTGS has special frames for transporting pallets of

of 60° will actually drill double the true stratigraphic thickness. core. If these are not available, the pallet should be packed as shown.

This is a Stratigraphic Efficiency of only 50%. Note that the labels face out and there are only two trays per layer.

The pallet needs to be strapped twice in both directions.

Material BQ NQ HQ PQ

clay 1.945 3.308 5.887 10.549

limestone 2.813 4.784 8.515 15.257

dolostone 2.677 4.553 8.103 14.519

sandstone 2.531 4.304 7.660 13.725

quartzite 2.761 4.696 8.356 14.973

gneiss 2.813 4.784 8.515 15.257

slate 2.928 4.980 8.863 15.881

basalt 3.137 5.336 9.496 17.015

granite 2.845 4.838 8.610 15.427

iron ore 4.497 7.648 13.611 24.388

porphyry copper ore 2.719 4.624 8.230 14.746

BHT lead-zinc ore 3.660 6.225 11.079 19.851

Table 4. Weights in kilograms per linear metre for each common core diameter and lithology. Other core diameters can be calculated using

proportions derived from the core diameters listed in Table 2. BHT = Broken Hill-type.

holes in our Core Library that have intersected oil bleeds without them being recognised at the time. NTGS intersected oil

bleeds in Victoria Basin despite this not previously being recognised as a petroleum basin. Never disregard a potential show

because of the host lithology. There are commercial oil reservoirs in fractured metamorphic and igneous rocks!

Signs of gas in core

Core of a gas-filled reservoir usually has a noticeable odour; there

Safety note

may a core flash (igniting gas being liberated from the core), or gas

The driller must be immediately notified of any

may bubble from drilling fluid or water surrounding the core (most

signs of oil or gas as this indicates a potentially

obvious when the core is removed from the inner tube but can persist

dangerous situation on a rig without spark-arrestors

surprisingly long – up to several days). Most gas reservoirs will also

or a blowout preventer.

have a bluish-white fluorescence under UV light.

Oil shows in core

% NTGS drilling is most likely to detect oil shows in core rather than cuttings. The following section explains how oil shows

are to be described and how the core is to be treated. The driller should be informed immediately that oil is detected.

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Oil bleeds

Most oil bleeds are obvious to the naked eye and have a distinctive odour. In natural light, oil can range in colour from opaque

very dark brown to colourless. Figure 12 shows typical oil bleeds from Cambrian rocks. Light oil is usually first detected in the

drilling fluid surrounding the core by iridescence and surface tension affects. In tight reservoirs, even light oil bleeds may not

appear until minutes or hours after the drilling fluid is wiped off. Such phenomena should be fully described, such as ‘straw

coloured live oil bleeding from hairline fracture at 456.8 m; first appeared 15 minutes after core removed from inner tube’.

Very light oil may dissipate even before the core is removed from the inner tube. Heavy oil is less mobile.

Sealing oil-soaked core

Core with significant live oil must be sealed to help prevent loss of volatiles and biodegradation.

This involves wrapping the

section of core in aluminium foil, several layers of plastic cling film and a final wrapping of aluminium foil held in place with

masking tape. Do not write directly on the core with marker pens – this may compromise any subsequent organic geochemical analyses. The masking tape is marked with the ubiquitous red and black lines and labelled with drillhole and depth. On a petroleum rig, these samples are then sealed by dipping in molten wax. Samples that contain live oil should be stored in a cool place.

Fluorescence

Benzol rings in a hydrocarbon cause it to fluoresce under UV light and a suitable lamp should be on site for all potentially

prospective holes. Three types of UV lights designed for use with hydrocarbons are available: a high-intensity hand-held UV

light, a box-mounted fluoroscope (also known as a viewing cabinet) and a corvascope (combination stereoscopic zoom micro-

scope and UV light). These operate in the range of 350-400 nm, nominally 365 nm, ideally with an intensity of $\geq 850 \text{ uw/cm}^2$.

Hand-held UV lamps designed for mineral prospecting come in various combinations of short (254 nm = 2537 Angstrom),

medium (302 nm) and long (365 nm) wavelengths. All will make hydrocarbons fluoresce, but the longest wavelength is

comparable with those designed for petroleum exploration. Most mineral prospecting lamps are of such low intensity that all

other background illumination has to be excluded. When using this method to examine core, it may be necessary to work at

night or under a blanket. Alternatively, a 'darkroom' viewing

cabinet can be used. Areas of fluorescence are marked with Safety note

chalk (not a solvent-based marking pen) so that they can be UV light will harm your eyes. Shorter wavelengths are the

relocated under natural light. more dangerous. Do not look directly into a source of UV

Colours of hydrocarbon fluorescence include green, gold, light and avoid working for more than 15 minutes at a time

orange, yellow, blue, blue-white or white. This colour range with reflected UV light. Normal eye glasses or transparent

broadly corresponds to heavy through to light oils. Oils of 35- safety glasses will reduce UV exposure.

45 API gravity fluoresce white to blue-white. Intensity is described

as bright, dull, pale or faint. Figure 13 shows typical oil-stained

core in UV light. Very heavy biodegraded oil may not fluoresce Safety note

in its natural state, but will after solvent extraction (see Cut and Various solvents have been used for rig-site hydrocarbon

solvent tests). Very light condensates may fluoresce beyond the detection. Those recommended in older manuals have been

human visible spectrum. Be wary of mineral fluorescence; calcite phased out because of health or environmental concerns.

and dolomite can both fluoresce blue-white, brown or red. Most Toluene, a common ingredient in petrol (up to 13% toluene),

greases and many other contaminants fluoresce (see glue and paint thinners, is currently the preferred choice.

Contamination). Toluene is variously known as C₇H₈, methylbenzene, methylbenzol, Methacide, phenylmethane, Toluol and Antisal

Cut and solvent tests 1A. Its international CAS Registry Number is 108-88-3.

Absorption through the skin or exposure to its vapour is

A 'crush cut' is the solvent extraction of hydrocarbons from a potentially dangerous. It can damage the nervous system and

crushed rock sample. Fluorescing cuttings or core chips are hand cause an irregular heartbeat. Workers with a history of asthma

picked and placed in a white porcelain dish (spot dish) and a few induced by solvent exposure should be warned. You can smell

drops of solvent added. Any hydrocarbons present will dissolve toluene when it reaches 290 ppb and it is considered safe to

and sometimes oil may be seen moving into the solvent even in work in levels up to 100 ppm. The Occupational Safety and

natural light. This is termed a 'streaming cut'. This phenomenon Health Administration has set a limit of 200 ppm of workplace

is best observed under UV light. In less than a minute, the cut air (690 times the detectable limit in humans). Toluene is

will impart a fluorescence to the solvent. As the solvent evaporates, heavier than air. Both liquid and vapour are flammable.

a residual ring of fluorescence is left in the spot dish. The solvent Contact with strong oxidisers may cause fire or explosion.

is also added to an identical spot dish without any cuttings. This The liquid can accumulate static charge by flow or agitation.

blind test, which should of course not fluoresce, checks for It should not be carried on aircraft. Proprietary detergent-

contamination in the solvent. It is also prudent to set up standards based substitutes (eg Sample Clean) are also available but

of all the possible contaminants on site. can be difficult to source.

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Figure 12. Examples of oil and bitumen in core. Scale bar = 1 cm.

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Figure 13. Oil-stained core photographed in normal light (bottom) and UV light (top).

Acid bounce test

Oil shows in a carbonate rock can be detected by immersing a suspect chip in dilute HCl. The CO₂ generated and the surface

tension of any oil will cause iridescent bubbles that buoy up the chip. As the bubbles burst at the surface the chip will sink

again. This test is overly sensitive; even slightly calcareous carbonaceous shale or drilling fluid contaminants may give a positive result.

Hot water extraction

Tipping very hot water (>75°C) over a suspected oil sample in core will extract some oil. After the resulting suspension has separated, a thin surface film of oil should fluoresce under UV light.

Standardised oil show description

NTGS uses a modification of the Wyman & Castano (1974) method of standardised oil show description. This method was developed for cuttings (chips) on a conventional petroleum well; but it also works with whole core and small chips broken from it.

% A description form (Appendix 5) must be completed on site. With core, the interval being described must be precisely stipulated. The following discussion will aid in the standardised show description.

Morphology

A textual description of the distribution of hydrocarbon indicators on a rock surface. Examples: irregular stain around vug, bleed from fracture, oil associated with bitumen on stylolite.

Mobility

The rate at which fluid emerges from the rock sample. Examples: immobile oil, live show bleeding when core was recovered and persisting for several hours.

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Percentage stain

What percentage of the fresh surface is stained when viewed in normal light?

% Stain Code

0 0

0-40 1

40-85 2

85-100 3

Hydrocarbon odour

How strong is the odour on the fresh surface? Strong would be the intensity of smelling the same amount of refined oil.

Odour Code

none 0

slight 1

fair 2

strong 3

Natural fluorescence

Note the amount, intensity and colour of hydrocarbon fluorescence under UV light. The percent fluorescence should be similar

to the percent staining recorded above. If fluorescence is restricted to fractures, vugs or mineral type, this should be noted.

% Fluorescence Code

0 0

0-40 1

40-85 2

85-100 3

The intensity of the fluorescence is qualitatively estimated. A benchmark reference sample should be retained.

Intensity of fluorescence Code

nil 0

weak 1

fair 2

strong 3

The colour of the natural oil fluorescence is rated as shown below. The colour approximates to its specific gravity.

Colour of natural fluorescence Code

none 0

brown 1

orange, gold, yellow 2

pale yellow, bluish-white 3

Colour of cut

Place equal volumes of chips and solvent in a non-fluorescing glass test tube. Shake and allow to settle before observing the

colour of the solvent in comparison to the chart in Figure 14.

Colour of cut Code

colourless 0

pale straw 0.5

straw 1

dark straw 1.5

light amber 2

amber 2.5

dark brown 3

very dark brown to opaque 3+

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Cut fluorescence

The fluorescence of the cut in the test tube is rated as below. It should reflect the code used above.

Cut fluorescence Code

none 0

slight (almost transparent when viewed through tube) 1

medium (translucent when viewed through tube) 2

strong (opaque when viewed through tube) 3

VERY

DARK

BROWN DARK LIGHT DARK PALE COLOUR-

OPAQUE BROWN AMBER STRAW STRAW STRAW LESS

3+ 3 2.5 2 1.5 1 .5 0

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Figure 14. Natural colour of oil and description code.

Show number

Wyman & Castano (1974) originally proposed that the mathematical averages of the codes for each show (their show number)

could be compared as a relative ranking.

Spurious hydrocarbon indications

It is easy to mistake other phenomena for indications of hydrocarbons. % The NTGS site

Representative must be aware of the

possibility of false hydrocarbon indications.

Drill gas

Drill gas is liberated from rock (generally a rich source rock) as it is crushed during drilling. The low circulating volume of a

cored slimhole means that quantities of gas can be sufficient to result in a quite rapid order of magnitude increase above

background. A small increase in drilling fluid relative density (RD) while drilling will not decrease drill gas, but if the hole is

circulated without drilling ahead, drill gas will decrease rapidly and may disappear entirely. That is, drill gas will not flow from

the drillhole.

Numerous slimhole well completion reports have credited drill gas as a show, but it should never be reported as such. As

Whittaker (1987) has said: gas flows from a well; it is not mined.

The situation can be further complicated since source rock responsible for drill gas also commonly contains unmigrated oil

and the drill gas will occur in association with heavier hydrocarbons that are genuine shows.

Trip gas and connection gas

When the circulation pump is stopped for any reason such as trips or connections, a greater amount of gas can flow into the

drilling fluid. Upon recommencement of circulation, this gas shows up at the surface as an increase. If such surges can be

lagged back to the resumption of circulation, trip gas or connection gas should be suspected. Such increases should be documented

but are not gas shows as such. Clear indications of connection gas or an increasing trend in trip gas are a sign of increasing

formation pressure and therefore of concern.

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Biogenic methane

A false gas show can occur when methane-bearing groundwater is intersected. As discussed separately in DRILLING PROBLEMS,

quantities of gas may be sufficient to be a safety concern. A few northern Queensland waterbores have considerable biologically

generated flammable headspace gas. Several people have lost a bet with a local publican (and the hair off their arm) when they ignite

the gas bubbling from a bore used for the pub ablution block. Since such gas is biogenic, not thermogenic, it is not strictly a gas show

in the sense used by petroleum explorers. Isotopes can be used to distinguish the two types of gas.

Bitumen

Bitumen, loosely any solid or semisolid hydrocarbon, is arguably a genuine hydrocarbon show and should be recorded. Bitumen

usually has nil to moderate-intensity dull brown (commonly spotted) natural fluorescence. It gives nil to moderate streaming white to

yellow white cut fluorescence.

Carbonaceous material

Carbonaceous matter may have faint natural fluorescence. It will not yield a crush cut with toluene. However, depending on the type

of organic matter, some superseded solvents such as acetone and carbon tetrachloride will give misleadingly intense cut fluorescence.

These are not shows but may be reported as such in well completion reports, particularly those from the 1960s and early 1970s.

Stylocumulate

Carbonate rocks commonly contain organic-rich stylocumulate. Comments similar to carbonaceous material (above) apply. In cuttings,

stylocumulate can be mistaken for carbonaceous shale. Stylocumulate can be misleading since it can have mineral fluorescence due

to secondary minerals and the organic matter itself may have a faint natural fluorescence.

Furthermore, zones of stylocumulate

commonly acted as migration pathways and may contain shows of residual hydrocarbons.

Contamination

Contamination is a common problem. There are numerous drilling fluid additives, lubricants, corrosion inhibitors and other substances,

both downhole and around a rig, which produce spurious hydrocarbon indications. Most insect repellents and sunscreens will contaminate

samples. Rod thread grease (pipe dope or DAFF) and core barrel and bit lubricants can be a serious problem. Some hammer oils are especially bad. Drillers may treat tight hole conditions by copious use of drilling lubricant and by greasing the outside of the drillstring.

In recent NTGS cored holes, this resulted in averages of 0.13 kg/m of grease and 1.0 L/m of bit lubricant. This is excessive contamination and would have made the onsite detection of any formation hydrocarbons impossible. Historically, diesel has also been added to the drilling fluid to combat various hole problems. When suspended in drilling fluid, diesel can scavenge other organic materials from the system and serious contamination will result. Older organic-based mud additives and any organic-based lost circulation material can become part of a biological cocktail as the drilling fluid ferments and methane is liberated. Bactericide treatment will be necessary.

Modern synthetic polymer drilling fluids are less prone to these problems but they do contain C17-C24 hydrocarbons, which are used as biomarkers in oil/source rock matching.

% Any possible contamination should be thoroughly documented to avoid later spurious assays. If any contamination problem is suspected, all potential contaminants should be sampled and submitted to the lab doing the organic geochemistry. Hands should be washed before handling core. Check the offsider getting the core out of the inner tube! You can almost guarantee that the piece of core he/she has gotten from the lifter will have his/her greasy finger marks on it. If a rack is used to lay out core it should be cleaned regularly. Washing core with a rag will leave fluorescent lint behind and the rag frequently becomes contaminated. The solvent being used to undertake onsite hydrocarbon studies should itself be regularly checked for contamination. A final word of warning: using detergent to clean equipment is also inadvisable, as most detergents fluoresce! Use plain water.

COLLECTION AND ASSAY OF NON-CORE SAMPLES

Most non-core lithological samples from NTGS drilling will be from precollars and waterbores. Geochemical samples from RC or RAB, as used for mineral exploration, would also fit into this category. Fluid samples are also important and should not be overlooked.

Rotary open hole samples

Sample collection and labelling

% Open hole samples should be collected at 1 m or 2 m intervals. These samples are normally laid out on the ground by the driller's assistant in rows corresponding to 10 m intervals.

The large volume of sample (a wheel barrow load per 2 m) from a percussion hole is normally put into large transparent plastic sample bags by the drillers. The NTGS site Representative will subsample this material using either a sample splitter or a hollow

spear made from PVC pipe. Subsamples are put into cloth bags and the remainder discarded on site. It is important that the subsample

is representative and uncontaminated (see Sample contamination).

Mud-drilled samples should be washed in a sieve to remove drilling fluid contamination. Cloth bags are used and wet

samples should be allowed to dry in the bags.

Individual cloth bags should always be labelled with the hole number and it should be clear what the composite interval is.

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Tedious as it is, best practice is to give from and to depths and drillhole name on each and every bag.

Cloth bags are put into a polyweave sack for transport. The necks of the polyweave sacks should be fastened with a special

wire-twisting tool. On no account should the drillhole name appear only on the polyweave sack.

This avoids the problem of

mixing of samples from different holes when the cloth bags are removed from the polyweave sack.

NTGS normally provides all bags and ties.

Sample contamination

Some mineral exploration companies are surprisingly casual in the way they treat samples that are expected to be representative

at ppm or ppb levels.

% Cloth bags containing samples for trace metal assay should not be transported or stored in contact with metal surfaces.

There is no point in assaying trace levels of metal in the contents of a sample bag that is rusted to the truck tray floor or in

contact with a corroded zinc-plated metal sample drum.

Some mineral exploration companies used plastic sample bags that were stapled shut. However, it was found that metal

staples were inadvertently included in the sample, either on site or when the bags were opened in the laboratory, and this

resulted in spurious metal assays. In other cases, a zinc anomaly was traced to a cloth sample bag that someone had used to

wipe up spilt sunscreen. High lithium levels were found to have come from grease on a sample bag. Even something as

seemingly benign as a few stray leaves can, depending on the sample preparation and analytical technique, seriously contaminate

a sample. Members of the pea family, for example, contain percentage levels of zinc. Tungsten analyses may be compromised

by fragments from the drill bit (most contain tungsten carbide) or if a tungsten mill is used in the laboratory. Diamonds can

easily be plucked from a diamond drill bit, especially if it is face-set.

Water samples

Water samples are important for groundwater studies and for the determination of resistivity of water (R_w) for wireline log

calibration. Duplicate samples are normally collected in 1 L screw-top plastic bottles. Glass bottles are required for trace

hydrocarbon analysis. Water samples have a limited shelf life and should be analysed as soon as possible after collection. Basic

measurements such as temperature, conductivity and pH should be made on site.

% Before sample collection, all sample bottles should be rinsed with dilute HCl and then repeatedly with the water to be

sampled. Samples should be kept cool as high ambient temperatures can influence subsequent hardness measurements. Bottles

should be permanently labelled with drillhole, depth, date and sampler's name. Air drilling is the best way to detect groundwater.

Water from the aquifer should be airlifted for sufficient time to enable an uncontaminated sample. If the only sample available

is still contaminated, allow it to settle before decanting into the sample container. During coring, it may not be possible to

detect minor groundwater influx and an aquifer may even be a lost circulation zone. It may not be practical to collect water

samples during continuous coring.

Oil samples

% Rinse the sample container repeatedly with the oil to be sampled. If sufficient is available, collect 2 L in glass bottles or tins.

McCartney bottles (screw-top bacterial culture bottles with vinyl, not rubber, seals) can be used for smaller quantities. Containers

should be permanently labelled with drillhole, depth, date and sampler's name.

ENGINEERING

The objective of any drillhole is to penetrate and adequately sample those formations required without risk and to be able to

wireline log the hole if so required. It is the Contractor's responsibility to drill the hole to engineering specifications stipulated

in the contract and detailed drillhole plan.

% The NTGS Representative is expected to monitor the Contractor and ensure that these objectives are met.

Engineering information in daily drilling reports

% The NTGS Representative must ensure that the driller's daily reports contain sufficient engineering information. This

includes details of casing, drilling fluids and hole trajectory. These are described individually below.

Casing

Casing is a hole liner used to curtain off unconsolidated near-surface material, to isolate other formations that cause drilling

difficulties, or to separate formations with different water chemistry or significantly different pressure gradients. On a petroleum

well or waterbore it also forms the conduit for delivery and supports the surface production equipment. Casing can be suspended from the surface, hung from an outer casing string, sat on the hole bottom, surface clamped, grouted or cemented in place. This will depend on the reasons the casing was run, the depth and pressure requirements.

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A conductor is surface casing used to contain unconsolidated material. On holes deeper than about 150 m this is followed

by other strings of casing of decreasing diameter nested inside each other. Once casing is run, a smaller-diameter hole is

necessary to continue drilling. This change in diameter is referred to as stepping down.

PVC pipe with solvent-cemented belled ends and, less commonly, fibreglass or threaded PVC casing are all used at relatively

shallow depths. Generally, steel casing is required elsewhere.

PVC comes in various numbered classes depending on the diameter and wall thickness (Table 5).

All such pipe used

downhole must comply with AS1477. Heavier duty is indicated by a higher class number. Class 9 can be used with care in

shallow situations such as a ≤ 6 m conductor, but Class 12 is preferable elsewhere. Only Type P primer and solvent is to be used

(AS3879). PVC casing joins can be secured with self-tapping screws while waiting for the solvent cement to cure. Care must

be taken to ensure that the screws do not protrude internally and only stainless steel screws are permissible in waterbore casing.

Normally, all PVC casing used in stratigraphic and exploration drilling is disposable and, subject to abandonment procedures

(see COMPLETION AND ABANDONMENT), is not retrieved. However, special threaded PVC casing may be run in multihole

downhole geophysical surveys so it can be reused in the next hole. Always carry extra PVC pipe to allow for damage in transit.

There are many different types of steel casing appropriate to various applications. Butt-welded steel casing, compliant with

AS1396, 1579 and 1836, is used in waterbores, but is non-retrievable and too labour intensive for stratigraphic drilling. Cored

mineral exploration and shallow stratigraphic holes are typically cased with retrievable threaded steel casing (Table 6) that

comes in 3 m lengths, or by using drill rods without a bit, or both (see Engineering design of typical cored stratigraphic

drillholes). Cored holes usually have proportionately more casing than conventional holes. Aside from the engineering aspects,

casing points depend on the rig capacity in each diameter and costs per metre.

% A rule of thumb for a typical cored stratigraphic hole in an unknown area is that after the conductor, casing should be run

in intervals of thirds of the proposed total depth (PTD).

Threaded heat-treated steel casing is used in oilfield and deep mineral holes where pressure-rating is a major issue in hole

engineering. This casing should only be transported with thread protectors and may not be cut or welded. Each section of

casing has a unique length and will be branded with this and API specifications including various grade designations. Pressure-

rated casing is most commonly referred to using a code in which a letter refers to the tensile strength (eg H indicates 60 000 psi;

J, 75 000 psi and N, 100 000 psi). A number indicates the minimum yield strength. For example, Grade J55 and K55 casing

both have a yield strength of 55 000 psi, whereas N80 designates 80 000 psi yield strength. There are numerous thread types,

given three-letter codes (eg EUE, LTC, BTC). The casing is also designated by its weight in lb/ft. In contrast to non-oilfield

applications, drift ID (ie actual ID as measured by prescribed tools) rather than nominal ID is commonly specified.

Adequate casing and a screen will be required if a hole is to be completed as a waterbore. If converting a slimhole to a

waterbore, bear in mind that casing used in waterbores less than 50 m deep must be a minimum of 100 mm diameter;

deeper bores require a minimum diameter of 125 mm.

% NTGS should ensure that the driller is carrying adequate casing for both waterbores and stratigraphic holes and that

holes are cased according to best practice as described above. All casing used in a drillhole must be described in the Daily

Operations Report and recorded in the Drillhole Completion Report. This should clearly indicate casing specifications, shoe

depth and which casing was retrieved and which was abandoned.

Nominal size (mm) Class ID (mm) OD (mm) OD bell

100 9 104.6 114.3 124

12 101.7 114.3 125

125 9 128.4 140.2 152

12 124.9 140.2 155

150 9 146.85 160.25 170

12 142.65 160.25 177

18 134.65 160.25 188

155 9 154.45 168.25 184

12 151.65 168.25 187

175 9 185.15 200.25 222

177 12 158.3 177.0 195

200 6 213.8 225.5 236

9 208.5 225.3 243

12 203.1 225.3 248

225 12 225.75 250.37 275

250 12 252.9 280.4 310

300 12 284.45 315.46 345

Table 5. Specifications of commonly used solvent weld-joint PVC casing. ID = inner diameter; OD = outer diameter. Data from Sinclair

Plastics.

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Steel casing size ID (mm / inch) OD (mm / inch)

AW 48.4 57.1

AX 50.8 57.1

BW 60.3 73.0

BX 62.7 73.0

EW 38.1 / 1.5 46.0

EX 41.3 46.4

H 100.0 114.3

HW 101.6 / 4.0 114.3

N 77.8 88.9

NW 76.2 / 3.0 88.9

PF 101.6 114.3

PW 127.0 / 5.0 139.7

RW 30.1 36.5

SW 152.4 / 6.0 168.2

XRT 28.8 30.2

API 5.5" K55 LTC 15.5 lb/ft 125.73 / 4.95 (4.85 drift) 139.7 / 5.5

API 2.375" J55 4.7 lb/ft 50.55 / 1.995 (1.901 drift) 60.325 / 2.375

Table 6. Specifications of common steel casing. ID = inner diameter; OD = outer diameter.

Drilling fluids

The drilling fluid (loosely called 'mud') performs the following functions, many of which are critical to the drilling process:

- Aid formation stability and productivity
- Clean the bottom of the hole
- Lift formation cuttings to the surface
- Suspend cuttings while circulation is stopped (eg trips)
- Permit cuttings removal at surface
- Control subsurface pressures
- Deposit a wall cake through porous and permeable formations to aid hole stability
- Cool the bit
- Lubricate the drillstring
- Assist in corrosion control of the drillstring
- Allow electrical logs to be run

- Limit any environmental damage caused by the discharge of the drilling fluids themselves or other materials in them

The selection of appropriate drilling fluids and maintenance of optimum fluid properties are essential to technical success, safety and cost effectiveness.

Drilling fluid parameters

On a deep hole, the following drilling fluid parameters may need to be routinely monitored:

- Relative density (RD), sometimes referred to as specific gravity (SG)* or mud weight (ppg is the API standard), which is

measured using a mud balance

- Filtration properties, measured with a standard API filter press (only for water-based muds)
- Flow properties, loosely termed 'viscosity', measured using the resistance to flow through a standard Marsh funnel with

the result reported in seconds, or by using a rotating rheometer that gives readings at different rotation speeds from which

'plastic viscosity' and 'yield point' are determined

- Gel development, measured with a rheometer
- Chemical composition and cation exchange capacity, measured as methylene blue capacity
- pH, measured with test papers or a meter
- Chlorides, measured with sulfuric acid, silver nitrate, phenolphthalein and potassium chromate

It is the responsibility of the driller (or specialist Contractor) to monitor and manage the drilling fluids within the parameters

set by NTGS. % NTGS should ensure that adequate tests are undertaken and that the results are reported on the Daily Operations

Report and recorded in the Drillhole Completion Report.

* To convert ppg (lbs/US gallon; API standard) to SG multiply by 0.12

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Drilling fluid additives

The numerous categories of drilling fluid additives are:

- Bactericides
- Calcium removers
- Corrosion inhibitors
- Defoamers
- Emulsifiers
- Filtrate reducers Safety Note
- Flocculants Many drilling fluid additives are corrosive and/
- Foaming agents or toxic. Materials Safety Data Sheets should
- Lost circulation materials be kept on site and special storage conditions
- Lubricants observed.
- pH control additives
- Shale control agents

- Surfactants
- Thinners, dispersants
- Viscosifiers
- Weighting materials

It is the responsibility of the driller (or specialist Contractor) to maintain adequate stocks on site. % NTGS should monitor the

inventory. The classifications, the proprietary name and the quantities used should be noted in the Daily Operations Report and

recorded in the Drillhole Completion Report. Such information is critical to assessment of drilling engineering, possible sample

contamination, environmental impact and cost control.

Engineering design of typical cored stratigraphic drillholes

The exact configuration of each hole will depend on the near-surface conditions, the nature of the formations to be intersected,

groundwater, depth, rig capacity, drillstring and bits available. Even when these are determined, multiple engineering solutions are

often possible. A typical cored stratigraphic slimhole (Figure 15a and b) drilled with a multipurpose rig would be:

- Auger or air drill into consolidated material (say, 3-6 m) and run a disposable conductor. Cement conductor in place if necessary but bear in mind that abandonment procedures (see COMPLETION AND ABANDONMENT) mean that no concrete or casing can be left within 30 cm of the surface
- Step down and air drill through the weathered zone into competent rock (say, 60-100 m). If sufficient water is found this hole can be developed as an onsite waterbore; the rig is then moved slightly and the precollar repeated, or the stratigraphic hole may be recommenced in PQ core if the ground is found to be suitable
- Run retrievable threaded steel HW casing in the stratigraphic hole
- Begin HQ3 coring, checking recovery; the driller may switch to normal HQ if there are no problems and if HQ offers a cost advantage. Take HQ or HQ3 to about one third of PTD, or no more than 90% rig capacity
- Run NW casing or open-ended HQ drillstring as casing
- Core NQ, or preferably NQ2, to PTD (600-1000 m)
- Retrieve steel casing
- Abandon hole in accordance with regulations

Hole orientation and deviation

The objective is to drill a straight hole but hole orientation will invariably change as drilling progresses. Variance from the planned

trajectory is recorded with respect to both horizontal and vertical frames of reference multiple times during drilling. Such directional

surveys are mandatory at intervals of not more than 200 m in petroleum wells in the NT and, as described below, NTGS has its own

specifications for stratigraphic holes.

Terminology can be confusing. The preferred terms are: 'azimuth', which is recorded as the horizontal direction of the hole

relative to magnetic north, and 'declination', which is the angle with horizontal. Declination may be expressed as a negative number

of degrees, a vertical hole having a declination of -90° . Unintentional changes in azimuth and declination are best known as drift and

deviation respectively. Figure 16 provides examples. Deflection, or deliberate directional drilling, uses these terms in a different

sense. Killeen et al (1995) gives a good review of hole orientation methods.

Don't expect a perfectly straight, perfectly vertical drillhole. There is no such thing. By virtue of the drilling process, all holes tend

to spiral. Even under ideal circumstances, rotary drilled holes will spiral clockwise ('walk to the right' in drilling parlance) and flatten

(trend towards vertical) with depth. Abrupt changes in drilling parameters or lithology can cause sudden changes in deviation and

drift. Maximum deviation will occur if the hole intersects a rock fabric (bedding, foliation) at about 40° . A 'dogleg' is usually defined

32

Figure 15a

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 G.L.
 DISPOSABLE CONDUCTOR 158.75(6.25") h
 6.25" 152.4(6") OD
 RETRIEVABLE OPEN-ENDED HQ RODS USED AS CASING
 RETRIEVABLE HW STEEL
 CUTTINGS
 135.25(5.352") ID
 3
 127(5") h
 114.3 OD
 5"
 101.6 ID
 60
 96.0 h
 HQ3 CORE
 88.9 OD
 HQ3
 77.8 ID
 61.1 HQ3 Core
 200
 NQ2 CORE
 75.7 h
 NQ2
 50.7 NQ2 Core
 600
 TD
 m04-057.dgn

Figure 15. Cross-sections of typical cored slimhole stratigraphic drillholes. Horizontal scale is accurate 1:1. Actual engineering will depend on surface conditions, nature of rocks to be drilled, PTD, abandonment requirements, rig capacity and specifications of bits and tubular elements. (a) A 6" OD 20 lb/ft steel pipe conductor in a 6.25" hole, followed by HW casing in a 5" hole that could be hammer, blade or air drilled, open-ended HQ rods used as casing in the HQ core section, and NQ2 core to TD. (b, overleaf) A 125 mm Class 9 PVC conductor, followed by PQ cored section cased with HW, HQ cored section cased with NW, and NQ core to TD.

33

Figure 15b

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158.75(6.25") h

140.2 OD

125mm CLASS 9 PVC

CUTTINGS

6.25" 152 bell

RETRIEVABLE HW STEEL

128.4 ID

6

RETRIEVABLE NW STEEL

122.6 h

PQ CORE
 114.3 OD
 PQ
 101.6(4.0") ID
 85.0 PQ Core
 100
 96.0 h
 88.9 OD
 HQ CORE
 HQ
 76.2(3.0") ID
 63.5 HQ Core
 300
 75.7 h
 NQ CORE
 NQ
 47.6 NQ Core
 1000
 TD
 m04-058.dgn
 34

as any deviation $> 3^\circ/30$ m. Rotary holes will steepen with less weight on the bit and flatten with high weight on the bit. A dull bit will

deviate more from vertical than a new bit. The high rotation speeds and less pendular weight involved in coring mean that deviation

is more of a tendency than with conventional drilling and that deviation becomes a more serious problem with decreasing hole

diameter. Deviation in excess of 10° is quite common in NQ cored holes in excess of 800 m and Hartley (1994) cites examples of up

to 10° deviation over 30 m for BQ in schist. Significant deviation means that the drillhole may:

- Miss the target (especially important in mineral exploration)
- No longer intersect bedding at 90° (not true stratigraphic thickness)
- Contain sections of tight hole and doglegs that increase drag and wear on the drillstring and lead to further drilling problems

% All stratigraphic holes must be surveyed for deviation and preferably for drift. The NTGS contract stipulates that

readings should be taken no more than 30 m apart. The actual depths will have to correspond to the end of core runs, and

depending on the lengths of collars, rods, subs and core barrel, this may not be exactly at absolute depths of 30 m, 60 m and so

on. A good rule of thumb is that the difference in declination between adjacent readings should not exceed 2°. Any more than

3° in 30 m is considered a dogleg. Insist that any spurious reading is repeated, if not before drilling is resumed, then no later

than the end of the next run. If deviation becomes a problem, surveys should be more closely spaced while remedial action is

underway. Penalties will apply if deviation from the proposed program is excessive. Typical penalties are shown below:

Depth interval Deviation from planned vertical trajectory Penalty

0-300 m greater than or equal to 5° 5% of metreage rate until the hole is within specification or enters another category

300-600 m greater than or equal to 7° 10% of metreage rate until the hole is within specification or enters another category

>600 m greater than or equal to 10° 20% of metreage rate until the hole is within specification or enters another category

any 30 m interval greater than or equal to 3° from previous reading 20% of metreage rate for the next 120 m

Since this is part of monitoring the driller's obligations under the contract, NTGS personnel must be familiar with

how deviation surveys are done and should check all measurements. These must be documented in the Daily Operations

Report.

Hole orientation can be recorded by numerous devices. These range from crude downhole acid-etch tubes (note that

NTGS forbids hydrofluoric acid (HF) on site), through combined compass and gravity tools and gyroscopic devices to

continuous computer telemetry. The recommended types of equipment are described below.

Electronic memory tools

Several tools have been developed to record hole azimuth and declination electronically. This information can be viewed

on a hand-held field computer and downloaded to a PC. The Flexit Multismart tool, for percussion holes, uses two non-

magnetic stainless steel drill rods located behind the hammer and an electronic-memory survey tool in an inner tube.

Data are recorded in 3 m increments as the rods are retrieved. Each reading takes less than 10 seconds. The Ranger

survey system is similar. The Tensor tool uses three magnetometers and two accelerometers to record hole declination,

azimuth and magnetic tool-face readings. It can be run in both rotary and diamond core holes and operated either in

single-shot or continuous mode. Electronic multishot surveys are the preferred technique to survey percussion precollars

since correct precollar alignment is critical for the diamond tail. Such tools are usually only operated by a specialist

Contractor and would not normally be required on fully cored NTGS holes.

Eastman camera

The single-shot Eastman camera is the most widely used in coring. This is a downhole wireline photoclinoimeter that

incorporates a timer, camera, compass and inclinometer. Rubber fingers are used to centre the instrument and it is seated

downhole in a special non-magnetic collar. The type employed in near-vertical holes uses a battery-powered light to

expose a small disc of photographic film resembling a compass face, which is developed back at the surface as a permanent

record of azimuth and declination. Although a special magnification device is available, few drillers have one and the

charts should be read with either a hand lens or a binocular microscope. Declination can be read to the nearest degree and

estimated to 0.1°. Azimuth is read to the nearest 5° and can be estimated to the nearest degree. Be wary when reading the

azimuth as most cameras photograph the card from underneath and so a reading east of north is read in a counterclockwise

direction (Figure 17). Some discs have 0-90° for each quadrant; others have a 0-360° compass scale. Note that the

35

75°

a

300m

15°

65°

MN

90

0m

80

165

270

0°

18

Planned hole

Actual trajectory

75°

b

300m

15°

65°

MN

20° 0m

° 80

70

165°

270

0°

18

Planned hole

Actual trajectory

m04-059.dgn

Figure 16. Explanation of hole orientation terminology. (a) This drillhole was intended to be vertical (90°), but at 300 m it has a declination

of -75°. This equates to a deviation, or build, of +15°. By 800 m, the deviation has increased to +25° (declination -65°). The azimuth of the

hole, towards 165°, is also the drift direction since the hole was planned to be vertical. Most operators would consider such variance from

planned as unacceptable. (b) In this example, the hole was planned for a declination of -70° (20° from vertical) and a heading of due south

(azimuth 180°). At 800 m, the hole was actually -65° from horizontal and so has a deviation from planned of +5° vertical. Its azimuth is 165°

instead of 180°, so it has drifted +25° and is eastnortheast of where it should be. Depending on the size of the target, this may be acceptable.

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exposed discs should be retained by NTGS in their individual envelopes and should not be exposed to sunlight and high

temperature.

Multishot Eastman cameras used in Australia are of the same vintage as the single-shot cameras and operate similarly

but are usually run at the completion of the hole. The multishot tool uses a timer to activate the camera and this requires

considerable skill by the driller to reposition the tool between readings.

Both multiple and single-shot Eastman survey tools can be equipped with angle units for use in holes closer to horizontal. The

declination is read from the long crosshair on a drum scale and the short crosshair records the azimuth on a circular compass scale

(Figure 17).

% Experience has shown that the Eastman camera is prone to failure. NTGS contracts stipulate that there should be two cameras

on site for deep holes. It is prudent to test both instruments at the surface and establish their relative precision. Remember that, at surface, the compasses will be affected by any nearby metal. Downhole, magnetic interference means that it is pointless taking a reading within 10 m of steel casing. Ensure that there are spare batteries, discs and developer and that the developer is kept refrigerated.

Mechanical controlled vertical drift indicators

Although very outdated technology, these inclinometers are more robust and reliable and less heat sensitive than Eastman cameras, but only measure deviation from vertical and not the azimuth of the hole. They are lowered into the hole on a wireline. The most common, a Totco, is routinely used for the upper section, at least, of onshore Australian petroleum wells. It uses a plumb bob to mark a paper bullseye. A timer activates the plumb bob twice, 60 seconds apart, and the bullseye is rotated 180° between the readings. If the survey was successful, both readings should be within 0.5° of each other (the precision) and exactly 180° apart. Instruments and bullseye charts are available in 0-1°, 3°, 5°, 7°, 8°, 14°, 16°, 21°, 24°, and 90° values. Accuracy is commonly taken to be $\pm 0.75^\circ$, but it decreases as the hole deviates further from vertical. % Be sure whether each concentric circle represents 0.5°, 1.0° or 2.0° etc deviation from the vertical.

Electric wireline surveys

Some wireline logging companies specialise in downhole surveying while others will offer it as part of one of their suites (commonly with the dipmeter). Data should only be recorded on the uphole (not downhole) run.

Plotting azimuth and declination / drift and deviation

The older convention was to plot declination (or deviation) and azimuth as being constant over the interval between the midpoints of adjacent surveys. For example:

Depth (m) Deviation (degrees from vertical)

Reading at 30 5

Plot 0-45 7.5

Reading at 60 10

Plot 45-75 12.5

Reading at 90 15

Plot 75-90 15

A better technique is to honour the readings and interpolate between them. There are various computer methods, including the

Angle Averaging Method, Balanced Tangential Method, Radius of Curvature Method (preferred by most petroleum operators)

and Minimum Curvature Method, details of which can be found in BIBLIOGRAPHY.

Calculating true vertical depth and horizontal displacement

% True vertical depth (TVD) and horizontal displacement in a planned vertical hole are calculated as follows:

$$\text{TVD} = \text{Driller's Depth} \times \cos(\text{deviation})^\circ$$

$$\text{HORIZONTAL DISPLACEMENT} = \sin(\text{deviation})^\circ \times \text{TVD}$$

As most holes are spiralled, the calculated TVD and horizontal displacement will be maximum values.

□ Example 3

A drillhole with a driller's depth of 1000 m and a deviation of 10° from vertical has a TVD of 984.81 m and a horizontal displacement of 171.1 m.

37

20

10 N 10 20 3

NE 30 MAG

0

40

40

5

50

50

NW

E

N

60

60

70 8 W

80 E 80 70

0

a

80 7

70

06

60

SW

0

SE

50

50

0 40
 04 3 20 1
 30
 10 20 0
 S
 10
 20
 30
 b 40
 50
 60
 70
 80
 20 1
 0
 N 0 20
 1
 2.0
 1.5
 1.0
 0.5
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Figure 17. Reading hole orientation and deviation surveys. Note that all are shown enlarged about four times actual size. (a) Eastman surveys

from a planned vertical hole. In the diagrammatic representation (left), deviation is read from the crosshair on the bullseye (5° from vertical).

Azimuth is read from the projection of the crosshair onto the circular compass scale around the circumference (058° NE). Note that the

compass face is reversed because the instrument photographs the disc from below and that the quadrants are, somewhat confusingly, each

numbered $0-90^\circ$. The example of an actual Eastman film (right) has a normal compass face with north at zero; concentric circles indicate 1°

increments, and displacement of the crosshair is in the direction of the deviation. The indicated deviation is therefore 2° and azimuth 245° . (b)

Eastman survey in an inclined hole. The long crosshair on the drum scale shows the angle relative to vertical (38°). Azimuth is read from the

short crosshair on the circular compass scale (due N in this case). (c) Chart from a double-acting mechanical drift indicator. The two readings

(arrowed dots) are 180° apart, indicating a successful survey. Assuming a vertical hole was planned, a deviation of 1.25° is read on the

bullseye scale. Adapted from ADITCL (1997).

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Survey data presentation in NTGS reports

% The original survey discs are to be retained by NTGS. The NTGS Technical Note and Drillhole Completion Report will each have tabulated data and a graphical representation of hole trajectory.

DRILLING PROBLEMS

Drilling problems specific to coring are discussed in DRILLING PROBLEMS WHILE CORING; others are discussed below.

DRILLING PROBLEMS ASSOCIATED WITH THE FORMATION

% Under the terms of the contract, NTGS is liable for the extra costs these problems cause, so the Geologist has to be familiar with

them. Unless otherwise specified, remedial action is at the discretion of the driller and this is not a good time to harass the Contractor.

Taking a kick and a blowout

The first influx of overpressured liquid or gas into a drillhole is termed taking a kick. Any uncontrolled flow of hydrocarbons from the drillhole is termed a blowout. Unexpectedly intersecting shallow gas has caused blowouts even on well engineered petroleum wells.

It is disastrous in an inadequately cased hole with a rig that has no blowout preventer, spark arrestors and no kill mud on site. The risk

of shallow gas can be evaluated from the overall petroleum prospectivity, the density of nearby waterbores and other drillholes that

have not intersected shallow gas.

Although shallow gas is a worst-case scenario, the same can be said for the potential of intersecting hydrocarbons (oil or gas)

under pressure at any depth. A diamond core rig also runs the risk of inducing a blowout if the inner tube is retrieved at excessive

speed. The lowered pressure inside the drillstring will allow the influx of formation fluid. This is termed swabbing.

% No NTGS drillhole will be knowingly located on a potential structural or stratigraphic trap and particular attention must be

paid to any hole proposed to intersect known or suspected hydrocarbon reservoir. The final responsibility for site location with

respect to the risk of blowout rests with the Director NTGS.

Abnormal formation pressure

Intersecting rock that is under abnormally high formation (as opposed to hydrostatic) pressure can never be treated lightly. Worldwide,

nearly one half of all onshore petroleum wells and more than one third of offshore wells have experienced trouble of one type or

another by drilling through overpressured formations. Although mostly confined to Mesozoic and younger sedimentary rocks, the

notion that overpressure doesn't affect 'old' rocks is a fallacy. In addition, basins of any age that contain sulphate evaporites must be treated with extra caution.

Abnormal pressure can be generated in several ways:

Compaction: pore water expands with increasing burial depth and increasing temperature, whereas the pore space is reduced by

increasing geostatic load. If pore water is prevented from escaping, the pore water will become overpressured. This is usually a

problem in thick, young shales. Compaction effects can be compounded by tectonic stress, salt or shale diapirism, etc.

Diagenesis: the following processes can all release free water, which if trapped in pore spaces, will give rise to abnormal pressure:

smectite to illite clay diagenesis, recrystallisation of carbonates, gypsum to anhydrite transformation and diagenesis of volcanic ash

(which also produces CO₂ and CH₄). In addition, diagenesis can produce impermeable barriers (eg stylolites) to further confine

formation fluids.

Differential density: when a pore fluid present in any non-horizontal structure has a density significantly less than the normal pore

fluid density in the area, abnormal pressures can be encountered in the updip portion of the structure.

Fluid migration: the upward migration of fluids can result in a shallow formation becoming pressured (or 'charged'). The path for

such migration can be natural or man-made (eg casing leaks, bad cement jobs).

Thermal cracking of hydrocarbons: approximately 85 m³ of gas can be generated from the thermal cracking of one barrel of oil

(Barker, 1990). The by-product organic compounds degrade further and become concentrated in the pore spaces, thus occluding

permeability and further contributing to the overpressure.

Tectonic effects: associated with faults.

Detecting overpressure

Detecting overpressure while drilling involves monitoring changes to lithology, drilling and circulation parameters as a transition

zone is intersected. The most obvious changes are:

- An increase in penetration rate, torque and drag through the transition zone into an overpressured shale
- An increase in drilling fluid density and viscosity
- Rotary drilling of an overpressured ('sloughing') shale will result in long crescent-shaped cuttings and a volume of cuttings seemingly greater than the volume drilled

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- Cored shale will swell and jam in the inner tube or swell and rapidly disintegrate into splinters in the core tray; it may

cause tight hole conditions or disintegrate and cause a washout

- A noticeable increase in flowline temperature

Overpressure problems will get worse the longer the formation is exposed to the drilling. A polymer and/or saline (usually

KCl) drilling fluid may inhibit shale expansion. Where the problem is serious, the formation may need to be cased off.

% The site Geologist should be aware of this problem and inform the driller if overpressure is suspected.

Artesian water

Several basins in the NT contain artesian aquifers. Water may be under considerable pressure and it is not uncommon for a fountain

higher than the mast when open hole drilling. Some artesian water is also very hot (may boil at the surface), which poses an additional threat.

The smaller annulus and higher pump pressure on a diamond core rig mean that the majority of pressure is contained inside

the drillstring. Water pressure may piston the inner tube up uncontrolled after the core is broken at the end of a run.

% Check nearby waterbores for the possibility of artesian water. If there is a possibility, ensure that the driller has the

appropriate licence and training and is informed in advance.

Lost circulation

Lost circulation is the most common and one of the most expensive drilling problems. Aside from the loss of expensive

downhole fluids, it frequently results in the drillhole being terminated prematurely and/or the loss of downhole equipment.

This section deals with a loss of returns to the formation and does not necessarily imply a total loss of returns; it includes partial

and seepage losses. Lost circulation is usually thought of as loss of the drilling fluid, but it also includes loss of cement when

cementing casing. Zones of lost circulation pose special problems in pressure control. There is the potential of an underground

blowout and, since no fluid level can be seen, it is impossible to detect swabbing (inducing a blowout).

Some instances of loss to porous formations are unavoidable. In other cases, lost circulation can be artificially induced due

to negligence. The causes of lost circulation can be grouped into the following categories:

- Natural formation porosity, especially in unconsolidated, cavernous or naturally fractured formations
- Artificially fractured formation, which can result from excessive drilling fluid RD, excessive pump pressure, a formation integrity

test taken too far, a poor cement job or pressure surges associated with drillstring movement (running in the hole too fast or

pulling out too fast with a blocked bit)

In both rotary drilling and diamond coring, lost circulation may be treated with various lost circulation materials (eg mica),

grouts and expanding foam fillers designed to occlude porosity and permeability. Lost circulation is often more difficult to

treat in diamond cored holes. Drilling ahead with no returns due to lost circulation is certainly not a desirable practice, but

fairly commonly undertaken (especially by mineral drillers) for short distances in the hope of regaining circulation. Water,

rather than drilling fluid, is generally used once the decision has been taken to drill blind. It may be necessary to run casing if

it is critical to penetrate below a major lost circulation zone.

% It is up to the driller to decide how best to treat lost circulation, but all action must be taken in consultation with the NTGS site

Representative.

Differential sticking

If the drilling fluid pressure exceeds the pore pressure of a permeable formation, mud filtrate is lost to the formation and

excessive mud filter cake builds up. Differential pressure sticking occurs when the drillstring adheres to the wall of the hole

through this zone. The drill string usually becomes stuck after having been stationary. The driller will be unable to rotate or

move the string up or down but circulation will not be restricted. The slick rods used in continuous diamond coring and collars

used in other drilling are much more prone to differential sticking than pipe with upsets. On continuously cored holes, the

drillstring is left stationary in the hole proportionately longer (while retrieving the core), which increases the risk. Only a few

tens of centimetres of bad formation can cause differential sticking and the situation will only get worse with time. Typical

delays are 24-36 hours. Prevention is far better than cure. Diamond drillholes add special lubricants to the drilling fluid as a

preventative measure. Do not allow drilling fluid RD to become excessive through porous and permeable formations and

maintain optimal hydraulic parameters.

Once the string is differentially stuck, there are several options. Because delays will exasperate the situation, many drillers

will use brute force as the first option. A conventional petroleum drillstring incorporates 'jars' that hammer the string up or

down a short distance specifically for this reason. Petroleum drillers may also 'spot a pill', meaning that they pump a slug of

specially mixed fluid so that it rests within the zone of differential sticking. The pill is designed to break down the filter cake,

thus reducing the bond between the pipe and the wall of the drillhole. There are no jars on a diamond core rig and differential

sticking can be a costly delay. The driller uses specialist drilling fluid additives, often treating the whole mud system rather

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than the specific target zone. The drillstring stick-up is marked and the driller attempts to work free, being careful not to exceed

the allowable stretch and torque. Excessive overpull is to be avoided. Some drillers will use the powerful rig leg jacks to lift the

entire rig (masts and drillstring get bent doing this!). If the first few attempts fail, the best solution is to reduce the hydrostatic

pressure (assuming that it is safe to do so, ie no hydrocarbons or abnormally pressured formations). This can be done by

reducing the RD of the drilling fluid, by swabbing and if necessary, by removal of fluids from the annulus. Although

counterintuitive, it helps if the drillstring is in compression, not tension, because this tends to move the effect of greatest stress

to the stuck point. Alternatively, differential sticking may be overcome by reverse circulation if this is possible with the

hardware available.

Spin out

A combination of formation amenable to the development of a thick mud cake, poor drilling fluid parameters, low fluid

velocities and the centrifugal effect resulting from the high rotation speed used in diamond coring can cause the mud cake to

become overly thick. The drillstring then differentially sticks. This has caused holes to be prematurely abandoned, with loss of

downhole equipment. The risk increases with depth. Experience has shown that problems can arise at only 130 rpm if the

drilling fluid is below par (Dunster 1991), and drilling fluid properties should be carefully monitored to avoid spin out. In a

conventional petroleum well, barite is added to the mud when a gas kick occurs (ie gas under pressure flows into the well). This

is a last resort in cored slimholes because the barite would spin out and prevent the resumption of drilling.

Solids in the annulus

Solid particles such as cuttings and cavings which are too large or too dense to be carried out of the annulus may wedge

between the drillstring and the walls of the hole. The main symptom is the inability to move the string up. It may be able to be

moved down (if off bottom), but then cannot be moved up as far as previously. This occurs in combination with rotation

becoming increasingly difficult. Circulation may also be compromised. Chert from near the surface and pebbles plucked from

a soft matrix conglomerate (especially tillite) are common sources of problems. Diamond cored holes are also prone to 'sanding in', whereby unconsolidated sand or poorly consolidated sandstone builds up above the core barrel and wedges it in the hole.

The drilling fluid should be conditioned to increase its lifting capacity, drilling lubricant should be added, the hole flushed and surged, while working the drillstring. If equipment is available, reverse circulation is another option.

Salt

Intersecting a substantial thickness of NaCl poses several problems. Firstly, it dissolves, meaning reduced or no core recovery

and an out-of-gauge hole. Secondly, it may deleteriously affect the drilling fluid. Polymer systems are generally tolerant to salt

contamination, but salt will reduce the yield point, meaning higher concentrations of polymer are required. Tens of metres of

salt have been cored satisfactorily by supersaturating the drilling fluid with KCl or NaCl.

DRILLING PROBLEMS ASSOCIATED WITH DOWNHOLE EQUIPMENT

These problems are the sole responsibility of the driller and, under most contracts, any delays they cause will be at no expense

to NTGS.

Key seating

This occurs where the drillstring wears into the wall of the hole at one or more doglegs or ledges. On a conventional rotary

hole, upsets or larger diameter tools such as collars, stabilisers or the hammer get stuck when being pulled out. Key seating is

less likely to be a problem on a diamond cored hole with a slick drillstring, but it does occur. Key seats need to be reamed out.

Drillstring washout

The abrasion of slick drillstring against casing or the hole wall (particularly at a dogleg) can weaken the walls or joints of drill

rods to the point of leakage. Of all the cored petroleum slimholes >1000 m deep drilled in Australia prior to 1991, one fifth

experienced at least one drillstring washout and several had multiple washouts (Dunster 1991).

Drillstring washout reduces

circulation to the bit and the driller must be vigilant to detect the decrease in pump pressure, in order to avoid the more

catastrophic downhole problems outlined below. Drillstring washout can be confirmed by circulating a tracer that will return to

the surface prematurely. A washout necessitates an immediate and very careful trip so as not to further compromise a weakened

drillstring.

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Rusty drillstring

An inappropriately stored drillstring will rust, particularly internally. When such drillstring is used, rust flakes from the inside of the string will block the jets and waterways of the bit or may stop the overshot attaching to the inner tube.

Bent rods or drillpipe

A slightly bent drill string will cause unacceptable whip and wobble and be abraded against the side wall of the drillhole. This is especially true at the high rotation speed used in diamond coring. If the bend occurs while drilling, it may be difficult to detect which element(s) of the string are at fault.

Parted drillstring

The drillstring may part by 'twisting off' when a tubular element shears or if a joint comes loose or breaks (Figure 18). If joints are not correctly tightened (make-up torque too low), the pin (male thread) typically fails at about one third from the widest point of the thread. Overtightened threads (make-up torque too high) can break at either the pin, the box (female thread) or both. Diamond drill rods are particularly prone to twisting off because of the combination of thin walls and damage to the outside of the female threaded ends by tongs. If one length is faulty, it is likely that others are too and multiple twist-offs can and do occur.

Figure 18. A twist off. Even the thickest-walled tubular elements, in this case a drill collar with hardened steel walls >3.0 cm thick, can fail. This was one of many such drillstring failures encountered in this conventional petroleum well. X-rays of the drillstring subsequently found numerous cracked and fatigued collars and heavy weight pipe that resulted from abuse on a previous contract. A new drillstring had to be mobilised for the next well in the program.

Figure 19. A new diamond drill bit (left) and a damaged counterpart (right) showing what happens when the bit it is not adequately cooled. Two simultaneous washouts in the drillstring meant that insufficient mud was passing through the bit and within a few metres it had lost its crown and was worn back to the shell. Friction generates considerable heat. The material shown between the bits is a piece of

melted matrix fused to dropped core of shale which itself has become hot enough to be deformed like plasticine before resolidifying. It is normally impossible to retrieve the damaged bit in this situation and the hole would have to be sidetracked or abandoned.

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Undergauge hole

Undergauge hole can be caused by a worn bit, formations that swell or splinter, or excessive wall cake. Remedial reaming back

to bottom can be time consuming, especially when diamond coring, as the bit is not designed to function in this way and

reaming will cause premature wear of the gauge diamonds. Undergauge hole may also cause problems with running casing or wireline logging.

Bit failure

A diamond bit can become 'polished' if the diamonds do not protrude from the matrix sufficiently to cut cleanly. This can be

caused by inappropriate bit selection, excessive rotation speed or too little weight on the bit.

Reaming, fractured ground or

overpressured formation will cause excessive wear on the outer edge of a diamond bit. Overdrilling dropped core will prematurely

wear the inner diamond pads. Blocked waterways are another common cause of diamond bit failure. This can usually be

remedied by better bit selection. Diamond bits will fail catastrophically if for some reason (such as drillstring washout) they are

not cooled (Figure 19). Roller bits can fail due to blocked jets, or the jets may wash out. Bearings are another weak point on

roller bits and can fail before the bit wears to the point it would normally be pulled. A worst-case scenario is for bearing failure

to result in the loss of one or more cones downhole.

Wireline breakage

The wireline that retrieves the core is run thousands of metres every shift. It is prone to kink and fray and has to be replaced

regularly as part of preventative rig maintenance. Broken wirelines are not unusual. They are designed to part at a weak point

at the connection with the overshot.

Fishing

A fish is any undesirable object in the hole that impedes further drilling, the running of casing or wireline logging. Common

fish include a parted drillstring, a failed drill bit, a wireline logging tool (see WIRELINE LOGGING) or foreign object ('junk') in the

hole. Broken overshot jaws can be a problem in diamond coring. Even a tiny fragment of metal can ring or wipe the face of a diamond bit. Attempting to drill ahead without knowing there is junk in the hole is responsible for over half of the premature deaths of diamond bits. The attempted retrieval of a fish (fishing) has been required on up to 20% of all petroleum drilling jobs and 80% of all workovers up to the mid-1980s (Kemp 1990). Basic fishing tools designed to tap into, or over, each of the diameters of casing and drillstring tubular elements should be carried as part of the rig inventory on deep drillholes. These taps and other useful fishing tools are shown in Figure 20.

% It will be up to the NTGS Representative to decide when to cut losses and abandon fishing attempts. This should be agreed with the driller before commencing fishing operations.

SURFACE EQUIPMENT FAILURE

The most common surface equipment failures on a multipurpose rig include mechanical breakdown, blown hydraulic or air hoses, leaks in the top drive swivel and worn pump liners.

WIRELINE LOGGING AND VERTICAL SEISMIC PROFILING

WIRELINE LOGGING

Wireline logging normally accounts for about 10-20% of the total cost of a petroleum well and can be as high as 30% of the cost of a mineral exploration hole. While it can be argued that fully cored holes obviate the need for sophisticated wireline logs, a minimum suite is probably still necessary to facilitate drillhole-to-drillhole correlation, especially to uncored petroleum wells. It is recommended that caliper, gamma and resistivity, at least, be run in all deep stratigraphic holes in petroleum basins and that logging be undertaken with the rig still over the hole. Gamma logging of core at surface (for example, with Geoscience Australia's shielded twin detector logger) is a poor substitute for downhole logs but, depending on freight costs, may be a cheaper alternative.

The downhole wireline logging equipment chosen will depend on the quality of the logs (a function of the Contractor's hardware and software), cost, length of the cable, diameter of the tools relative to the hole, and access to the site. Downhole temperature may also be a limiting factor in unusual circumstances. Waterbore and coal logging equipment will be limited by depth. Most conventional petroleum logging tools will not fit into narrow-diameter slimholes, but several service companies have special small-bore tools. Large petroleum logging units are usually mounted on conventional-drive highway trucks, but slimhole units are typically on multiwheel drives.

% The NTGS site Representative is expected to QC the wireline logging operations.

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Top Sub

Cup

Steel Ball

Seat

a

Barrel

d

Catcher

Shoe

c

b m04-067.dgn

Figure 20. Fishing tools. (a) Carrot tap designed to screw into the drillstring to retrieve pipe or rods. (b) Bell tap that screws over tubular

elements. (c) Reverse circulation junk basket for retrieval of small metal objects. Once the tool is in position, a steel ball is dropped inside the

drillstring. It seats in the tool and reverses the circulation at the face of the bit. Face-set tungsten carbide is used to mill over the junk, which

is flushed into one or more internal basket catchers. A magnet can also be incorporated into the catcher. (d) Poor-boy fishing tool that can be

fabricated on site using the pin on a sub and piece of slotted casing. It is designed to twist down over the fish and the fingers bend closed

underneath it. Images (a)-(c) courtesy of Diamond Boart, Gotco International and Logan Oil Tools.

Preparations

Wiper trip

A wiper trip or dummy trip means pulling the string at least part way out, running back in, then pulling out completely. It is used

routinely by some operators in the oil industry in the belief that this removes any obstructions in the hole, creates a uniform

mud cake and increases the chances of getting to bottom. This practice is unusual in cored drillholes in hard rock. Recommended

NTGS procedure is to wiper trip through any of the following:

- Lost circulation zones
- Thick porous and permeable clastic zones
- Tight hole (doglegs, sloughing shale, etc)
- Where any nearby drillholes experienced trouble wireline logging.

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Conditioning and sampling the drilling fluid prior to wireline logging

Correct sampling and analysis of the drilling fluid in the hole during logging is critical to the evaluation of resistivity and

self-potential (SP) logs. The drilling fluid is sometimes 'sweetened up' (particularly in terms of viscosity) prior to logging.

This practice should be resisted for the following reasons:

- If the drilling fluid could be improved, it should have been done during active circulation while drilling
- Changes to the physical properties of the drilling fluid are invariably accompanied by chemical changes
- Downhole conditions will still be equilibrating when logging commences and will vary during logging
- The drilling fluid that has penetrated the formation may not be physically displaced by any newly conditioned

drilling fluid, but cation exchange will be operating

The recommended drilling fluid preparation prior to logging is simply to circulate bottoms up, preferably twice. The

time circulation stopped and the final flowline temperature should appear on log headers. The last mud sample circulated

off bottom should be sampled for filtrate and mud cake evaluation. Mud resistivity measurements should be made twice,

once immediately after collection and again about midway through the entire logging process.

Ensure that the correct

mud data, including the temperature, appear on the log header. A quick check for unweighted mud is:

R_{mf} should be 0.75-0.88 times R_m

R_{mc} should be 1.11-1.50 times R_{mf}

where the various resistivities are: R_m - mud, R_{mf} - filtrate and R_{mc} - mud cake.

Recording datum and depth

The header should show the datum being used and both driller's and wireline total depths. These depths are rarely the

same. The wireline depth is usually less, often because cavings prevent the tool getting to bottom. There are other

reasons for discrepancy. Even assuming that the measuring wheel is accurately calibrated and there is no slippage, twisting

can shorten the cable by a few percent. In the opposite sense, cable has an elastic stretch of up to 0.8 m per 1000 m. The

wireline depth should be calibrated against casing shoe depth. In the case of oilfield API casing, the casing depth will be

known to within millimetres and should be more precise than the wireline.

SAFETY

% If logging is to be undertaken with the rig still over the hole, ensure that the work area is free of obstructions. The

drillers will assist the logging Contractor to rig up. All other personnel not directly involved must be kept away from the area around the drill floor and from any logging tools laid out on the surface. The hole shall be covered at all times unless there is a tool in the hole and a slotted cover should be used while logs are being run. All sheaves should be properly guarded and loads must be moved across the cable when logging operations are underway. Other general safety procedures outlined in SAFETY AND EMERGENCY PROCEDURES shall be followed.

Logging suite

Most operators will be able to provide preliminary hard copy and final digital data on site. Typical vertical scales generated on site are 1:200 and 1:500. It is handy to have a hardcopy at the same vertical scale as the site lithological log. The digital version should include LIS and/or LAS ≥ 1.0 format.

Tools with a radioactive source are never to be the first logs run. Logs which do not read through casing need only be run 5-10 m into it. Repeat intervals vary from tool to tool but 5-10 m should be considered a minimum.

There are dozens of ever-changing, impressive-sounding proprietary names for the various wireline logging tools.

The basic generic types are described below.

Mechanical caliper

Principle

Measures the vertical change in diameter and sometimes shape of hole with mechanical arms; holds other instruments against the hole wall

Uses

QC of depth on other wireline logs, detection of tight hole or washouts in hole wall that might compromise quality of other log data, calculation of hole volume for cementing

Display

Use metric even if the bit is in inches; the bit diameter may be shown as a dotted line for comparison with hole gauge

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Calibration

Calibrated against rings of known diameter or casing ID

Limitations

Low repeatability, particularly of single arm tool, because of elliptical hole; badly out of gauge holes will exceed caliper

reach

Checks

Check casing reads correctly

Natural gamma ray

Principle

Scintillometric measurement of natural radioactivity, usually as total counts; spectral tools will record separate K, Th

and U

Uses

Definition of bed boundaries, lithological identification, correlation, establishment of 'shale' base line, detection of

radioactive ore deposits and feldspathic sandstones, matching of separate logging runs

Display

API units, 0-100 API or 0-200 API units, linear, increasing to left

Calibration

Test pit

Limitations

Tool must be centred; absolute reading will depend on detector type, logging speed and time constant; thick mud cake

may influence readings; largely unaffected by casing and will read through casing, but is attenuated; barite, KCl and/or

mica (added to combat lost circulation) in drilling fluid will affect readings, subject to temporal variation (drift); vertical

resolution of a basic tool is about 0.5 m so thin beds tend to 'smear' and inflection points appear displaced

Checks

Be sure that logging speed is recorded and that the time constant used was appropriate for that speed; repeatability varies

by lithology and due to random drift; some detectors are temperature sensitive; zero readings are impossible (lowest in

evaporites)

Spontaneous potential (SP)

Principle

Measures difference between the electrochemical potential of a downhole movable electrode and another fixed at the

surface to detect relative difference in salinities of connate water (R_w) and mud filtrate (R_{mf}). Uses the same geometry

as the resistivity tool and is commonly run in combination

Uses

Detect permeable beds, bed boundaries; determine R_w and R_{mf} , definition of shale base line; gives qualitative indication

of 'shaliness'

Display

Millivolts (mV), 10 mV per division, linear scale, negative to left

Limitations

Limited by salinity of fluids, if water salinity and mud filtrate are close, as normally is the case in a freshwater aquifer;

the SP curve will be too flat and of limited use; some filtrate invasion must be present; useless in cased hole; needs

correction for bed thickness, resistivity and hole diameter; sensitive to electrical interference

Checks

Must have good earth connection for surface electrode; check for unacceptable baseline drift and cycling due to magnetism

and bimetallism, telluric affects and cable noise

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Resistivity and conductivity

Principle

Aside from some ore minerals, there are relatively few highly conductive materials in the subsurface, so the conductivity

of sedimentary formations is often a function of the salinity of the formation fluid. Various types of tools have been

designed to measure the comparative resistivities (the inverse of conductivity) at different depths of wall invasion. Electrical

conductivity is usually measured with an induction probe

Uses

Resistivity is a good direct indicator of hydrocarbons, to obtain numerical measures of R_w and R_t for use in the Archie equation to

determine S_w (water saturation), determine oil/water contacts; short-spaced tools good for drillhole-to-drillhole correlation. Conductivity

measurements are important for calibrating airborne EM data

Display

Resistivity 0.2-200 or 0.2-2000 ohm, four-decade logarithmic, shallowest reading shown as a solid line, intermediate as dashed and

deep as dotted

Calibration

Precision resistors

Limitations

Dependant on instrument pad/mud cake contact; has to be corrected for temperature, hole and adjacent bed effects; useless in air-

filled hole, in non-conductive fluids (use IP instead) or cased hole; long- and short-spaced resistivity have different vertical resolution

and respond differently to borehole effects; tool is physically long and needs significant sump

Checks

All resistivity readings should be at or near zero in casing; run repeat section over zones of interests

Induced polarisation (IP)

Principle

A transmitter loop charges the formation with a high alternating current. Any conductive particles will become the most

highly charged with eddy currents. The magnitude of the currents, proportional to the media conductivity, is monitored

using a detector loop. Modern tools have four or more transmitter/receiver pairs. May be run with resistivity tool

Uses

Primarily used in minerals exploration for disseminated sulfides such as volcanic-hosted massive sulfides and porphyry

copper deposits; also used in coal exploration. Some application to uranium exploration in detecting zones of alteration

and redox fronts. Downhole IP may help calibrate surface IP surveys, although it can be difficult to detect if a drillhole

targeted on an IP anomaly has actually intersected it

Display

Various, normally inverted and displayed on a logarithmic scale

Limitations

Will not work in casing; will work in air and non-conductive fluids; temperature dependant

Checks

Compare with other electrical tools, particularly resistivity

Magnetic susceptibility

Principle

Measures the ratio of the intensity of natural magnetisation to the intensity of an applied magnetic field; operates in a

similar way to an IP tool; maghemite, ilmenite and pyrrhotite will all register high responses

Uses

Mineral exploration, detecting palaeochannels, crude lithology determination

Display

Since it is a ratio, it is strictly dimensionless

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Limitations

Measurements in highly conductive material need to be corrected for conductivity affects

Checks

Compare with electrical tools

Sonic

Principle

Measures the speed of pulsed compressional sound waves in subsurface formations (acoustic travel time); both first arrival and

full waveform tools are available. The tool usually comes in two parts, with a lower transmitter separated from one or more upper receivers

Uses

Lithology determination, especially in carbonate and evaporite rocks; porosity evaluation; generation of synthetic seismograms; abnormal pressure indicator

Display

Delta t microseconds/metre, range 500-100 $\mu\text{sec/m}$, usually linear, Φ s 45 to -15 percent porosity

Calibration

Calibrated against a quartz crystal clock, function checked at surface, but a bent tool could pass all checks and produce erroneous logs

Limitations

Tools are usually compensated for changes in hole size and tool tilt; assumes uniform intergranular porosity; cannot recognise secondary porosity; may not show gas in consolidated rock; useless in casing; not a pad-type tool so will work in out of gauge holes but may cycle-skip

Checks

Look for cycle skipping or spurious spikes and slow the tool if more than two spikes per 50 m; sonic in liquid-filled casing

should read 187.0 $\mu\text{sec/m}$ ($= 57 \mu\text{sec/ft}$); values $<131 \mu\text{sec/m}$ or $>623 \mu\text{sec/m}$ are unusual; sonic should register closely with

gamma

Density

Principle

Measurement of radioactivity loss between an emitting source (usually 2 curie ^{137}Cs) and return of gamma rays to one or more

detectors. Loss of energy is by collision with electrons, the electron density being proportional to the bulk density. Compensated

tools use short- and long-spaced detectors to correct for mud cake. Both source and detectors are heavily shielded to minimise

borehole effects

Uses

Porosity evaluation; determination of absolute bulk density, shale content; gas detection

Display

1.90-2.90 g/cm^3 or 1.95-2.95 g/cm^3 , linear; may be displayed relative to the selected matrix (limestone or sandstone)

Calibration

Shop calibrated in blocks of known density; onsite tool verification using two small gamma sources

Limitations

Residual hydrocarbons, especially gas, will give erroneously high porosity; susceptible to shaliness; grain density must be

known quite accurately to calculate porosity, eg a variation of $\pm 0.05 \text{ g/cm}^3$ can produce an error of 2.6% porosity; is a pad tool

and so dependant on hole gauge; relatively slow logging speed so be wary in unstable holes - not a tool to get stuck in the hole!

Checks

Confirm logging speed; repeat section in good hole should be within 0.05 g/cm^3 ; check for drift; check density of known

uniform lithology (eg halite 2.032 g/cm^3 , granite $2.5\text{-}2.8 \text{ g/cm}^3$)

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Neutron

Principle

A neutron source (commonly 20 curie AmBe) bombards the formation with neutrons which are captured by atoms in both the

formation fluid and matrix. Because of the similarity in mass, most neutrons are captured by hydrogen atoms in water or

hydrocarbons which in turn reflect formation porosity. The compensated neutron tool uses two thermal detectors as sensors

Uses

Delineation of porous zones, determination of formation fluid type, direct indication of gas, evaluation of shale content; in

combination with density to identify lithology

Display

Either limestone, sandstone or less commonly dolostone porosity units; usually displayed with density

Calibration

Shop calibration in water-filled tank; on site using two small radioactive sources

Limitations

Elements such as Cl, B and Li interfere; cannot distinguish between water and medium density oil; is influenced by changes in

hole diameter; will work through steel casing but PVC (which contains H) will cause a problem; temperature dependant; shale

content needs to be verified using another log; not a tool to get stuck in the hole!

Checks

Cross-check with density and gamma

Dipmeter

Principle

Shows bedding by comparing the depth shift in four or more microresistivity sensors

Uses

Dip determination, fracture identification; can replace other microresistivity tools

Display

Needs onsite processing to generate tadpole diagrams

Calibration

As per resistivity

Limitations

One or more arms not contacting the wall (floating arm problem)

Checks

As per resistivity

Downhole temperature

Principle

Record sufficient downhole temperature data for correction of wireline logs and drilling fluid resistivity and allow estimation

of geothermal gradient

Uses

Many of the wireline logs, notably resistivity, vary as a function of temperature. Correct values of R_m , R_w , R_{mf} , R_{mc} and K all

require the temperature of the zone under investigation. Flowline temperature should have been recorded prior to cessation of

circulation. Digital temperature logs, where available, are usually run on the way down as one of the first logs and again as one

of the last runs. Normal procedure is to record the bottom hole temperature using a maximum reading thermometer on each

run. Simple as this may sound, experience has shown that up to a quarter of such readings are unreliable. Failure to reset the

thermometer after leaving the tool sitting in the sun doesn't help. Downhole temperature should be recorded at every opportunity

and the time since circulation stopped is also important to calculate a time-dependant correction

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Display

Degrees Celsius

Calibration and checks

It is recommended that several downhole thermometers be available on site and that they are cross-checked at the surface

Limitations

Will not work in an air-drilled hole and unreliable inside casing

Velocity survey and vertical seismic profiling

These techniques, and variations of them, measure acoustic waves between a surface source and a wireline geophone/hydrophone at

various depths in the drillhole. These data are used to establish a tie between subsurface stratigraphy and a surface reflection seismic

section. Downhole techniques are higher resolution than surface seismic. The most basic survey is termed a Velocity Survey or a

Check Shot Survey. These only record first-breaks (first arrival signals). Recording depths are chosen from variations in the sonic log

and are typically dozens or hundreds of metres apart. Vertical seismic profiles (VSPs) are much higher resolution and record the full

wavelet signature of both downgoing and upgoing wavelets and include reflections below the total depth of the drillhole. Receiver

spacing is typically a few tens of metres or less.

These surveys are normally very expensive and undertaken by a specialist contractor. GSWA has undertaken VSPs in some of its

stratigraphic holes. It would only be required in NTGS drillholes that have a sophisticated downhole logging suite (a high-resolution

sonic log is required) and where such drillholes are located on seismic lines. A review of the VSP technique is listed among the online

resources (Appendix 1).

DOWNHOLE LOSS OF WIRELINE EQUIPMENT

Loss of a wireline logging tool is a special case of fishing. The wireline logging contractor should carry a cable spear and a fishing tool

specially designed to tap onto the top of the wireline logging tool. Downhole loss of a tool with a radioactive source is a worst-case

scenario.

COMPLETION AND ABANDONMENT

A stratigraphic drillhole may be completed as a waterbore or, after appropriate plugs etc, may be abandoned. Current NT Government

recommendations for waterbore construction and the abandoning of drillholes are quite onerous and described below.

% The NTGS site Representative must ensure that the drillers leave the hole in an appropriate condition.

Subsurface plugs

Subsurface plugs are necessary to partition aquifers, hydrocarbon-bearing formations or formations with different hydrostatic pressures.

They also prevent reentry of the drillhole. The plugging program will be specific to each drillhole and both waterbore and petroleum

drilling regulations may be applicable.

Most drillholes will require more than one plug. In general, each aquifer is to be cement capped if the hole is to be abandoned.

Tens of metres of cement are required for each plug. For plugs placed off bottom in an open hole (balanced cement plugs), the caliper

log should be used to locate an in-gauge portion of hole. Displacement, bridge, van Ruth, CW or bull nose plugs may used in addition

to cement in special applications.

% NTGS must ensure that the drillers put adequate subsurface plugs in place and that these are properly documented.

Surface plugs for open holes

All holes are to be capped. Stratigraphic holes will have a permanent marker.

Open holes, collared or uncollared, particularly if eroded and cratered, are a danger to wildlife, livestock and vehicles. The

Department requires that all drillholes shall be closed and that the closure shall be permanent. Hence, holes as a minimum must be

plugged with concrete in the manner described in the cross-section diagrams (Figure 21), or backfilled completely using drill chips

or concrete. Note that the use of plastic or PVC caps or plugs such as 'occy' plugs is not allowed.

% NTGS must ensure that the drillers put adequate surface plugs in place and that these are properly documented.

Surface plugs for collared drillholes including NTGS stratigraphic holes

Holes are to be plugged with concrete at least 0.3 m below ground level using a concrete cone plug. The plug will be fitted with

a length of wire rope that extends to the surface, and a labelled metal tag as an indicator. The hole above the plug is to be filled

with compacted earth (Figure 21a). Be wary of concrete too near the surface on land that is likely to be cultivated or graded.

Plugs may be made in moulds such as polyethylene flowerpots or 'witch's hats'. Flowerpot moulds may be reused if the

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drainage holes are blocked with masking tape and the inside wiped with diesel or the mould may be left on the plug. PVC

collars may be readily cut below ground level using a powered brushcutter modified with a diamond masonry blade. If necessary,

the cut section of collar may be removed from the hole using chain tongs or an oil filter remover.

Plugs for uncollared RAB holes

Holes are to be plugged at least 1.0 m below ground level with a concrete cone plug. The plug is to be at least 50 mm larger

than the diameter of the original drillhole but, depending on the nature of the ground, must be of sufficient size as to remain

firmly in position. To enable the placement of the plug the hole may be reamed out with hand tools or counterbored by the

drillrig with a larger drill bit. The hole above the plug is to be backfilled with compacted earth and mounded over at the surface.

(Figure 21b). The intention is that water shall not enter the hole and cause it to erode and reopen, and particular care is

required to ensure the long-term effectiveness of the plug.

Waterbore completion

Waterbore completion must follow NT Government recommendations. The top of the production casing must be a minimum 500 mm above ground level. Should any other casing be installed it must be a minimum 300 mm above ground level. The waterbore will be completed with a locked cap. A reinforced concrete slab surrounds the base of the casing(s). The concrete slab must have a minimum thickness of 100 mm and a minimum diameter of 1 m. The bottom of the cement must be a minimum of 25 mm below the natural ground surface and the top level 75 mm above the natural ground surface. The concrete must be a suitable mix of cement, gravel and sand. The surface of the slab must slope away from the casing(s). A bore marker post 2.3 m long and ≥ 50 mm diameter must be placed approximately 6 m from the completed or abandoned bore. This post will be inserted to a depth of 450 mm and cemented. The bore marker shall have the Registered Number welded clearly on the post and the post shall be painted white. % NTGS must ensure that the drillers complete waterbores in accordance with the appropriate legislation as described above.

compacted fill

_ 1.0 m

0.3 m minimum

reamed-out <

concrete plug

a b

Figure 21. Hole abandonment details. (a) PVC-collared drillhole with concrete plug. (b) Uncollared drillhole with subsurface concrete plug.

SITE RESTORATION

% NTGS is responsible for site restoration. The site should be restored to its original contours except that the area around the

collar must not be a depression. If runoff could erode the site, it should be ripped along the contours and spoon drains constructed.

It is not normally necessary to replant the site, except on agricultural land. The NTGS site Representative must obtain before

and after photographs of the drill location for inclusion in the Drillhole Completion Report.

ACKNOWLEDGEMENTS AND DISCLAIMERS

This manual is intended for informal internal use within NTGS and comes with a standard disclaimer found at the start of

this document. Figures and photographs reproduced here are with the permission of the relevant agencies. This does not

mean that NTGS or the NT Government endorses these products or suppliers. This manual was proofread by Masood

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Comprehensive wellsite manual for onshore conventional petroleum drilling, used by several international joint ventures.

Fraser K, 1991. Managing drilling operations. Elsevier Applied Science, London, New York.

A succinct text from the United Kingdom that provides a good overview of petroleum drilling operations.

Hartley JS, 1994. Drilling - tools and programme management. AA Balkema Publishers, Rotterdam.

A very useful Australian overview of drilling with good treatment of continuous diamond coring, aimed at wellsite geologists.

Heinz WF, 1994. Diamond drilling handbook. Third edition. Sigma Press, Halfway House (Ganteng province, South Africa).

A comprehensive, well illustrated and well written textbook on coring; some engineering aspects are unique to South

Africa.

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Available at: <http://www.mgls.org/95Sym/Papers/Killeen/> Includes a comparison of five different techniques in the same hole.

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A practical guide that includes core logging, orientation and sampling; evolved from the Australian Manual of geological practise used in many university courses.

McPhater D and MacTeirnan B, 1983. Well-site geologist's handbook. McKinlay-Smith International Limited, Penn Well Books, Tulsa.

A basic, very practical wellsite guide for petroleum drilling.

Moore PL, 1986. Drilling practices manual. Second edition. Penn Well Books, Petroleum Publishing Company, Tulsa.

A standard text directed at conventional rotary petroleum drilling; numerous worked examples of calculations.

Seubert BW, 1995. PT, PetroPEP Nusantara, Jakarta. The wellsite guide. An introduction to geological wellsite operations.

Version 5 September 2000. http://www.petropep.de/W_Guide.pdf

A manual for petroleum mudlogging.

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Reviews common methods of testing for shows in cores, sidewall samples and ditch samples, together with a simple method of recording these observations in a consistent manner.

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APPENDIX 1 – ONLINE RESOURCES

GLOSSARIES OF DRILLING TERMS

General

<http://www.doir.wa.gov.au/minerals/glossary.html>

Glossaries of oilfield and drilling terms

<http://www.glossary.oilfield.slb.com/>

http://www.akita-drilling.com/glossary_of_drilling_terms.htm

<http://www.ukooa.co.uk/ukooa/glossary.htm>

http://www.osha.gov/SLTC/etools/oilandgas/glossary_of_terms/glossary_of_terms_a.html

<http://www.mms.gov/glossary/index.htm>

Ocean Drilling Program dictionary

<http://www-odp.tamu.edu/publications/dictionary/dict-intro.html>

Drilling for groundwater

http://www.agwt.org/info/R_Terms.asp

Glossary of casing and pipe terms

<http://www.inter-mountain.com/glossary.htm>

SAFETY MANUALS

Diamond drilling safety manual

<http://www.boartlongyear.com/catalogues/PDFS/diamond.pdf>

Geotechnical drilling operations and inspection

<http://www.doh.dot.state.nc.us/safety/wpsm/ch11b/37.html>

Queensland exploration safety guidelines

<http://www.nrm.qld.gov.au/mines/exploration/pdf/explsafe.pdf>

WIRELINE LOGGING

Wireline logging and log analysis

<http://www.reeves-wireline.com/pdfmineralscatalog/LogAnalMinApps.PDF>

Vertical seismic profiling

http://www.mines.edu/students/p/pjbrown/vsp_paper.pdf (and links therein)

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APPENDIX 2 – GLOSSARY OF COMMONLY USED ABBREVIATIONS

AAPG American Association of Petroleum POOH pull out of hole

Geologists ppf pound per foot (use lb/ft)

APE authorisation for expenditure ppg pounds per gallon (ppg is API standard)

API American Petroleum Institute PTD proposed total depth

BBL US barrel (approximately 159 L); BBL PV plastic viscosity (mud)

is API standard, bbl is preferred by QC quality control

most journals RC reverse circulation

BHA bottom hole assembly RD relative density

BHT Broken Hill-type RDMO rig down, move out

BOP blowout preventer REC recovered / recovery

C&C condition and circulate RIH run in hole

C&F cartage and freight Rm resistivity of mud
 CIF cartage, insurance, freight Rmc resistivity of mud cake
 CIRC circulate Rmf resistivity of mud filtrate
 CSD casing shoe depth Ro residual oil saturation
 DC drill collar ROP rate of penetration
 DCDMA Diamond Core Drill Manufacturers RPM revolutions per minute
 Association Rt resistivity of invaded zone
 DEMOB demobilisation RTSTM rate too small too measure
 DDR daily drilling report Rw formation water resistivity
 DOR daily operations report Rxo resistivity of flushed zone
 DP drillpipe SG specific gravity
 DST drillstem test Sw water saturation
 ECD equivalent circulating density SW saltwater
 EOH end of hole SWC sidewall core
 EMW equivalent mud weight SWL standing water level
 F filtrate SX sacks
 F formation factor TCI tungsten carbide insert (bit)
 FIS free in store TD total depth
 FIT formation integrity test TDS total dissolved salts / solids
 FLT flowline temperature TOC total organic carbon
 FW freshwater Tr trace
 GL ground level TS thin section
 GNC gross nominal capacity TSTM too small too measure
 ID inside diameter TVD true vertical depth
 IP induced polarisation (log) TWT two-way time
 K absolute permeability UDR universal drill rig
 KB (depth below) Kelly bushing US unservicable
 LC lost circulation UV ultraviolet
 LCM lost circulation material VSP vertical seismic profile
 MIRU move in, rig up WL water loss (mud)
 MOB mobilisation WOB weight on bit
 NB new bit WOC wait on cement
 NFG not functional WOD wait on daylight
 NR no returns WOO wait on orders
 OD outside diameter WOT wait on tools
 OH open hole WOW wait on weather
 P&A plugged and abandoned WTS water to surface
 PBTD plugged back total depth X/O crossover (in drillstring)
 PDC polycrystalline diamond compact Y/P yield point (mud)
 PDQ pretty damn quick

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APPENDIX 3 – DAILY OPERATIONS REPORT

The following form is for use on deep cored stratigraphic holes. It can be modified to suit other types of drilling. % This or a

similar form must be completed and dispatched to the office every 24 hours for the time the rig is on site or in transit to another

NTGS drillhole. If no fax is available, normal practice is to dictate the contents of the form over the phone.

Cum = cumulative

Est = estimated

PTD = proposed total depth

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NTGS STRATIGRAPHIC DRILLING – DAILY REPORT

Operations Report – Status at ____ hrs

Drillhole: Date: Shift Hrs: Shift m: Cum Depth:

Contractor: Cum Days: Cum Hrs: Cum m: PTD:

Core Dia: Core Recovery: Deviation: Water Truck:
(\$)(hrs)(km)

Consumables: Other Charges: Est Cum Cost:

Budget Remaining:

Safety Issues: NTGS Personnel:

Summary of Activities:

Geological Report

Current Formation: Prognosed Top: m Actual Top: m

Prognosed Base: m

Maximum dip in core: Formation Problems:

Lithological Summary:

Mineral / Petroleum Occurrences:

Other Remarks

Report Prepared By: Sent/Relayed By:

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APPENDIX 4 – NTGS DRILLING LOG

This is a simple generic form. It can be modified to suit the needs of specific projects.

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NORTHERN TERRITORY GEOLOGICAL SURVEY Scale 1:

Project: Drillhole: Date: / /

Location: Geologist: Sheet: of

Sample/Photo

Recovery

Box No.

Bed TK

Metres

Dip

i Lithology & Grainsize Description and Comments

P G V C M F V Z C

ntgs core logging form. m04-055.dgn

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APPENDIX 5 – HYDROCARBON SHOW LOG

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